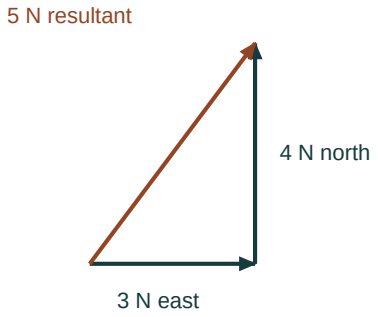
01

Mechanics and Heat

A field guide to understanding, working and checking.

Resolve first. Then add.



Horizontal component: 3 N

Vertical component: 4 N

The diagonal combines both.

Motion, vectors, forces, work, energy, temperature and heat. New merged lectures are the main mechanics source; the older heat lecture supplies thermal topics.

10 lessons / 2 practice tests / Worked solutions

English edition / November 2026

How to use this handout

1. Read the concept and check the symbol meanings before choosing a formula.
2. Cover the worked steps and reproduce the reasoning yourself.
3. Attempt the quick check and the two subject tests without notes.
4. Mark your work; record whether the error was concept, method, algebra, units or reading. Retry after a delay.

Your study route

1. Choose units and describe motion
2. Constant acceleration and graphs
3. Resolve vectors before adding
4. Projectiles: one time, two directions
5. Newton laws and friction
6. Circular motion and connected objects
7. Work and kinetic energy
8. Energy conservation and springs
9. Temperature, calorimetry and phase changes
10. Heat transfer and thermal expansion

LESSON 01 / 10

Choose units and describe motion

After this lesson, you can

- Distinguish distance from signed displacement.
- Calculate average speed and average velocity from a route.

Start with: Add and subtract lengths; divide by elapsed time; convert minutes to seconds.

THEORY

Position tells you where an object is. Displacement is final position minus initial position; distance counts the entire path. Average velocity uses displacement, while average speed uses distance. A negative velocity gives direction, not a negative speed.

RULE

$$\bar{v} = \Delta x / \Delta t$$

$$\text{average speed} = \text{distance} / \Delta t$$

Δx final position minus initial position (m)

Δt elapsed time (s)

$\bar{v}$ average velocity (m/s)

A method that earns marks

1. Choose east or right as positive and sketch the route.
2. Add every path length for distance; subtract start position from finish for displacement.
3. Divide distance and displacement separately by the same total time.
4. Attach a direction to velocity and a unit to every result.

WORKED EXAMPLE

A student walks 30 m east and 10 m west in 20 s. Distance is 40 m, displacement is 20 m east, average speed is 2 m/s, and average velocity is 1 m/s east.

1. Take east as positive: the two legs are +30 m and -10 m.
2. Distance counts both legs: 30 + 10 = 40 m; displacement is 30 - 10 = +20 m.
3. Over 20 s, average speed is 40/20 = 2 m/s and average velocity is +20/20 = +1 m/s, east.

Common mistake
Do not replace displacement with distance in a velocity question.

Exam habit: Write the two numerators before dividing: distance for speed, displacement for velocity. This earns method marks even if arithmetic slips.

SELF-CHECK

Can average speed be nonzero while average velocity is zero?

Answer: Yes. A return trip has zero displacement but a positive total distance.

LESSON 02 / 10

Constant acceleration and graphs

After this lesson, you can

- Select a constant-acceleration equation from the known quantities.
- Interpret slope and area on a velocity-time graph.

Start with: Signed velocity; rectangle and triangle areas; rearranging a simple equation.

THEORY

Acceleration measures the change of velocity per second. On a position-time graph, slope is velocity. On a velocity-time graph, slope is acceleration and signed area is displacement. The constant-acceleration equations require acceleration to remain constant.

RULE

$$v = u + at$$

$$s = ut + \tfrac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

u initial velocity (m/s)

v final velocity (m/s)

a constant acceleration (m/s^2)

s displacement (m)

t elapsed time (s)

When this applies

The equations in this lesson assume constant acceleration throughout the stated interval.

A method that earns marks

1. List the known quantities with signs and identify the unknown.
2. Check that the acceleration is constant before using the equations.
3. Choose an equation without an extra unknown, then substitute with units.
4. Use graph slope or area as an independent interpretation of the result.

WORKED EXAMPLE

A car starts from rest and accelerates at 3 $\mathrm{m/s^2}$ for 4 s. Its speed is 12 m/s and displacement is 24 m. The triangular area under its velocity-time graph gives the same displacement.

1. Starts from rest means u = 0; the example gives a = 3 m/s^2 and t = 4 s.
2. The velocity gain is at = 12 m/s, so v = 12 m/s.
3. The displacement is one half times 3 times 4^2 = 24 m.
4. On the velocity-time graph the triangle has base 4 s and height 12 m/s; its area is also 24 m.

Common mistake
A straight horizontal velocity-time graph means zero acceleration, not zero velocity.

Exam habit: For a graph, state which axes you used before calculating slope or area; their units identify acceleration or displacement.

SELF-CHECK

A car has velocity +8 m/s and acceleration -2 m/s^2. Is it initially speeding up?

Answer: No. Acceleration opposes its current velocity, so its speed initially decreases.

LESSON 03 / 10

Resolve vectors before adding

After this lesson, you can

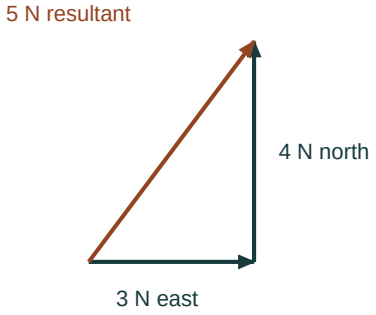
- Add perpendicular vectors using signed components.
- Give a resultant direction in the correct quadrant.

Start with: Right-triangle trigonometry; Pythagoras; compass directions.

THEORY

A vector has size and direction. Two vectors must be added through their signed components, not by simply adding magnitudes. The resultant points from the initial tail to the final head in a head-to-tail drawing.

Resolve first. Then add.



Horizontal component: 3 N

Vertical component: 4 N

The diagonal combines both.

Perpendicular forces add as vectors. A 3 N east component and 4 N north component give a 5 N resultant; the balancing force points the other way.

RULE

$$A_x = A\cos\theta$$
$$A_y = A\sin\theta$$
$$R = \sqrt{R_x^2 + R_y^2}$$

A_x horizontal vector component (same as vector)

A_y vertical vector component (same as vector)

R resultant magnitude (same as vector)

θ angle from the stated axis (degrees or radians)

A method that earns marks

1. Draw axes and identify the reference axis for each angle.
2. Resolve each vector and assign signs to its components.
3. Add horizontal components and vertical components separately.
4. Calculate magnitude and use component signs to choose the direction.

WORKED EXAMPLE

A 3 N force east and a 4 N force north produce a 5 N resultant, 53.1 degrees north of east. The balancing force is 5 N in the opposite direction.

1. East and north are perpendicular, so the example has Rx = 3 N and Ry = 4 N.
2. The magnitude is the square root of $3^2 + 4^2$, giving 5 N.
3. The angle satisfies tan θ = 4/3, so θ = 53.1° north of east.
4. A balancing force reverses both components: 3 N west and 4 N south, with the same 5 N magnitude.

Common mistake
The inverse tangent alone can return the wrong quadrant.

Exam habit: Report both an angle and its reference direction; an angle alone can leave the quadrant ambiguous.

SELF-CHECK

Does a vector with negative horizontal and positive vertical components point northeast?

Answer: No. It points northwest, in quadrant II.

LESSON 04 / 10

Projectiles: one time, two directions

After this lesson, you can

- Separate a projectile into horizontal and vertical motions.
- Use the shared flight time to calculate horizontal range.

Start with: Constant-acceleration motion; squaring and square roots; horizontal and vertical directions.

THEORY

Ignoring air resistance, horizontal acceleration is zero and vertical acceleration is downward. Horizontal and vertical motion share the same time. At the top, only the vertical velocity becomes zero; horizontal velocity continues.

RULE

$$v_x = u\cos\theta$$
$$v_y = u\sin\theta - gt$$
$$y = u\sin\theta\, t - \tfrac{1}{2}gt^2$$

u_x initial horizontal velocity (m/s)

u_y initial vertical velocity (m/s)

g gravitational acceleration magnitude (m/s^2)

t flight time (s)

When this applies

Ignore air resistance and take gravity as uniform; launch and landing heights need not match.

A method that earns marks

1. Draw the launch point and select a positive vertical direction.
2. Write horizontal motion with zero acceleration and vertical motion with gravity.
3. Use the height information to find the positive physical flight time.
4. Use this same time in the horizontal displacement equation.

WORKED EXAMPLE

A ball leaves a horizontal table at 3 m/s and takes 0.4 s to land. Range = 1.2 m and drop = 0.5 times 9.8 times 0.4 squared = 0.784 m.

1. A horizontal launch has initial vertical velocity zero; its horizontal velocity remains 3 m/s.
2. The stated time is 0.4 s, so horizontal range is 3 times 0.4 = 1.2 m.
3. The vertical drop is one half times 9.8 times 0.4^2 = 0.784 m.
4. The unequal range and drop are consistent because the two directions obey different motion equations.

Common mistake
Do not set the total velocity to zero at the highest point.

Exam habit: If you choose upward positive, the downward displacement and acceleration must both carry negative signs.

SELF-CHECK

At the highest point of an angled projectile, which velocity component is zero?

Answer: Only the vertical component; horizontal velocity remains constant in this model.

LESSON 05 / 10

Newton laws and friction

After this lesson, you can

- Construct a free-body diagram for one selected object.
- Use net force and friction to find acceleration.

Start with: Signed addition; mass and weight; perpendicular directions.

THEORY

A free-body diagram shows only forces acting on the selected object. Net force causes acceleration. Action-reaction forces act on different objects and therefore do not cancel on one free-body diagram. Static friction adjusts up to its maximum; kinetic friction applies during sliding.

RULE

$$\sum F = ma$$
$$W = mg$$
$$f_k = \mu_k N$$
$$f_s \leq \mu_s N$$

m object mass (kg)

N normal contact force (N)

μ_k coefficient of kinetic friction (dimensionless)

a acceleration (m/s^2)

When this applies

The example treats the block as sliding and the given 4 N as kinetic friction.

A method that earns marks

1. Isolate one object and draw only the forces acting on it.
2. Choose axes and use the perpendicular force balance to obtain the normal force if needed.
3. Choose static or kinetic friction from the stated motion.
4. Add signed forces along the motion and divide the net force by mass.

WORKED EXAMPLE

A 2 kg block on a horizontal surface is pulled with 10 N. With kinetic friction 4 N, the net force is 6 N and acceleration is 3 $\mathrm{m/s^2}$.

1. Choose the 10 N pull as positive; the 4 N sliding friction acts in the opposite direction.
2. The horizontal net force is 10 - 4 = 6 N.
3. For the 2 kg block, a = 6/2 = 3 m/s^2 in the direction of the pull.
4. Weight and normal force balance vertically under the horizontal-surface assumptions, so they do not enter the horizontal sum.

Common mistake
The normal force is not always mg: angled pulls and inclines change it.

Exam habit: Show the signed force equation before substitution; do not write the applied force alone as ma.

SELF-CHECK

A sliding block has a 7 N pull and 7 N friction. Must it be at rest?

Answer: No. Zero net force gives zero acceleration; it may continue moving at constant velocity.

LESSON 06 / 10

Circular motion and connected objects

After this lesson, you can

- Identify the real forces that supply inward acceleration.
- Set up separate force equations for connected masses.

Start with: Newton second law; force diagrams; uniform circular motion directions.

THEORY

Changing direction is acceleration even at constant speed. Centripetal means inward; it names the required net force, not an extra physical force. In an ideal taut string over a massless frictionless pulley, connected objects share acceleration magnitude.

RULE

$$a_r = v^2/r$$

$$\sum F_{\text{inward}} = mv^2/r$$

v speed along the circle (m/s)

r circle radius (m)

a_r inward acceleration (m/s^2)

T string tension (N)

When this applies

Connected-object formulas assume a taut massless string and an ideal frictionless pulley when stated.

A method that earns marks

1. For circular motion mark the centre and select inward as a positive radial direction.
2. Sum the actual radial force components and equate them to mv^2/r.

3. For connected objects draw a diagram for each mass and use compatible acceleration directions.
4. Use the ideal string constraint to share acceleration magnitude, then solve for acceleration and tension.

WORKED EXAMPLE

A 2 kg object moves at 3 m/s in a circle of radius 1.5 m. Inward acceleration = 6 $\mathrm{m/s^2}$ and inward net force = 12 N.

1. In the circular example, v = 3 m/s and r = 1.5 m.
2. Squaring speed first gives ar = 9/1.5 = 6 $\mathrm{m/s^2}$, directed toward the centre.
3. Multiplying by the 2 kg mass gives 12 N inward net force.
4. The problem would need a physical setup to decide whether tension, gravity or another force provides those 12 N.

Common mistake
Do not add a separate centripetal force to tension or gravity.

Exam habit: Do not put tension and an extra centripetal force on the same diagram; centripetal describes the inward net force.

SELF-CHECK

If the speed remains fixed and radius doubles, what happens to inward acceleration?

Answer: It halves because v^2/r is inversely proportional to radius at fixed speed.

LESSON 07 / 10

Work and kinetic energy

After this lesson, you can

- Determine the sign of work from force and displacement directions.
- Calculate net work from a change in kinetic energy.

Start with: Cosine of common angles; squaring speed; positive and negative energy transfers.

THEORY

Work transfers energy when a force has a component along displacement. A perpendicular force does no work. Net work equals the change in kinetic energy. For a varying force along a line, work is the signed area under the force-position graph.

RULE

$$W = Fd\cos\theta$$
$$K = \tfrac{1}{2}mv^2$$
$$W_{\text{net}} = K_f - K_i$$

W work transferred by a force (J)

F force magnitude (N)

d displacement magnitude (m)

K kinetic energy (J)

A method that earns marks

1. Identify the displacement and the angle between it and the force.
2. Use only the force component along displacement to calculate work.
3. Add work from all forces for net work, or use the kinetic-energy change.
4. Check that positive net work corresponds to an increase in speed.

WORKED EXAMPLE

A 10 N force pulls through 3 m at 60 degrees to displacement. Work = 10 times 3 times 0.5 = 15 J.

1. The example states an angle of 60° between the 10 N force and 3 m displacement.
2. Only 10 cos 60° = 5 N acts along the displacement.
3. Work by this force is 5 times 3 = 15 J.
4. The remaining perpendicular component contributes zero work; 15 J is not automatically net work if other forces also do work.

Common mistake
Use the force-displacement angle, not an unrelated angle in the drawing.

Exam habit: When using the energy theorem, square each speed before subtracting; (vf - vi)^2 is not vf^2 - vi^2.

SELF-CHECK

Can a nonzero force do zero work during a displacement?

Answer: Yes. A force perpendicular to the displacement does zero work.

LESSON 08 / 10

Energy conservation and springs

After this lesson, you can

- Track kinetic, gravitational and spring energy between two states.
- Identify when a loss term is required in an energy balance.

Start with: Work and kinetic energy; choosing a height reference; spring compression or extension.

THEORY

Energy can move between kinetic, gravitational and elastic stores. Mechanical energy stays constant only when the model excludes losses or other transfers. Choose a height reference; only height changes affect the result.

RULE

$$K_i + U_i + W_{\text{other}} = K_f + U_f$$

$$U_g = mgh$$

$$U_s = \tfrac{1}{2}kx^2$$

U_g gravitational potential energy (J)

U_s elastic potential energy (J)

k spring stiffness (N/m)

x spring deformation from natural length (m)

When this applies

The falling-mass example neglects air resistance and other energy transfers; spring-energy formulas require the proportional linear range.

A method that earns marks

1. Name initial and final states and choose a zero-height level.
2. Write the energy balance before removing terms that are zero.
3. Include friction or other transfers if the model contains them.
4. Solve for the unknown and check the physical sign and units.

WORKED EXAMPLE

A mass drops 5 m from rest without losses. With g = 10 $\mathrm{m/s^2}$, its final speed follows v squared = 2gh = 100, giving 10 m/s independent of mass.

1. Set the final height to zero; the initial height is then 5 m.
2. The object starts from rest, so its initial energy is gravitational: mg times 5.
3. With no losses, mg times 5 equals one half mv^2; dividing by mass shows why mass cancels.
4. Using g = 10 m/s^2 gives v^2 = 100 and speed v = 10 m/s.

Common mistake
Spring extension is measured from natural length, not total length.

Exam habit: For spring energy convert centimetres to metres before squaring the deformation.

SELF-CHECK

Would the same loss-free 5 m fall give a heavier object a larger final speed?

Answer: No. Mass cancels from mgh = one half mv^2.

LESSON 09 / 10

Temperature, calorimetry and phase changes

After this lesson, you can

- Separate temperature change from phase change in a heating problem.
- Apply an energy balance with consistent mass and temperature units.

Start with: Energy in joules; mass conversion; subtraction of temperatures.

THEORY

Temperature describes thermal state; heat is energy transferred because of a temperature difference. During a phase change of a pure substance at fixed pressure, energy can transfer without a temperature change. In an insulated mixing model, heat lost equals heat gained.

RULE

$$Q = mc\Delta T$$

$$Q = mL$$

$$T_K = T_C + 273.15$$

Q energy transferred as heat (J)

c specific heat capacity (J/(kg K))

L specific latent heat (J/kg)

ΔT temperature change (K or °C difference)

A method that earns marks

1. Identify the initial and final phase and temperature.
2. Split the process into warming, phase-change and further-warming stages where needed.
3. Use mcΔT for each warming stage and mL for each phase change.
4. Add the energy requirements or balance heat gained and lost in an insulated mixture.

WORKED EXAMPLE

Heating 0.20 kg of water by 10 degrees C needs Q = 0.20 times 4200 times 10 = 8400 J. Melting ice requires a separate latent-heat term.

1. The water remains liquid while warming, so the relevant relation is Q = mcΔT.
2. A rise of 10°C is a temperature difference of 10 K.
3. For 0.20 kg and c = 4200 J/(kg K), Q = 0.20 times 4200 times 10 = 8400 J.
4. If the process also melts ice, that extra stage needs mL; it is not included in the 8400 J.

Common mistake
A temperature difference of 10 degrees C equals 10 K; an absolute temperature does not.

Exam habit: On a heating curve label sloping segments mcΔT and horizontal phase-change segments mL.

SELF-CHECK

Does liquid water warming from 20°C to 30°C have ΔT = 283.15 K?

Answer: No. ΔT = 10 K. The Celsius-to-kelvin offset cancels when subtracting two temperatures.

LESSON 10 / 10

Heat transfer and thermal expansion

After this lesson, you can

- Classify heat transfer as conduction, convection or radiation.
- Predict rate changes and calculate small linear expansion.

Start with: Direct and inverse proportionality; temperature differences; metres and millimetres.

THEORY

Conduction transfers energy through matter without bulk transport; convection uses fluid motion; radiation uses electromagnetic waves and can cross vacuum. Most solids expand on warming. A thermal insulator reduces transfer but does not create cold.

RULE

$$\Delta L = \alpha L_0 \Delta T$$

$$P_{\text{cond}} = kA\Delta T/L$$

$$P_{\text{rad}} = \epsilon\sigma A(T^4 - T_s^4)$$

P heat-transfer rate (W)

k thermal conductivity (W/(m K))

A transfer surface area (m^2)

L wall thickness (m)

α linear expansion coefficient (K^{-1})

When this applies

The wall comparison uses steady one-dimensional conduction with unchanged material and temperature difference.

A method that earns marks

1. Identify the physical mechanism and select the matching relation.
2. For a comparison, hold every unstated quantity fixed and change only the named factor.
3. For expansion use the original length and the temperature rise.
4. Use kelvin for absolute temperatures in radiation and convert the final length if requested.

WORKED EXAMPLE

Doubling the thickness of an otherwise unchanged wall halves its conductive heat-transfer rate. Doubling the area doubles that rate.

1. For conduction, the same wall material and temperature difference give P proportional to A/L.
2. Doubling thickness replaces L by 2L, so P becomes P/2.
3. Doubling area instead gives 2P.
4. If both area and thickness double together, the factors cancel and the rate is unchanged.

Common mistake
Use kelvin in fourth-power radiation equations; Celsius produces incorrect results.

Exam habit: For a ratio question explicitly state what stays constant; otherwise the proportionality is incomplete.

SELF-CHECK

A wall is made three times thicker with all other quantities unchanged. What is its new conduction rate?

Answer: One third of the original rate.

Before the tests

These original questions practise the lesson objectives. They are not predictions of an exam paper. Use the separate practice book if you want all papers together.

Mechanics and Heat / Test A

25 marks | Suggested time: 25 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. A runner completes one lap and returns to the start. Average velocity is:

   A. Always positive

   B. Impossible to define

   C. Zero

   D. Equal to average speed

2. The area under a velocity-time graph gives:

   A. Displacement

   B. Force

   C. Speed at the final instant

   D. Acceleration

3. Two perpendicular forces of 6 N and 8 N have resultant magnitude:

   A. 10 N

   B. 48 N

   C. 2 N

   D. 14 N

4. At the top of a projectile path, acceleration is:

   A. Zero

   B. Horizontal

   C. Upward

   D. Downward with magnitude g

5. An object moving at constant velocity has:

   A. A forward net force

   B. Increasing acceleration

   C. Zero net force

   D. No forces acting at all

Part II: Show your reasoning

1. A cart travels 12 m right then 4 m left in 4 s. Find distance, displacement, average speed and average velocity.

2. A car slows from 20 to 8 m/s in 4 s with constant acceleration. Find acceleration and displacement.

3. Find the magnitude and direction of a vector with components -3 m and +4 m.

4. A ball is projected horizontally at 5 m/s from a height of 4.9 m. Ignore air resistance and use g = 9.8 $\mathrm{m/s^2}$. Find flight time and horizontal range.

5. A 5 kg block slides to the right on a horizontal floor with coefficient of kinetic friction 0.2. It is pulled horizontally to the right with 20 N. Use g = 10 $\mathrm{m/s^2}$. Find normal force, friction and acceleration.

Mechanics and Heat / Test B

25 marks | Suggested time: 25 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. Doubling speed at fixed radius changes centripetal acceleration by a factor of:

   A. Two

   B. Four

   C. Eight

   D. One half

2. A force perpendicular to displacement does work equal to:

   A. Fd

   B. Zero

   C. -Fd

   D. Half of Fd

3. If the extension of a Hooke-law spring doubles while k stays constant, its stored energy becomes:

   A. Four times as large

   B. Twice as large

   C. Unchanged

   D. Half as large

4. During melting at fixed pressure, added heat mainly:

   A. Raises temperature continuously

   B. Changes phase

   C. Decreases mass

   D. Disappears

5. Which heat-transfer mechanism can operate through a vacuum?

   A. Fluid mixing

   B. Convection

   C. Conduction through air

   D. Radiation

Part II: Show your reasoning

1. An ideal Atwood machine has masses 2 kg and 3 kg. Use g = 10 $\mathrm{m/s^2}$. Find acceleration and tension.

2. A 2 kg object speeds up from 3 to 5 m/s. Find the net work done.

3. A spring of constant 200 N/m is compressed by 0.10 m and releases a 0.50 kg block on a frictionless horizontal surface. Find its speed when the spring returns to natural length.

4. How much energy heats 0.50 kg of water from 20 to 30 degrees C? Use c = 4200 J/(kg K).

5. A rod is 2 m long and has expansion coefficient 0.000012 per K. It warms by 50 K. Find the increase in length in metres and millimetres.

Mechanics and Heat / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **C.** The net displacement is zero even though the distance travelled is not.
2. **A.** Area has units (m/s) times s = m and includes the sign of velocity.
3. **A.** The Pythagorean calculation gives the square root of 36+64 = 10 N.
4. **D.** Gravity continues to act at the top.
5. **C.** Constant velocity means zero acceleration, so forces balance.

Part II: Written solutions

1. Distance = 16 m; displacement = +8 m; speed = 16/4 = 4 m/s; velocity = 8/4 = +2 m/s. Award one mark for each quantity.
2. a = (8-20)/4 = -3 $\mathrm{m/s^2}$. s = [(20+8)/2] times 4 = 56 m. Two marks for each calculation, including the sign and unit.
3. Magnitude = 5 m. The vector is in quadrant II: 126.9 degrees counterclockwise from positive x, or 53.1 degrees north of west. Two marks for magnitude and two for a correct direction.
4. 4.9 = 0.5 times 9.8 times t^2, so t = 1 s. Range = 5 times 1 = 5 m. Two marks per result with the equation shown.
5. N = 50 N, friction = 0.2 times 50 = 10 N, net force = 10 N, a = $1\frac{0}{5}$ = 2 $\mathrm{m/s^2}$. One mark for each step.

Mechanics and Heat / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **B.** The speed is squared.
2. **B.** The cosine of 90 degrees is zero.
3. **A.** Spring energy is proportional to extension squared.
4. **B.** Energy changes the phase while temperature remains at the melting point.
5. **D.** Electromagnetic radiation needs no material medium.

Part II: Written solutions

1. Net driving force on the two-mass system is (3-2) times 10 = 10 N. Total mass = 5 kg, so a = 2 $\mathrm{m/s^2}$. For the rising 2 kg mass, T-20 = 2 times 2, hence T = 24 N. Two marks per result.
2. Net work = 0.5 times 2 times (25-9) = 16 J. Give marks for the energy theorem, both squared speeds, substitution and unit.
3. Spring energy = 0.5 times 200 times 0.10 squared = 1 J. Set 1 = 0.5 times 0.50 times v squared to get v = 2 m/s. Two marks for each stage.
4. Temperature rise = 10 K. Q = mc times temperature rise = 0.50 times 4200 times 10 = 21000 J. Four marks: model, temperature difference, substitution, result/unit.
5. Increase = 0.000012 times 2 times 50 = 0.0012 m = 1.2 mm. Award marks for formula, substitution, metres and conversion.

Source and coverage notes

Motion, vectors, forces, work, energy, temperature and heat. New merged lectures are the main mechanics source; the older heat lecture supplies thermal topics.

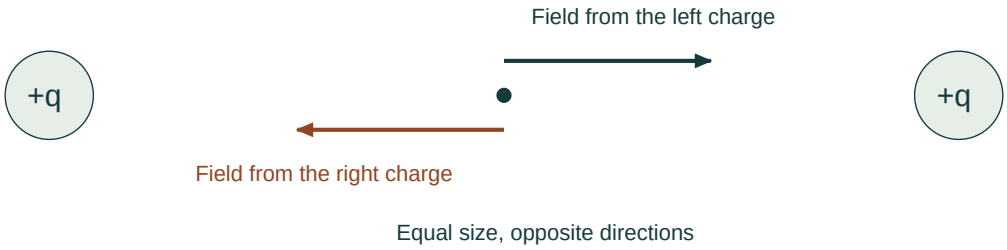
Source folder: Open original course folder The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. This revised study aid does not replace lecturer-specific derivations or instructions. All practice tests and suggested marking guidance are original study material.

02

Electricity and Magnetism

A field guide to understanding, working and checking.

Equal positive charges: fields cancel at the midpoint



Concept-first coverage of the course plan. Chapters 23-31 have lecture support; inductance, AC and EM waves are course-plan extensions without separate lecture files in this folder. Advanced field integrals are recognition topics, not worked derivations.

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12 lessons / 2 practice tests / Worked solutions

English edition / November 2026

How to use this handout

1. Read the concept and check the symbol meanings before choosing a formula.
2. Cover the worked steps and reproduce the reasoning yourself.
3. Attempt the quick check and the two subject tests without notes.
4. Mark your work; record whether the error was concept, method, algebra, units or reading. Retry after a delay.

Your study route

1. Charges, conductors and Coulomb force
2. Electric field and superposition
3. Flux and Gauss law
4. Symmetry and conductors
5. Potential is not field
6. Capacitors and dielectrics
7. Current, resistance and electrical power
8. DC networks, Kirchhoff rules and RC behaviour
9. Magnetic force: direction before calculation
10. Sources of magnetic fields
11. Induction and Lenz law
12. Inductance, AC and electromagnetic waves

LESSON 01 / 12

Charges, conductors and Coulomb force

After this lesson, you can

- Predict attraction or repulsion from charge signs.
- Use simple Coulomb-force ratios without unnecessary arithmetic.

Start with: Positive and negative charge; squares; micro means one millionth.

THEORY

Charge is conserved and occurs in integer multiples of the elementary charge. Ordinary charging transfers electrons. Like charges repel and unlike charges attract. Conductors let charge move readily; insulators restrict it. A neutral object can still be attracted by polarization.

RULE

$$F = k|q_1 q_2|/r^2$$

$$k = 8.99 \times 10^9\ \mathrm{N\,m^2/C^2}$$

F electrostatic force magnitude (N)

q_1, q_2 source charges (C)

r separation of charges (m)

When this applies

Coulomb calculations here use point charges, or a stated model equivalent to point charges.

A method that earns marks

1. Mark the signs and positions of both charges.
2. Predict attraction or repulsion before calculating magnitude.
3. For a ratio, compare the charge product and the square of the separation.
4. For a numerical force, convert charges and lengths to SI units first.

WORKED EXAMPLE

Two charges at distance r exert force F. Doubling their separation gives $\frac{F}{4}$. Doubling one charge doubles the force.

1. Keep both charges fixed and replace the separation r by 2r.
2. The denominator becomes $(2r)^2 = 4r^2$, so the force becomes F/4.
3. If instead one charge doubles at the original separation, its product with the other charge doubles, so force becomes 2F.
4. Changing separation or charge magnitude alone does not change attraction into repulsion.

Common mistake
A negative charge does not mean a negative force magnitude.

Exam habit: A magnitude answer is positive; state attraction or repulsion separately to communicate direction.

SELF-CHECK

Tripling separation with fixed charges changes the force by what factor?

Answer: It becomes one ninth as large.

LESSON 02 / 12

Electric field and superposition

After this lesson, you can

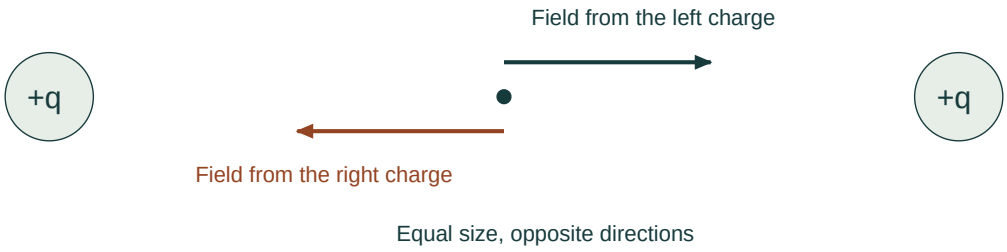
- Combine electric fields by their directions at an observation point.
- Explain why a negative test charge feels force opposite the field.

Start with: Vector addition; positive and negative charges; multiplication.

THEORY

The electric field gives force per unit positive test charge. Field arrows point away from positive sources and toward negative sources. The field exists whether or not a test charge is present. Add fields as vectors, so equal magnitudes can cancel.

Equal positive charges: fields cancel at the midpoint



At this midpoint, the net field and force on any test charge are zero. Where the net field is nonzero, a negative test charge feels force opposite to it.

RULE

$$E = F/q_{\text{test}}$$

$$E = k|Q|/r^2$$

$$\vec{F} = q\vec{E}$$

E electric field magnitude (N/C)

q charge experiencing the force (C)

F force magnitude (N)

A method that earns marks

1. Choose the observation point and draw the field arrow from each source there.
2. Point away from a positive source and toward a negative source.
3. Add signed collinear fields or use vector components when needed.
4. Only after finding the net field, use the test-charge sign to obtain force direction.

WORKED EXAMPLE

At the midpoint between equal positive charges, the two electric fields oppose and cancel. Between equal opposite charges they reinforce toward the negative charge.

1. At the midpoint between equal positive charges, both source distances are the same, so field magnitudes match.
2. The left positive charge points the field right; the right positive charge points it left. The net field is zero.
3. Replace the right source by an equal negative charge: its field now points toward it, also to the right.
4. The two fields therefore reinforce, pointing from the positive source toward the negative source.

Common mistake
Zero net field at a point does not mean each source field is zero.

Exam habit: Draw field arrows before multiplying by a test charge; otherwise a negative test charge can lead to two direction reversals.

SELF-CHECK

If the net field is zero at a point, is a charge placed there forced in either direction by that field?

Answer: No. The electric force qE is zero there in the stated source-field model.

LESSON 03 / 12

Flux and Gauss law

After this lesson, you can

- Use the surface normal to determine electric flux.
- Distinguish enclosed charge from outside charge in Gauss law.

Start with: Surface area; normal means perpendicular; cosine at 0° and 90°.

THEORY

Electric flux measures the field passing through an oriented surface. The angle is measured from the surface normal, not the surface itself. Net flux through a closed surface depends on enclosed charge. An outside charge can affect the field at the surface but contributes zero net closed-surface flux.

RULE

$$\Phi = EA\cos\theta$$

$$\Phi_{\text{closed}} = Q_{\text{enclosed}}/\epsilon_0$$

Φ electric flux (N m^2/C)

E electric field magnitude (N/C)

A surface area (m^2)

Q_{enclosed} net charge inside a closed surface (C)

A method that earns marks

1. Decide whether the question concerns one flat surface or a complete closed surface.
2. For a flat surface, draw the normal and identify its angle to the field.
3. For a closed surface, add only the enclosed charges with signs.
4. Interpret positive closed flux as net outward flux and negative as net inward flux.

WORKED EXAMPLE

A cube enclosing +q has net outward flux $\frac{q}{\varepsilon_0}$. Moving q off-centre changes local field values but not the total flux.

1. The cube is a closed surface enclosing net charge +q.
2. Gauss law fixes its total outward flux as q divided by ϵ0.

3. Moving q off-centre leaves enclosed charge unchanged, so total flux is unchanged.
4. The nearby faces can now have stronger local fields; Gauss law does not say each face receives an equal share after the move.

Common mistake
Zero net closed-surface flux does not imply zero field everywhere.

Exam habit: The angle in EA cos θ is to the normal; a field parallel to the surface gives zero flux.

SELF-CHECK

Can an external charge produce a nonzero field on a box while contributing zero net flux through it?

Answer: Yes. Field enters and leaves; the total closed-surface flux from that external charge cancels.

LESSON 04 / 12

Symmetry and conductors

After this lesson, you can

- State the electrostatic properties of conducting material.
- Use the inverse-square field rule outside a spherical conductor.

Start with: Electric field; conductor versus insulator; inverse-square ratios.

THEORY

Gauss law is always valid, but it easily gives field magnitude only when symmetry makes the surface calculation simple. In electrostatic equilibrium the field inside conducting material is zero and excess charge lies on its surface. A conductor is an equipotential. For the outside-sphere inverse-square rule, assume an isolated conducting sphere with no external fields.

RULE

$$E_{\text{outside sphere}} = k|Q|/r^2$$
$$(r > R)$$

Q net sphere charge (C)

R sphere radius (m)

r distance from sphere centre (m)

E electric field magnitude (N/C)

When this applies

The sphere is isolated with spherical symmetry and is in electrostatic equilibrium.

A method that earns marks

1. Check that the conductor is in electrostatic equilibrium.
2. Distinguish a point in the conducting material from a point outside the sphere.
3. For an outside point measure distance from the centre, not from the surface.
4. Use the point-charge field or a simple ratio, then describe the radial direction.

WORKED EXAMPLE

A charged conducting sphere behaves outside like a point charge at its centre. At twice the outside radius the field is one quarter as large.

1. Outside a spherically symmetric charged conducting sphere, the field has the same form as a charge at its centre.
2. Compare two outside distances r and 2r.
3. The square of the distance grows by a factor of four, so the field magnitude becomes E/4.
4. This comparison uses centre distances; doubling the gap from the surface would generally give a different ratio.

Common mistake
Do not extend the zero-field-in-metal statement to every cavity containing a charge.

Exam habit: Write r > R beside the outside-sphere formula to show that its domain has been checked.

SELF-CHECK

A field is 80 N/C at 0.20 m from the centre. What is it at 0.40 m if both points are outside?

Answer: 20 N/C: the centre distance doubles and the field is divided by four.

LESSON 05 / 12

Potential is not field

After this lesson, you can

- Add scalar potentials with their charge signs.
- Relate potential difference to energy change and field work.

Start with: Scalar versus vector; signed energy changes; potential energy.

THEORY

Electric potential is energy per unit charge, a scalar. Electric field is a vector. Add potentials with charge signs, but add fields with directions. Work by the electric field equals the decrease in electric potential energy.

RULE

$$V = kQ/r \text{ (zero at infinity)}$$

$$\Delta U = q\Delta V$$

$$W_{\text{field}} = -\Delta U$$

V electric potential (V = J/C)

ΔV final minus initial potential (V)

ΔU change in potential energy (J)

q moving charge (C)

A method that earns marks

1. Separate a field question, requiring direction, from a potential question, requiring signed scalar addition.
2. Choose the potential reference supplied by the problem.
3. Calculate energy change from charge times final-minus-initial potential.
4. Reverse the sign of that energy change to obtain work by the electric field.

WORKED EXAMPLE

Equal positive charges can give zero field at their midpoint but positive potential there. Cancellation rules differ for scalars and vectors.

1. At the midpoint of two equal positive charges, equal field arrows point in opposite directions and

cancel.

2. Each positive charge contributes a positive potential at that point when zero potential is chosen at infinity.
3. The two potentials add rather than cancel because potential has no directional arrow.
4. Thus zero net field and positive potential can occur at the same point without contradiction.

Common mistake
A charge moving to lower potential does not always lose potential energy; a negative charge does the opposite.

Exam habit: Write ΔV = Vfinal - Vinitial before calculating; swapping the endpoints reverses both energy and work signs.

SELF-CHECK

A +2 C charge moves through ΔV = +3 V. What are ΔU and work by the field?

Answer: ΔU = +6 J and work by the field = -6 J.

LESSON 06 / 12

Capacitors and dielectrics

After this lesson, you can

- Distinguish series and parallel capacitor connections.
- Calculate stored charge and identify the effect of a dielectric.

Start with: Potential difference; charge; reciprocals and addition.

THEORY

A capacitor stores separated charge and electric-field energy. Capacitance describes charge stored per volt. Parallel capacitors share voltage and their capacitances add. Series capacitors carry equal charge magnitude and have a smaller equivalent capacitance.

RULE

$$Q = CV$$

$$U = \tfrac{1}{2}CV^2$$

$$C_{\parallel} = C_1 + C_2$$

$$1/C_s = 1/C_1 + 1/C_2$$

C capacitance (F)

Q stored charge magnitude on either plate (C)

V potential difference between plates (V)

U stored electric energy (J)

A method that earns marks

1. Identify shared nodes: parallel capacitors share voltage, while ideal series capacitors carry equal charge magnitudes.
2. Add capacitances for parallel or add their reciprocals for series.
3. Use Q = CV with compatible prefixes such as µF times V giving µC.
4. For dielectric questions first check whether voltage or charge is held fixed.

WORKED EXAMPLE

Two 6 μF capacitors give 12 μF in parallel and 3 μF in series. Inserting a dielectric increases capacitance.

1. For two 6 µF capacitors in parallel, capacitances add: 6 + 6 = 12 µF.
2. For series, 1/Cs = 1/6 + 1/6 = 1/3 in reciprocal µF, so Cs = 3 µF.
3. The series result is smaller than either individual capacitor, providing a quick plausibility check.
4. Inserting a suitable dielectric increases capacitance; the resulting charge or voltage change depends on the connection to a battery.

Common mistake
Do not use resistor series/parallel rules for capacitors.

Exam habit: For series, invert the reciprocal sum at the end; stopping at 1/C is a common lost result mark.

SELF-CHECK

What charge magnitude is stored by 4 µF at 12 V?

Answer: 48 µC, from Q = CV.

What changes a capacitor's capacitance?

These simple comparisons cover the geometry and dielectric ideas in the lecture without a derivation.

- Use the first formula for parallel plates with a small gap compared with plate size, so edge effects can be ignored.
- Here A is overlapping plate area, d is separation, and ε describes the material filling the gap.
- Doubling area doubles capacitance. Doubling separation halves it.

- A dielectric filling the gap increases capacitance. Battery attached: voltage stays fixed. Disconnected and isolated: charge stays fixed.

$$C = \frac{\varepsilon A}{d}$$
$$Q = CV$$

An isolated charged capacitor has capacitance $2\,\mu\mathrm{F}$ and voltage $12\,\mathrm{V}$. A dielectric triples its capacitance. Find the new charge and voltage.

1. Initial charge: $Q = CV = 2 \times 12 = 24\,\mu\mathrm{C}$.
2. Isolation keeps the charge at $24\,\mu\mathrm{C}$.
3. New capacitance: $6\,\mu\mathrm{F}$; new voltage: $V = 24/6 = 4\,\mathrm{V}$.

Charge stays $24\,\mu\mathrm{C}$; voltage falls to $4\,\mathrm{V}$.

LESSON 07 / 12

Current, resistance and electrical power

After this lesson, you can

- Find current, voltage and power for an ohmic resistor.
- Predict how wire length and area change resistance.

Start with: Charge flow; multiplication and division; circuit voltage.

THEORY

Current is the rate of charge flow. Conventional current follows positive-charge flow, opposite electron drift in a metal. Resistance depends on material and geometry. Ohm law describes an Ωic element under fixed physical conditions.

RULE

$$I = \Delta Q / \Delta t$$
$$V = IR$$
$$R = \rho L / A$$
$$P = VI$$

I conventional current (A)

R resistance (Ω)

V potential difference (V)

P electrical power (W)

ρ resistivity (Ω m)

When this applies

Ohm law is used for an ohmic resistor under unchanged physical conditions.

A method that earns marks

1. Label voltage across the resistor and current through it.
2. Use V = IR to find the missing electrical quantity.
3. Calculate power using VI or an equivalent resistor relation.
4. For wire comparisons hold material and temperature fixed before changing length or area.

WORKED EXAMPLE

A 6 Ω resistor across 12 V carries 2 A and dissipates 24 W. A longer wire of the same material and area has greater resistance.

1. The 6 Ω resistor has 12 V across it, so I = 12/6 = 2 A.
2. Its electrical power is VI = 12 times 2 = 24 W.
3. Power of 24 W means 24 joules of energy are transferred each second.
4. A longer wire with unchanged material, area and temperature has larger resistance because R is proportional to length.

Common mistake
More resistance lowers power at fixed voltage but raises power at fixed current.

Exam habit: Do not confuse power in watts with energy in joules; multiply power by time when energy is requested.

SELF-CHECK

What happens to resistance if a wire length doubles but material, area and temperature stay fixed?

Answer: Resistance doubles.

LESSON 08 / 12

DC networks, Kirchhoff rules and RC behaviour

After this lesson, you can

- Reduce simple resistor networks and recover branch currents.
- Describe the initial and final behaviour of an RC charging circuit.

Start with: Ohm law; series and parallel connections; charge and energy conservation.

THEORY

Series resistors share current; parallel resistors share voltage. Kirchhoff junction rules express charge conservation, and loop rules express energy conservation. An initially uncharged capacitor charges gradually through a resistor and eventually blocks steady DC in the ideal model.

RULE

$$R_s = R_1 + R_2$$

$$1/R_p = 1/R_1 + 1/R_2$$

$$\tau = RC$$

R_{eq} equivalent resistance (Ω)

I current (A)

V voltage (V)

τ RC time constant (s)

A method that earns marks

1. Identify series elements by shared current and parallel elements by shared voltage.
2. Reduce the network and calculate total source current.
3. Return to the original circuit to calculate branch currents or voltage drops.
4. Check junction current sums and loop voltage sums; for RC questions distinguish transient from long-time behaviour.

WORKED EXAMPLE

Two 6 Ω resistors in parallel give 3 Ω. Across 12 V the total current is 4 A, split equally into 2 A in each branch.

1. Both 6 Ω resistors are connected across the same two nodes, so they are parallel.

2. Their equal-resistor equivalent is 6/2 = 3 Ω.
3. The 12 V source therefore supplies 12/3 = 4 A total.
4. Each branch still has 12 V and carries 12/6 = 2 A; the branch currents sum to the 4 A source current.

Common mistake
Adjacent components in a drawing are not necessarily in series.

Exam habit: After finding an equivalent resistance, redraw or relabel the original circuit before assigning individual voltage drops.

SELF-CHECK

After a long time, does an ideal capacitor in a DC series RC circuit carry a steady current?

Answer: No. Once charged, it blocks steady DC in the ideal model.

One time constant: the 63% and 37% rules

Recognize how quickly a capacitor charges or discharges without deriving an exponential.

- Use an ideal series resistor-capacitor circuit with constant R and C. The time constant τ is measured in seconds.
- Charging from zero: after one time constant, capacitor voltage and charge reach about 63% of their final values.
- Discharging: after one time constant, about 37% of the starting voltage and charge remain.
- A larger resistance or capacitance makes the change slower.

$$\tau = RC$$

$$1 - e^{-1} \approx 0.632$$

$$e^{-1} \approx 0.368$$

A capacitor charges from zero toward $10\,\mathrm{V}$. Its time constant is $2\,\mathrm{s}$. Estimate its voltage after $2\,\mathrm{s}$.

1. The elapsed time is one time constant.
2. Multiply final voltage by 0.632: $10 \times 0.632 = 6.32\,\mathrm{V}$.

About $6.3\,\mathrm{V}$. It is not fully charged after one time constant.

Kirchhoff signs and a real battery

Choosing consistent signs is the main skill. A negative current answer tells you the actual direction.

- At a junction, charge does not keep building up: add entering and leaving currents separately.
- Choose a loop direction. Across a resistor, travelling with the assumed current gives $-IR$; travelling against it gives $+IR$.

- Across a battery, going from its negative to positive terminal gives $+\mathcal{E}$; the reverse gives $-\mathcal{E}$.
- For a battery supplying current, internal resistance r causes the terminal voltage to be less than its emf $\mathcal{E}$.

current in $=$ current out

$\sum \Delta V = 0$

$V_{\text{terminal}} = \mathcal{E} - Ir$ (battery delivering current)

A battery has emf $12\,\text{V}$ and internal resistance $1\,\Omega$. It supplies a $5\,\Omega$ resistor. Find current and terminal voltage.

1. Use one loop: $12 - I(1) - I(5) = 0$.
2. Total resistance is $6\,\Omega$, so $I = 12/6 = 2\,\text{A}$.
3. Terminal voltage is $12 - (2)(1) = 10\,\text{V}$. Check the external resistor: $IR = (2)(5) = 10\,\text{V}$.

Current $2\,\text{A}$; terminal voltage $10\,\text{V}$.

LESSON 09 / 12

Magnetic force: direction before calculation

After this lesson, you can

- Determine when magnetic force vanishes or is largest.
- Explain circular motion without a change in kinetic energy.

Start with: Perpendicular directions; circular acceleration; charge sign.

THEORY

A magnetic field exerts force on a moving charge when its motion has a perpendicular component. The force is perpendicular to velocity and therefore does no work on a point charge. It can change direction without changing speed.

RULE

$$F = |q|vB\sin\theta$$

$$F_{\text{wire}} = BIL\sin\theta$$

q particle charge (C)

v particle speed (m/s)

B magnetic field magnitude (T)

F magnetic force magnitude (N)

θ angle between velocity and field (degrees or radians)

A method that earns marks

1. Decide whether the object is a moving charge or a current-carrying wire.
2. Identify the angle to the magnetic field and the perpendicular motion or wire length.
3. Use the right-hand rule for positive charge or conventional current; reverse for negative charge.
4. Calculate magnitude and check that the force is perpendicular to motion.

WORKED EXAMPLE

A positive charge moving perpendicular to a uniform magnetic field follows a circular path if no other force acts. The field supplies the inward force.

1. For the positive charge in the example, velocity is perpendicular to the uniform field, so the magnetic force magnitude is qvB.
2. At each point the force is perpendicular to velocity and turns the particle inward.
3. Because the force does no work, speed and kinetic energy remain constant.
4. With no other forces, the continual turning gives a circular path in the plane perpendicular to the field.

Common mistake
Do not apply the right-hand direction unchanged to an electron.

Exam habit: A direction answer needs the charge sign; a right-hand-rule result for a positive charge must be reversed for an electron.

SELF-CHECK

Does a charge moving exactly along a magnetic field feel magnetic force?

Answer: No. The angle is 0° or 180°, so its sine is zero.

LESSON 10 / 12

Sources of magnetic fields

After this lesson, you can

- Relate a current to the direction of the magnetic field it produces.
- Use simple distance and current ratios for a long wire.

Start with: Current; inverse proportionality; circular field-line diagrams.

THEORY

Currents produce magnetic fields. Around a long straight wire the field forms circles. Inside an ideal long solenoid the field is approximately uniform. For two parallel wires, currents in the same direction attract and opposite currents repel.

RULE

$$B_{\text{wire}} = \mu_0 I/(2\pi r)$$

$$B_{\text{solenoid}} = \mu_0 n I$$

B magnetic field magnitude (T)

I wire current (A)

r perpendicular distance from wire (m)

n solenoid turns per unit length (m^{-1})

When this applies

The wire is sufficiently long and the observation point sufficiently far from its ends for the long-wire approximation.

A method that earns marks

1. Identify whether the source is a straight wire, a solenoid or a pair of wires.
2. For a straight wire, use the right-hand grip rule to draw field circles.
3. Apply the matching field relation; for ratios hold unstated factors fixed.
4. For paired wires determine attraction or repulsion from relative current directions.

WORKED EXAMPLE

Doubling distance from a long wire halves its magnetic field; doubling current doubles the field.

1. For the long wire, B is proportional to I/r.
2. Moving from r to 2r at the same current gives half the original field.
3. Doubling current at fixed r instead doubles the field.
4. If current and distance both double, the two factors cancel and field magnitude stays the same.

Common mistake
The wire field varies as inverse distance, not inverse distance squared.

Exam habit: Use 1/r for the long-wire field; the $1/r^2$ rule belongs to a different source geometry.

SELF-CHECK

A wire gives 30 µT at 2 cm. What field does it give at 6 cm with the same current?

Answer: 10 µT because distance triples.

LESSON 11 / 12

Induction and Lenz law

After this lesson, you can

- Identify which change produces an induced emf.
- Use Lenz law to choose the induced current direction.

Start with: Magnetic field direction; loop area; clockwise versus counterclockwise viewed from a stated side.

THEORY

An induced emf requires a change in magnetic flux, which can arise from changing field, area or orientation. Lenz law sets the direction: the induced effect opposes the change of flux. A static magnet near a stationary loop does not produce a sustained induced current.

RULE

$$|\mathcal{E}| = N|\Delta\Phi_B|/\Delta t$$

$$\Phi_B = BA\cos\theta$$

Φ_B magnetic flux through one turn (Wb)

N number of coil turns (dimensionless)

$\mathcal{E}$ induced emf (V)

Δt time interval of flux change (s)

A method that earns marks

1. State the original flux direction and whether its magnitude increases or decreases.
2. Choose an induced field that opposes that change.
3. Use the grip rule to convert the required field to current direction, naming the viewing side.
4. For magnitude, multiply the flux change per turn by the number of turns and divide by elapsed time.

WORKED EXAMPLE

If outward flux increases, the induced field points inward. The current must therefore circulate clockwise when viewed from that side.

1. In the example, outward flux through the loop is increasing.
2. The induced effect must oppose the increase, so its own field points inward.
3. Viewed from the side toward which the original flux points outward, the inward field requires clockwise current.
4. If the outward flux were decreasing instead, the induced field and current direction would both reverse.

Common mistake
Opposes the change is not the same as always opposes the original field.

Exam habit: Say oppose the change in flux, not oppose the flux: a decreasing field produces a reinforcing induced field.

SELF-CHECK

Does a stationary loop near an unchanging stationary magnet have a sustained induced emf?

Answer: No. Its magnetic flux is not changing.

LESSON 12 / 12

Inductance, AC and electromagnetic waves

After this lesson, you can

- Convert sinusoidal peak voltage to RMS and calculate resistive heating.
- Relate electromagnetic frequency to vacuum wavelength.

Start with: Square roots; frequency and wavelength; electrical power.

THEORY

Course-plan extension: an inductor resists changes of current and stores magnetic energy. AC reverses direction periodically. RMS values describe equivalent heating for resistive loads. In vacuum an EM wave has mutually perpendicular electric field, magnetic field and travel direction.

RULE

$$U_L = \tfrac{1}{2}LI^2$$

$$V_{\text{rms}} = V_{\text{peak}}/\sqrt{2} \text{ (sinusoidal)}$$

$$c = f\lambda \text{ (vacuum)}$$

V_{peak} maximum sinusoidal voltage magnitude (V)

V_{rms} root-mean-square voltage (V)

L inductance (H)

f wave frequency (Hz)

λ wavelength (m)

When this applies

The peak-to-RMS conversion assumes a sinusoid; the wavelength calculation assumes propagation in vacuum.

A method that earns marks

1. Identify whether the question concerns an inductor, sinusoidal AC or an electromagnetic wave.
2. For sinusoidal AC divide peak voltage by the square root of two.
3. Use RMS values for average power in a pure resistor.
4. For vacuum waves calculate wavelength as c divided by frequency and convert metres to the requested unit.

WORKED EXAMPLE

A sinusoidal supply with peak voltage 141.4 V has RMS voltage 100 V. Light of frequency $6 \times 10^{14}\ \mathrm{Hz}$ has vacuum wavelength 500 nm.

1. For the stated sinusoidal supply, 141.4 divided by $\sqrt{2}$ is approximately 100 V RMS.
2. This RMS value matches the heating effect of 100 V DC in the same pure resistor.
3. For the separate light example, wavelength is 3.00×10^{8} divided by $6 \times 10^{14} = 5 \times 10^{-7}$ m.
4. Since one nanometre is 10^{-9} m, the wavelength is 500 nm.

Common mistake
An ideal transformer does not continuously transform steady DC after switching transients.

Exam habit: Keep this extension conceptual: use RMS for resistor power and $c = f\lambda$ for waves; do not mix peak and RMS values.

SELF-CHECK

A 20 V peak sinusoid is applied to 10 Ω. What is the average power?

Answer: 20 W: $V_{rms}^2/R = (20^2/2)/10$.

Before the tests

These original questions practise the lesson objectives. They are not predictions of an exam paper. Use the separate practice book if you want all papers together.

Electricity and Magnetism / Test A

30 marks | Suggested time: 30 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. A neutral object loses electrons. It becomes:

   A. Negative

   B. Positive

   C. Uncharged forever

   D. Magnetic only

2. A negative test charge in a field directed right feels force:

   A. Left

   B. Zero

   C. Right

   D. Up

3. A charge outside a closed empty box gives net flux through the box of:

   A. $\frac{q}{\varepsilon_0}$

   B. Always positive

   C. Always negative

   D. Zero

4. Inside the material of a conductor in electrostatic equilibrium, the electric field is:

   A. Equal to its surface value

   B. Infinite

   C. Zero

   D. Always outward

5. For equal and opposite point charges, take potential as zero at infinity. At their midpoint the potential is:

   A. Zero

   B. Undefined

   C. Positive

   D. Negative

6. Adding a dielectric to an isolated charged capacitor increases C. Its voltage:

A. Decreases

B. Increases

C. Must stay fixed

D. Becomes infinite

Part II: Show your reasoning

1. Two charges exert 12 N at separation r. What is the force at separation 2r, and at r if one charge is tripled?

2. A field of 200 N/C points east. Find the force on a charge of -0.003 C.

3. A uniform field of 50 N/C passes through a flat area of 0.20 square metres. The chosen surface normal points in the same direction as the field. Find flux. What if the normal is perpendicular to the field?

4. An isolated charged conducting sphere, with no external fields, has field magnitude 80 N/C at r = 0.20 m. Find the field at r = 0.40 m. Both distances are from the centre and outside the sphere.

5. A +2 C charge moves through a potential difference of +3 V. Find its potential-energy change and the work done by the field.

6. A 4 μF capacitor is connected to 12 V. Find the charge in microcoulombs.

Electricity and Magnetism / Test B

30 marks | Suggested time: 30 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. At fixed voltage, doubling resistance makes power:

   A. Unchanged

   B. Double

   C. Half

   D. Four times

2. After a long time in an ideal DC charging circuit, capacitor current is:

   A. Maximum

   B. Negative forever

   C. Zero

   D. Equal to the battery voltage divided by the resistor resistance

3. A charge moving parallel to a magnetic field experiences magnetic force:

   A. qvB

   B. Twice qvB

   C. Opposite its velocity

   D. Zero

4. Two parallel wires carrying currents in the same direction:

   A. Repel

   B. Attract

   C. Have no interaction

   D. Must carry zero field

5. Which situation produces an induced emf?

   A. No change of area, field or angle

   B. Changing magnetic flux

   C. A stationary unchanging field only

   D. Constant flux

6. In vacuum, electric and magnetic fields of a plane EM wave are:

   A. Parallel to each other

B. Longitudinal only

C. Zero

D. Perpendicular to each other and to travel

Part II: Show your reasoning

1. A 3 Ω resistor carries 2 A. Find voltage and power.

2. A 4 Ω and 8 Ω resistor are in series across 24 V. Find total current and each voltage drop.

3. A straight 0.50 m wire carries 2 A perpendicular to a 0.30 T field. Find force magnitude.

4. The field is 30 microtesla at 2 cm from a long straight wire. Find the field at 6 cm at the same current.

5. A 20-turn coil has flux per turn changing by 0.03 Wb in 0.10 s. Find average emf magnitude.

6. A sinusoidal voltage has peak value 20 V. Find its RMS value, then the average power in a 10 Ω resistor.

Electricity and Magnetism / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **B.** Removing negative charge leaves a positive net charge.
2. **A.** The negative charge reverses the field direction in $F = qE$.
3. **D.** No charge is enclosed, so total entering and leaving flux cancels.
4. **C.** Free charges redistribute until the internal field is cancelled.
5. **A.** The signed scalar potentials cancel when distances and charge magnitudes match.
6. **A.** Charge remains fixed and $V = \frac{Q}{C}$.

Part II: Written solutions

1. At 2r: 12/4 = 3 N. Tripling one charge at r: 36 N. Two marks for each ratio and result.
2. Magnitude = 0.003 times 200 = 0.60 N. Direction is west because the charge is negative. Two marks for magnitude and two for direction with reasoning.
3. Parallel normal: flux = 50 times 0.20 = 10 N m squared/C. Perpendicular normal: cosine 90 degrees = 0, so flux = 0. Two marks per case.
4. Inverse-square scaling gives 80 times (0.2$\frac{0}{0}$.40) squared = 20 N/C. Credit the outside-sphere condition and inverse-square reasoning.
5. Energy change = 2 times 3 = +6 J. Work by field = -6 J. Two marks each; the signs are essential.
6. Q = CV = 4 times 12 = 48 microcoulombs. In SI this is 0.000048 C. Credit formula, substitution, prefix and result.

Electricity and Magnetism / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **C.** $P = \frac{V^2}{R}$, so doubling the denominator halves it.
2. **C.** After a long time, the capacitor voltage equals the battery voltage. The resistor voltage and circuit current are zero. During charging, charge moves in the wires onto and off the plates; it does not cross the ideal dielectric.
3. **D.** The sine of the angle is zero.
4. **B.** Each current experiences the magnetic field of the other wire.
5. **B.** Faraday law depends on the rate of flux change.
6. **D.** A plane electromagnetic wave in vacuum is transverse.

Part II: Written solutions

1. V = IR = 6 V. P = VI = 12 W (or I^2R). Two marks per result.
2. Total resistance = 12 Ω; current = 2 A; drops are 8 V and 16 V. One mark each; the drops must sum to 24 V.
3. F = BIL = 0.30 times 2 times 0.50 = 0.30 N. Four marks for formula, perpendicular condition, substitution and unit.
4. B is inversely proportional to r, so B = 30 times 2/6 = 10 microtesla. Credit model, ratio, arithmetic and unit.
5. Magnitude = 20 times 0.03/0.10 = 6 V. The direction needs the direction of the flux change; it cannot be inferred from these magnitudes alone.
6. RMS = 20/square root of 2 = 14.14 V. Average power = RMS squared/10 = 20 W. Two marks per stage.

Source and coverage notes

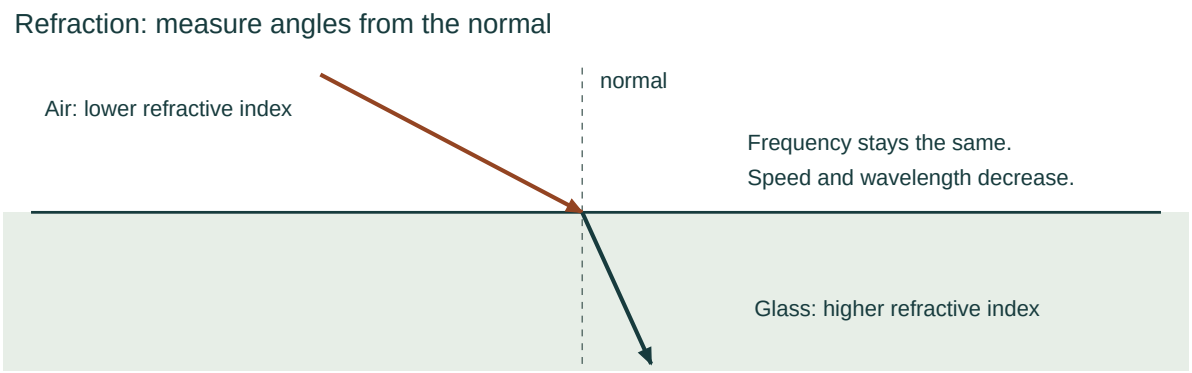
Concept-first coverage of the course plan. Chapters 23-31 have lecture support; inductance, AC and EM waves are course-plan extensions without separate lecture files in this folder. Advanced field integrals are recognition topics, not worked derivations.

Source folder: Open original course folder The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. This revised study aid does not replace lecturer-specific derivations or instructions. All practice tests and suggested marking guidance are original study material.

03

Optics and Waves

A field guide to understanding, working and checking.



The introduction and four current sections: wave motion; electromagnetic theory, photons and light; geometrical optics; physical optics. Derivations are replaced by plain-language interpretation and short exam calculations.

12 lessons / 2 practice tests / Worked solutions

English edition / November 2026

How to use this handout

1. Read the concept and check the symbol meanings before choosing a formula.
2. Cover the worked steps and reproduce the reasoning yourself.
3. Attempt the quick check and the two subject tests without notes.
4. Mark your work; record whether the error was concept, method, algebra, units or reading. Retry after a delay.

Your study route

1. Wave language and simple harmonic motion
2. Read a travelling wave without deriving it
3. Wavefronts, spreading and group velocity
4. Superposition, standing waves and beats
5. Light, photons and the electromagnetic picture
6. Reflection, refraction and total internal reflection
7. Mirrors, lenses and image signs
8. Optical systems, prisms and instruments
9. Interference and diffraction
10. Polarization and Malus law
11. Scattering, apparent depth and lens power
12. Recognize advanced optics without long derivations

LESSON 01 / 12

Wave language and simple harmonic motion

After this lesson, you can

- Convert an oscillation count into frequency and period.
- Relate displacement, speed and acceleration at key SHM positions.

Start with: One oscillation is one complete cycle; reciprocal quantities; displacement from equilibrium.

THEORY

A wave carries a disturbance and energy. Mechanical waves need a medium; electromagnetic waves can travel through vacuum. Transverse displacement is perpendicular to travel; longitudinal displacement is parallel. Simple harmonic motion has a restoring acceleration proportional and opposite to displacement.

RULE

$$T = 1/f$$

$$f = N/t$$

$$a = -\omega^2 x$$

N number of complete cycles (dimensionless)

t total elapsed time (s)

f cycles per second (Hz)

T time for one cycle (s)

ω angular frequency (rad/s)

A method that earns marks

1. Count complete cycles rather than individual crossings of equilibrium.
2. Divide cycle count by total time for frequency.
3. Find period as total time per cycle or the reciprocal of frequency.
4. For SHM compare the point with equilibrium and use restoring acceleration toward equilibrium.

WORKED EXAMPLE

If 20 oscillations take 5 s, frequency is 4 Hz and period is 0.25 s. At an extreme of SHM, speed is zero and acceleration magnitude is maximum.

1. Twenty complete oscillations in 5 s means 20/5 = 4 cycles per second, so f = 4 Hz.
2. One cycle takes 5/20 = 0.25 s, giving T = 0.25 s.
3. At an extreme the object must turn around, so its instantaneous speed is zero.
4. Its displacement magnitude is largest there; restoring acceleration therefore has maximum magnitude and points toward equilibrium.

Common mistake
The direction of particle motion is not necessarily the direction of wave travel.

Exam habit: Write f = N/t and T = t/N side by side to prevent interchanging hertz and seconds.

SELF-CHECK

At SHM equilibrium is acceleration magnitude maximum?

Answer: No. Displacement and restoring acceleration are zero; speed is maximum.

LESSON 02 / 12

Read a travelling wave without deriving it

After this lesson, you can

- Read amplitude, wavelength and frequency from a travelling-wave expression.
- Use v = fλ and distinguish pattern motion from particle motion.

Start with: Sine-wave shape; π; coefficients of x and t; SI units.

THEORY

Amplitude is maximum displacement, wavelength is crest-to-crest separation, and frequency counts cycles per second. Wave speed is the speed of the pattern, not the oscillating particle. In a sine wave, the coefficients of position and time reveal wavelength and period.

RULE

$$v = f\lambda$$

$$k = 2\pi/\lambda$$

$$\omega = 2\pi f$$

$$y = A\sin(kx - \omega t + \phi)$$

A amplitude or maximum displacement (m)

k wave number (rad/m)

ω angular frequency (rad/s)

λ wavelength (m)

f frequency (Hz)

When this applies

The example takes x and y in metres and t in seconds.

A method that earns marks

1. Match the wave to y = A sin(kx - ωt + ϕ).

2. Read A directly, then divide 2π by k to obtain wavelength.
3. Divide ω by 2π to obtain frequency.
4. Multiply frequency by wavelength for pattern speed and check the units.

WORKED EXAMPLE

For $y = 0.02\sin(4\pi x - 20\pi t)$, amplitude is 0.02 m, wavelength is 0.50 m, frequency is 10 Hz and speed is 5 m/s.

1. Compare y = 0.02 sin(4πx - 20πt) with the standard form: A = 0.02 m, k = 4π rad/m and ω = 20π rad/s.
2. Wavelength is $2\pi/(4\pi)$ = 0.50 m.
3. Frequency is $20\pi/(2\pi)$ = 10 Hz.
4. Pattern speed is 10 times 0.50 = 5 m/s; this is not the maximum transverse speed of an individual particle.

Common mistake
The coefficient of time is angular frequency, not ordinary frequency.

Exam habit: Identify coefficients before calculating; treating k as wavelength or ω as ordinary frequency loses the factor 2π.

SELF-CHECK

Is the coefficient 20π in the example the frequency in hertz?

Answer: No. It is angular frequency; divide it by 2π to get 10 Hz.

Phase means position within a cycle

A phase difference tells whether two oscillations are at the same or different stages of a cycle.

- One full cycle is 2π radians. Half a cycle is π radians.
- At a fixed time, points one wavelength apart are in phase. Points half a wavelength apart are opposite in phase.
- At one place, a full period brings the oscillation back to the same phase.
- These formulas give the size of the phase difference. Its sign depends on the order of comparison and the travelling-wave convention.

$$\Delta\phi = 2\pi\frac{\Delta x}{\lambda}$$ (phase-difference magnitude at one time)

$$\Delta\phi = 2\pi\frac{\Delta t}{T}$$ (phase-change magnitude at one place)

Two points on a sinusoidal wave are separated by $\lambda/4$. Find their phase-difference magnitude at the same time.

1. The separation is one quarter of a full spatial cycle.
2. Therefore $\Delta\phi = 2\pi \times (1/4) = \pi/2$.

$\pi/2$ radians, or $90°$. They are neither in phase nor exactly opposite in phase.

LESSON 03 / 12

Wavefronts, spreading and group velocity

After this lesson, you can

- Predict intensity and amplitude changes for spherical spreading.
- Distinguish wavefront, ray, phase velocity and group velocity.

Start with: Surface area grows with radius squared; square roots; amplitude versus intensity.

THEORY

A ray indicates propagation direction and is normal to a wavefront. Plane wavefronts are flat, spherical wavefronts spread from a point source, and cylindrical wavefronts spread around a line source. Phase velocity tracks a crest; group velocity tracks the envelope of a narrow wave packet.

RULE

$$I \propto A^2$$

$$I_{\text{spherical}} \propto 1/r^2$$

I energy-transfer rate per unit area (W/m^2)

A wave amplitude (depends on disturbance)

r distance from point source (m)

When this applies

The spreading example assumes an isotropic point source, no absorption and unchanged medium properties.

A method that earns marks

1. Identify whether the source produces plane, spherical or cylindrical spreading.
2. For an isotropic point source use intensity proportional to 1/r^2.
3. Convert an intensity ratio to an amplitude ratio by taking its square root.

4. In description questions, identify a crest for phase velocity and the packet envelope for group velocity.

WORKED EXAMPLE

For an isotropic point source without absorption, doubling distance reduces intensity to one quarter and amplitude to one half.

1. The same source power crosses a sphere around the source at each distance.
2. Doubling radius makes the sphere area four times as large.
3. The available power per square metre is therefore one quarter as large.
4. Because intensity is proportional to amplitude squared, the amplitude ratio is $\sqrt{(1/4)} = 1/2$.

Common mistake
The ideal plane-wave model does not have spherical inverse-square spreading.

Exam habit: Do not apply the inverse-square intensity rule automatically to a plane wave or line source.

SELF-CHECK

An intensity of 18 W/m^2 is measured at r. What is it at 3r?

Answer: 2 W/m^2: divide by $3^2 = 9$.

LESSON 04 / 12

Superposition, standing waves and beats

After this lesson, you can

- Calculate beat frequency from two nearby frequencies.
- Connect fixed-string harmonics with allowed wavelengths.

Start with: Adding displacements; wavelength; frequency; fixed endpoints.

THEORY

Overlapping disturbances add algebraically. In-phase equal waves reinforce; opposite-phase equal waves cancel. A standing wave forms when matching waves travel in opposite directions. Nodes stay at zero displacement. Beats are slow changes in loudness from nearby frequencies.

RULE

$$f_{\text{beat}} = |f_1 - f_2|$$

$$\lambda_n = 2L/n \text{ (string fixed at both ends)}$$

f_{beat} number of loudness cycles per second (Hz)

L vibrating string length (m)

n harmonic number, 1, 2, 3, … (dimensionless)

λ_n wavelength of harmonic n (m)

A method that earns marks

1. For superposition add instantaneous signed displacements at the same point.
2. For beats take the positive difference of the two nearby frequencies.
3. For a string fixed at both ends draw nodes at both ends.
4. Fit n half-wavelengths into the length, then use wave speed to find each frequency.

WORKED EXAMPLE

Two tones at 256 and 260 Hz produce 4 beats per second. A fixed string in its fundamental mode contains one half-wavelength.

1. The tones have frequencies 256 Hz and 260 Hz.
2. Their frequency difference is 4 Hz, so there are four loudness beats each second.
3. For the separate string example, the fundamental has one loop between two end nodes.
4. That loop occupies half a wavelength, so $\lambda 1 = 2L$ and $f1 = v/(2L)$.

Common mistake
A node is not a point where two travelling waves are individually absent.

Exam habit: A beat frequency is a difference, not the average of the two tone frequencies.

SELF-CHECK

For a fixed string, how does the second-harmonic frequency compare with the fundamental?

Answer: It is twice the fundamental frequency in the ideal uniform-string model.

LESSON 05 / 12

Light, photons and the electromagnetic picture

After this lesson, you can

- Compare photon energy using frequency or vacuum wavelength.
- Distinguish beam brightness from energy per photon.

Start with: Direct and inverse proportionality; wave frequency; energy transfer.

THEORY

Light behaves as an electromagnetic wave in propagation and exchanges energy in photons. Photon energy increases with frequency. In the photoelectric effect, emission requires enough energy per photon; increasing brightness alone cannot overcome a frequency below threshold.

RULE

$$E_\gamma = hf = hc/\lambda_0$$

$$\lambda_0 = \text{vacuum wavelength}$$

$$c = 3.00 \times 10^8\ \text{m/s}$$

E_γ energy of one photon (J)

h Planck constant (J s)

f light frequency (Hz)

λ vacuum wavelength (m)

A method that earns marks

1. Decide whether the question asks about one photon or the whole beam.
2. Use Eγ proportional to frequency, or inversely proportional to vacuum wavelength.
3. For brightness comparisons separate photon energy from the number arriving each second.
4. For photoelectric emission compare energy per photon with the threshold requirement.

WORKED EXAMPLE

Blue light has shorter wavelength than red light and therefore higher photon energy. A brighter red beam need not have more energetic photons.

1. Blue light has a shorter vacuum wavelength than red light.

2. Since $E\gamma = hc/\lambda$, the smaller wavelength corresponds to a larger energy per photon.
3. Increasing red-beam brightness at the same colour increases the energy arriving per second, usually through more photons.
4. Each red photon still has the same lower energy; brightness alone does not change its frequency.

Common mistake
Higher intensity does not mean higher energy per photon. In E = hc/lambda, use the vacuum wavelength. At an ordinary stationary boundary, frequency and photon energy stay the same even though wavelength in the medium changes.

Exam habit: Use the phrase energy per photon when discussing threshold effects; total beam energy does not answer the same question.

SELF-CHECK

If light A has twice light B frequency, how do photon energy and vacuum wavelength compare?

Answer: A has twice the photon energy and half the vacuum wavelength.

LESSON 06 / 12

Reflection, refraction and total internal reflection

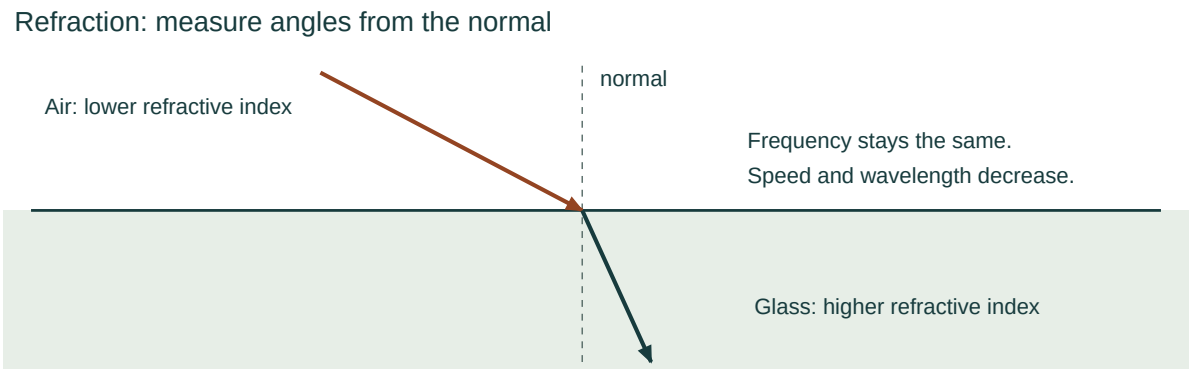
After this lesson, you can

- Track speed, frequency and wavelength across a refracting boundary.
- Recognize the two conditions for total internal reflection.

Start with: Normal to a surface; sine; reciprocal refractive index.

THEORY

Angles of incidence, reflection and refraction are measured from the normal. At a stationary interface frequency stays the same, while speed and wavelength may change. When light enters a higher refractive index at a nonzero angle, the ray bends toward the normal. Total internal reflection needs travel from higher to lower index and an incidence angle greater than the critical angle.



A ray entering glass from air bends toward the normal at oblique incidence. The ray sketch is qualitative; frequency is unchanged while speed and wavelength decrease.

RULE

$$\theta_i = \theta_r$$

$$n = c/v$$

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

$$\sin\theta_c = n_2/n_1$$

- n refractive index (dimensionless)
- c vacuum light speed (m/s)
- v light speed in medium (m/s)
- θ_c critical incidence angle (degrees or radians)

When this applies

The boundary is stationary and the media are treated as uniform; air index is approximated as 1.

A method that earns marks

1. Draw the surface normal and measure all ray angles from it.
2. Use v = c/n to compare speeds; frequency stays fixed at a stationary boundary.
3. Use wavelength = speed/frequency to compare wavelengths.
4. For total internal reflection check higher-to-lower index travel, then compare incidence with the critical angle.

WORKED EXAMPLE

From air into glass with index 1.5, light slows to $2 \times 10^8\ \mathrm{m/s}$. Its frequency stays fixed and its wavelength decreases.

1. Air is approximated by n = 1 and glass has n = 1.5.
2. In glass, speed is $3.00 \times 10^8 / 1.5 = 2.00 \times 10^8$ m/s.
3. The frequency is set by the source and remains the same across the boundary.
4. Because speed drops while frequency stays fixed, wavelength in glass is two thirds of its air value.

Common mistake
The critical angle is an incidence angle in the higher-index medium.

Exam habit: At the critical angle the refracted ray runs along the boundary; total internal reflection occurs above that angle.

SELF-CHECK

Can light travelling from air into glass undergo total internal reflection at that interface?

Answer: No. Total internal reflection requires travel from higher to lower refractive index.

LESSON 07 / 12

Mirrors, lenses and image signs

After this lesson, you can

- Use the stated image-distance sign convention consistently.
- Infer image reality, orientation and relative size from di and m.

Start with: Reciprocals; positive and negative numbers; a simple ray sketch.

THEORY

A real image is formed where rays actually meet and can be projected on a screen. A virtual image is where rays appear to originate. Use one sign convention consistently: real object distance positive, real image distance positive; converging focal length positive and diverging focal length negative.

RULE

$$1/f = 1/d_o + 1/d_i$$

$$m = -d_i/d_o$$

f signed focal length (cm or m consistently)

d_o object distance (same length unit as f)

d_i signed image distance (same length unit as f)

m transverse magnification (dimensionless)

When this applies

Use the lesson convention: real object and real image distances positive; converging focal length positive.

A method that earns marks

1. Write the sign convention and assign f and do before substitution.
2. Rearrange the lens equation to isolate 1/di and then invert.
3. Calculate m = -di/do.
4. Use di sign for real or virtual, m sign for orientation, and magnitude of m for size.

WORKED EXAMPLE

For a converging lens with focal length 10 cm and object distance 30 cm, image distance is 15 cm and magnification is -0.5: real, inverted, reduced.

1. A converging lens has f = +10 cm and the real object has do = +30 cm.
2. Then 1/di = 1/10 - 1/30 = 1/15 per cm, so di = +15 cm.
3. Magnification is -15/30 = -0.5.
4. Positive image distance means real; negative magnification means inverted; magnitude 0.5 means half the object height.

Common mistake
Do not call every image behind a lens virtual; interpret actual ray convergence and your convention.

Exam habit: After subtraction, invert 1/di to get di; include all three requested image descriptions.

SELF-CHECK

What does m = +2 tell you about an image?

Answer: It is upright and twice the object height; the sign convention and lens setup determine its image position.

LESSON 08 / 12

Optical systems, prisms and instruments

After this lesson, you can

- Explain the different roles of objective and eyepiece.
- Calculate telescope angular magnification for normal adjustment.

Start with: Focal length; image formation; ratios of lengths.

THEORY

A prism refracts light at two surfaces; dispersion separates colours because refractive index depends on wavelength. In a multi-lens system, the first image becomes the object for the next lens. A microscope and a telescope both use an objective and eyepiece, but they serve different object distances.

RULE

$$m_{\text{total}} = m_1 m_2$$

$$|M_{\text{telescope}}| = f_o / f_e \text{ (normal adjustment)}$$

f_o objective focal length (cm or m consistently)

f_e eyepiece focal length (same length unit as fo)

M angular magnification (dimensionless)

m_1, m_2 transverse magnifications of successive stages (dimensionless)

When this applies

The telescope is in normal adjustment, giving the final image at infinity.

A method that earns marks

1. Identify the instrument and whether it views a distant or nearby object.
2. For a normal-adjustment astronomical telescope divide objective focal length by eyepiece focal length for magnification magnitude.
3. State image orientation separately.
4. For a general multi-lens system locate the first image before using it as the next object.

WORKED EXAMPLE

An astronomical telescope with objective focal length 80 cm and eyepiece 4 cm has angular magnification magnitude 20. Its normal image is inverted.

1. The telescope objective focal length is 80 cm and the eyepiece focal length is 4 cm.
2. Their ratio is 80/4 = 20, so the angular magnification magnitude is 20.
3. An astronomical telescope with this usual two-converging-lens arrangement gives an inverted image.
4. The ratio needs no length conversion because both focal lengths use the same unit.

Common mistake
Linear image magnification and angular magnification are different quantities.

Exam habit: Use the telescope focal-length ratio only for the stated telescope model, not as a universal magnification formula.

SELF-CHECK

Keeping the objective fixed, does doubling the eyepiece focal length increase or decrease telescope magnification?

Answer: It halves the magnification magnitude.

LESSON 09 / 12

Interference and diffraction

After this lesson, you can

- Predict how slit geometry changes an interference or diffraction pattern.
- Distinguish double-slit bright fringes from single-slit dark fringes.

Start with: Wavelength; coherent sources; nanometres and millimetres; small angles.

THEORY

Interference occurs when overlapping waves combine. A stable visible pattern needs a fixed phase relationship (coherence). Diffraction is spreading associated with an opening or obstacle. Narrower slits spread light more. Young double-slit bright fringes are evenly spaced when the angles are small.

RULE

$$\Delta y = \lambda L/d$$

$$d\sin\theta = m\lambda \text{ (double-slit bright)}$$

$$a\sin\theta = m\lambda \text{ (single-slit dark, } m \neq 0\text{)}$$

Δy double-slit fringe spacing (m)

L slit-to-screen distance (m)

d separation of double slits (m)

a single-slit width (m)

λ wavelength (m)

When this applies

Even double-slit spacing uses the small-angle, distant-screen approximation and coherent illumination.

A method that earns marks

1. Identify whether the setup is double-slit interference or single-slit diffraction.
2. Label d as slit separation and a as slit width; they describe different distances.
3. For double-slit spacing use λL/d with all lengths in consistent units.
4. For a single slit remember that a smaller width moves the first minima outward and broadens the centre.

WORKED EXAMPLE

Doubling screen distance doubles double-slit fringe spacing. Halving a single-slit width broadens its central diffraction maximum.

1. Double-slit spacing is proportional to screen distance when wavelength and slit separation stay fixed.
2. Replacing L by 2L therefore doubles fringe spacing.
3. In the separate single-slit comparison, the first dark direction obeys sin θ = λ/a.
4. Halving a makes that angle larger, so the bright central maximum broadens.

Common mistake
The double-slit bright condition is not the single-slit bright condition.

Exam habit: Label whether you are locating a bright double-slit fringe or a dark single-slit fringe before using the order number.

SELF-CHECK

If double-slit separation doubles with screen distance and wavelength fixed, what happens to fringe spacing?

Answer: It halves.

Recognize gratings, Michelson paths and thin films

All three appear in the physical-optics section. Start by identifying which arrangement the question describes.

- A diffraction grating has many equally spaced slits. It makes sharp bright directions that separate wavelengths.
- A Michelson interferometer splits light into two arms. Each beam travels to a mirror and back, so mirror movement changes a round-trip path.
- A thin film produces interference between reflections from its two surfaces. Reflection can add a phase reversal, so do not copy the ordinary double-slit bright rule.
- For fringe counting in a simple Michelson setup, each one-wavelength change in path difference shifts the pattern by one fringe.

$\Delta\text{path} = 2\Delta x$ (one Michelson mirror moves)

In a Michelson interferometer using $500\,\text{nm}$ light in air, one mirror moves $250\,\text{nm}$ along its arm. What path change and fringe shift result?

1. Round-trip path change: $2 \times 250 = 500\,\text{nm}$.
2. This equals one wavelength.

The path changes by $500\,\text{nm}$, so one fringe passes a fixed observation point.

LESSON 10 / 12

Polarization and Malus law

After this lesson, you can

- Apply the initial half-intensity rule to unpolarized light.
- Apply Malus law using the angle between polarization and analyzer axes.

Start with: Cosine of common angles; intensity versus field amplitude; sequential fractions.

THEORY

Polarization describes the electric-field direction of a transverse light wave. An ideal polarizer transmits half the intensity of unpolarized light. An analyzer then transmits a fraction set by the angle between its axis and the incident polarization. Linear, circular and elliptical polarization describe different field-tip motions.

RULE

$$I = I_0 \cos^2 \theta$$

$$I_{\text{after first}} = I_{\text{unpolarized}}/2$$

I_0 intensity incident on the analyzer (W/m^2 or stated units)

I intensity transmitted by analyzer (same as I0)

θ angle between incident polarization and analyzer axis (degrees or radians)

When this applies

The calculation assumes ideal polarizers and initially unpolarized illumination.

A method that earns marks

1. Check whether the first incident beam is unpolarized or already linearly polarized.
2. For unpolarized light through an ideal first polarizer divide intensity by two.
3. Use that transmitted intensity as I0 for the analyzer.
4. Multiply by cos^2 θ, where θ is between the two relevant axes.

WORKED EXAMPLE

Unpolarized intensity 100 units passes a polarizer then an analyzer at 60 degrees. After the first: 50; after the second: 50 times 0.25 = 12.5 units.

1. The initial 100 intensity units are unpolarized, so the first ideal polarizer transmits 50 units.
2. The analyzer axis is at 60° to the first polarizer axis.
3. Cos 60° is 0.5 and its square is 0.25.
4. The analyzer therefore transmits 50 times 0.25 = 12.5 units, one eighth of the original intensity.

Common mistake
Do not halve already polarized intensity again before applying Malus law.

Exam habit: Square the cosine, not the angle; and do not apply an extra half factor to an already polarized analyzer input.

SELF-CHECK

Two ideal polarizers have perpendicular axes. What does the second transmit from the first?

Answer: Zero intensity in the ideal model, because $\cos^2 90° = 0$.

LESSON 11 / 12

Scattering, apparent depth and lens power

After this lesson, you can

- Calculate apparent depth near normal viewing and signed lens power.
- Explain why small-particle scattering favours shorter wavelengths.

Start with: Refraction; inverse relationships; centimetres-to-metres conversion.

THEORY

Small particles compared with wavelength produce Rayleigh scattering, which favours shorter wavelengths strongly. Droplets comparable to or larger than visible wavelengths can scatter a broad range of colours, helping explain white clouds. Refraction makes a submerged object appear shallower when viewed near the normal from air.

RULE

$$I_{\text{Rayleigh}} \propto 1/\lambda^4$$

$$d_{\text{apparent}} \approx d_{\text{real}}/n$$

$$P_{\text{lens}} = 1/f \text{ (f in metres)}$$

d_{real} actual depth (m)

d_{apparent} apparent depth (m)

n liquid refractive index relative to air approximation (dimensionless)

P lens optical power (D = m^{-1})

f signed focal length (m)

When this applies

The apparent-depth approximation assumes viewing from air near the normal; the Rayleigh rule applies to particles much smaller than wavelength.

A method that earns marks

1. Select the relevant model: small-particle scattering, near-normal depth or lens power.

2. For depth viewed from air divide real depth by the liquid index.
3. For power convert focal length to metres before taking its reciprocal.
4. Retain the focal-length sign so converging power is positive and diverging power negative.

WORKED EXAMPLE

A pool 2 m deep in water of index 1.33 appears about 1.50 m deep from air near the normal. A converging lens with f=0.25 m has power +4 dioptres.

1. For the pool, near-normal apparent depth is $2/1.33 \approx 1.50$ m, so the bottom appears shallower.
2. The index does not change the real 2 m depth; it changes the directions of the emerging rays.
3. For the separate converging lens, f = +0.25 m.
4. Its power is $1/0.25 = +4\ m^{-1}$, written +4 dioptres.

Common mistake
A focal length of 25 cm is not 25 m; using it directly gives the wrong lens power.

Exam habit: A reciprocal of a focal length in centimetres does not directly give dioptres; convert to metres first.

SELF-CHECK

A diverging lens has focal length -50 cm. What is its power?

Answer: -2 D, since -50 cm = -0.50 m.

LESSON 12 / 12

Recognize advanced optics without long derivations

After this lesson, you can

- Recognize what a quarter-wave plate changes about polarized light.
- Use Brewster angle and plane-wave relations at recognition level.

Start with: Linear and circular polarization; phase as position within an oscillation cycle; refractive index.

THEORY

The wave equation describes a disturbance propagating in space and time. Complex notation is a book-keeping method for amplitude and phase; the measured field is real. Maxwell laws connect electric and magnetic fields. Birefringent materials have different refractive behaviour for different polarizations. At Brewster incidence the reflected component can be linearly polarized.

RULE

$$\tan\theta_B = n_2/n_1$$

$$E = cB \text{ (plane wave in vacuum)}$$

θ_B Brewster incidence angle from the normal (degrees or radians)

n_1, n_2 incident and transmitted medium indices (dimensionless)

E plane-wave electric-field magnitude (V/m)

B plane-wave magnetic-field magnitude (T)

When this applies

The wave plate is operating at its intended wavelength; Brewster law here uses ordinary transparent nonmagnetic media.

A method that earns marks

1. Identify whether the question concerns phase delay, interface polarization or vacuum wave fields.
2. For a quarter-wave plate, compare the two orthogonal component amplitudes and their phase difference.
3. For Brewster incidence use the index ratio with inverse tangent.
4. For a plane vacuum wave connect field magnitudes using E = cB without a long derivation.

WORKED EXAMPLE

A quarter-wave plate can turn suitable linearly polarized light into circular polarization by introducing a quarter-cycle phase difference between equal orthogonal components. Unequal components generally give elliptical polarization.

1. Resolve suitable incident linear polarization along the plate principal axes.
2. Equal component amplitudes are required for the stated circular-polarization result.
3. A quarter-wave plate adds a quarter-cycle, or $\pi/2$, relative phase delay.
4. Equal amplitudes with this delay give circular polarization; unequal amplitudes generally give elliptical polarization.

Common mistake
These recognition notes do not replace a lecturer-assigned derivation; return to the source if such derivations are explicitly assessed.

Exam habit: For a recognition answer name the condition and consequence; deriving Maxwell equations is unnecessary for these exercises.

SELF-CHECK

Is a quarter-cycle phase delay alone enough to guarantee circular polarization?

Answer: No. The two orthogonal components must also have equal amplitudes.

Before the tests

These original questions practise the lesson objectives. They are not predictions of an exam paper. Use the separate practice book if you want all papers together.

Optics and Waves / Test A

30 marks | Suggested time: 30 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. Which wave can propagate through vacuum?

   A. A water wave

   B. Light

   C. A stretched-string wave

   D. Sound in air

2. With wave speed fixed, doubling frequency makes wavelength:

   A. Half as large

   B. Unchanged

   C. Four times as large

   D. Twice as large

3. Group velocity describes the motion of:

   A. The wave-packet envelope

   B. The source itself

   C. One particle only

   D. A stationary node

4. Two tones at 440 and 443 Hz produce a beat frequency of:

   A. 1320 Hz

   B. 883 Hz

   C. 3 Hz

   D. 441.5 Hz

5. At the same medium, which has greater photon energy?

   A. Lower-frequency light

   B. Both always equal

   C. Higher-frequency light

   D. Longer-wavelength light

6. Light enters glass from air. Which quantity remains unchanged?

   A. Ray direction in every case

B. Speed

C. Wavelength

D. Frequency

Part II: Show your reasoning

1. A source makes 50 complete oscillations in 10 s. Find frequency and period.

2. A wave has frequency 8 Hz and wavelength 0.75 m. Find speed. If frequency becomes 12 Hz in the same nondispersive medium, find wavelength.

3. At distance r an isotropic source has intensity 18 W per square metre. Find intensity at 3r, ignoring absorption.

4. A 1 m string fixed at both ends supports waves at 40 m/s. Find its fundamental frequency and second harmonic.

5. Light A has twice the frequency of light B. Compare photon energy and vacuum wavelength.

6. Find the critical angle for glass of index 1.5 into air of index 1.0.

Optics and Waves / Test B

30 marks | Suggested time: 30 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. An image with magnification -2 is:

   A. Upright and smaller

   B. Inverted and twice as tall

   C. Virtual in every case

   D. Upright and twice as tall

2. In a simple astronomical telescope, the objective usually has:

   A. A longer focal length than the eyepiece

   B. Zero focal length

   C. The same role as a polarizer

   D. No effect on the image

3. If double-slit separation increases while wavelength and screen distance stay fixed, fringe spacing:

   A. Becomes negative

   B. Increases

   C. Stays fixed

   D. Decreases

4. Two ideal crossed polarizers transmit from an initially unpolarized beam:

   A. One quarter

   B. Zero

   C. All its intensity

   D. Half

5. The unit of optical lens power is:

   A. Hertz

   B. Watt

   C. Dioptre

   D. Newton

6. A quarter-wave plate acts primarily by changing the relative:

   A. Charge of light

B. Speed of the source

C. Mass of photons

D. Phase of orthogonal components

Part II: Show your reasoning

1. A converging lens has focal length 12 cm and an object at 24 cm. Find image distance, magnification and image type.

2. A telescope objective has focal length 100 cm and eyepiece 5 cm. Find angular magnification magnitude. What happens if the eyepiece is replaced by 10 cm?

3. In a double-slit experiment, wavelength is 500 nm, screen distance 2 m and slit separation 0.50 mm. Find fringe spacing.

4. Unpolarized intensity 80 W per square metre passes through a polarizer and an analyzer at 45 degrees. Find the final intensity.

5. A diverging lens has focal length -50 cm. Find its optical power.

6. For light from air into glass of index 1.5, estimate Brewster angle.

Optics and Waves / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **B.** Light is electromagnetic and does not require a material medium.
2. **A.** Wavelength equals speed divided by frequency.
3. **A.** The group is the envelope formed by combining nearby frequencies.
4. **C.** Beat frequency is the absolute difference.
5. **C.** Photon energy is proportional to frequency.
6. **D.** The boundary does not change the frequency of the incident oscillation.

Part II: Written solutions

1. Frequency = 50/10 = 5 Hz. Period = $1\frac{0}{5}0$ = 0.20 s. Two marks each.
2. Speed = 8 times 0.75 = 6 m/s. New wavelength = 6/12 = 0.50 m. Two marks each.
3. Inverse-square spreading gives 18/9 = 2 W per square metre. The distance triples, so the area over which power spreads increases ninefold.
4. Fundamental wavelength = 2 m; frequency = 40/2 = 20 Hz. Second harmonic = 40 Hz. Two marks per frequency with reasoning.
5. A has twice the photon energy and half the vacuum wavelength. Two marks for each relationship.
6. Sine of critical angle = $\frac{1}{1}$.5 = 0.6667, so critical angle = 41.8 degrees. Total internal reflection occurs above this angle. Credit condition, equation, value and interpretation.

Optics and Waves / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **B.** The minus sign indicates inversion and magnitude 2 gives the size ratio.
2. **A.** The long-focus objective and short-focus eyepiece give a large angular-magnification ratio.
3. **D.** Fringe spacing is inversely proportional to slit separation.
4. **B.** Their axes differ by 90 degrees and $\cos^2 90^\circ$ is zero.
5. **C.** Lens power is inverse focal length in metres, expressed in dioptres.
6. **D.** A controlled phase delay changes the polarization state.

Part II: Written solutions

1. 1/di = $\frac{1}{1}$2 - 1/24 = 1/24, so di = 24 cm. m = -1. Image is real, inverted and equal size. One mark each for distance, magnification, real and inverted/equal size.
2. Initial magnitude = 10$\frac{0}{5}$ = 20. New magnitude = 100/10 = 10. Doubling eyepiece focal length halves angular magnification.
3. Convert wavelength to 0.0000005 m and separation to 0.0005 m. Spacing = (0.0000005 times 2)/0.0005 = 0.002 m = 2 mm. Credit both conversions, calculation and result.
4. First polarizer gives 40. Analyzer gives 40 times cosine squared 45 degrees = 20 W per square metre. Two marks for each stage.
5. Convert to -0.50 m. Power=1/(-0.50)=-2 dioptres. Credit conversion, formula, sign and unit.
6. Tangent of Brewster angle=1.5/1.0. The angle is arctan(1.5)=56.3 degrees. Credit correct ratio, inverse tangent, result and interpretation as incidence angle.

Source and coverage notes

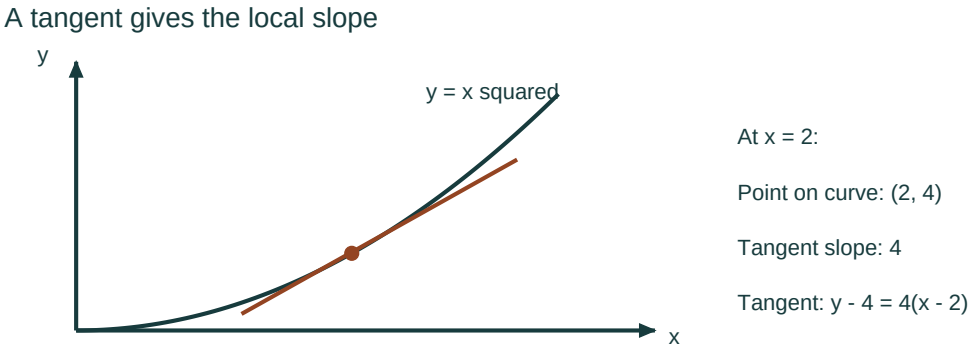
The introduction and four current sections: wave motion; electromagnetic theory, photons and light; geometrical optics; physical optics. Derivations are replaced by plain-language interpretation and short exam calculations.

Source folder: Open original course folder The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. This revised study aid does not replace lecturer-specific derivations or instructions. All practice tests and suggested marking guidance are original study material.

04

Calculus I

A field guide to understanding, working and checking.



MATH 203: derivative rules, implicit/logarithmic/exponential derivatives, L Hopital, graphing and extrema, optimization, Rolle and mean-value theorems, integration and the fundamental theorem. The full lecture syllabus is the main source.

12 lessons / 2 practice tests / Worked solutions

English edition / November 2026

How to use this handout

1. Read the concept and check the symbol meanings before choosing a formula.
2. Cover the worked steps and reproduce the reasoning yourself.
3. Attempt the quick check and the two subject tests without notes.
4. Mark your work; record whether the error was concept, method, algebra, units or reading. Retry after a delay.

Your study route

1. Limits, continuity and the derivative definition
2. Basic derivatives and tangent lines
3. Product and quotient rules
4. Chain rule and trigonometric functions
5. Implicit differentiation
6. Logarithmic, exponential and inverse-trig derivatives
7. L Hopital and indeterminate forms
8. Increasing, decreasing and concavity
9. Graphing, asymptotes and absolute extrema
10. Optimization and mean-value theorems
11. Antiderivatives and substitution
12. Definite integrals, signed area and the fundamental theorem

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LESSON 01 / 12

Limits, continuity and the derivative definition

After this lesson, you can

- Distinguish a limit from a value at the point.
- Find a polynomial derivative from the difference quotient.

Start with: Expanding a square; factoring and cancelling a nonzero common factor.

THEORY

A limit describes nearby values, not necessarily the function value at the point. Continuity requires the function value and the two-sided limit to exist and agree. The derivative is the limiting slope of secant lines and gives instantaneous rate of change.

RULE

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

h small nonzero input increment before the limit (same unit as x)

$f'(x)$ instantaneous rate of change (output unit per input unit)

A method that earns marks

1. Write f(x+h) and subtract the complete original f(x).
2. Expand brackets carefully and collect terms.
3. Factor out h and cancel only while h is nonzero.
4. Take the limit h approaching zero in the simplified expression.

WORKED EXAMPLE

For $f(x) = x^2$, the difference quotient becomes $\frac{2xh+h^2}{h} = 2x + h$. Letting h approach zero gives derivative 2x.

1. For f(x) = x^2, the incremented value is $(x+h)^2 = x^2 + 2xh + h^2$.
2. Subtracting f(x) leaves $2xh + h^2$.
3. Dividing by nonzero h gives 2x + h.
4. As h approaches zero, the secant slope approaches 2x, so f′(x) = 2x.

Common mistake
Zero over zero is an indeterminate form, not a final answer of zero.

Exam habit: Place brackets around f(x+h) and f(x) before subtraction, especially when either contains negative terms.

SELF-CHECK

Why should h not be set to zero before simplifying the difference quotient?

Answer: It would produce division by zero; the limit concerns nonzero h approaching zero.

LESSON 02 / 12

Basic derivatives and tangent lines

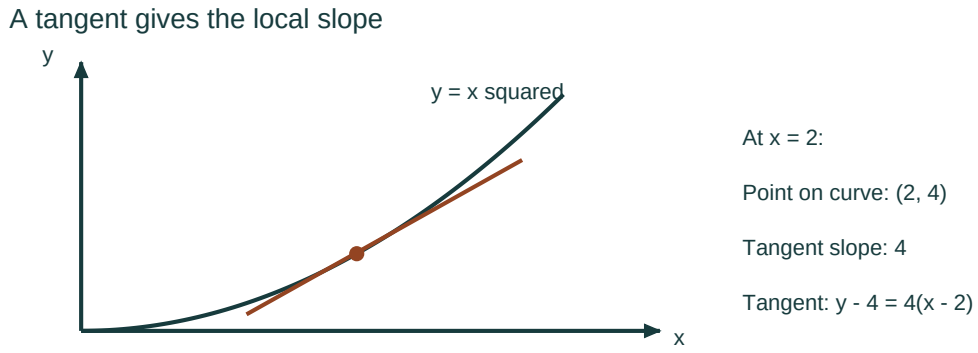
After this lesson, you can

- Differentiate sums and powers correctly.
- Build a tangent line from the original point and derivative slope.

Start with: Power rule; substituting an x value; point-slope form of a line.

THEORY

Differentiate a sum term by term. Constants disappear. The derivative gives the slope at a point, but the tangent line must also pass through the point on the original function. Rewrite roots and reciprocal powers before applying the power rule.



The tangent to $y = x^2$ at (2,4) has slope 4. The axes use different display scales; slope is calculated from coordinate values, not from the visual angle.

RULE

$$\frac{d}{dx}x^n = nx^{n-1}$$

$$y - f(a) = f'(a)(x - a)$$

a input coordinate of tangent point (input unit)

$f(a)$ output coordinate of tangent point (output unit)

$f'(a)$ tangent slope (output unit per input unit)

A method that earns marks

1. Differentiate the original function term by term.

2. Substitute x = a into the derivative for the slope.
3. Substitute x = a into the original function for the point.
4. Use y - f(a) = f′(a)(x-a), then simplify only if requested.

WORKED EXAMPLE

For $f(x) = x^2 + 1$ at $x = 2$, the point is (2,5) and slope is 4. The tangent is $y - 5 = 4(x - 2)$, or $y = 4x - 3$.

1. For f(x) = x^2 + 1, f′(x) = 2x because the constant contributes zero.
2. At x = 2 the slope is 4, while the original function gives y = 5.
3. The tangent must therefore pass through (2,5) with slope 4.
4. Its equation is y - 5 = 4(x-2), which expands to y = 4x - 3.

Common mistake
Do not use $f'(a)$ as the y-coordinate.

Exam habit: Keep the unsimplified point-slope equation visible: it shows both required ingredients for method marks.

SELF-CHECK

Does substituting x = 2 into f′ give the y coordinate of the tangent point?

Answer: No. It gives the slope; use f(2) for the y coordinate.

LESSON 03 / 12

Product and quotient rules

After this lesson, you can

- Differentiate a product using both changing factors.
- Preserve term order and the squared denominator in the quotient rule.

Start with: Basic polynomial and sine derivatives; multiplication and fractions.

THEORY

A product has two changing factors, so both contribute to the derivative. A quotient rule numerator must preserve its order; reversing it changes the sign. Do not differentiate numerator and denominator separately to differentiate a fraction.

RULE

$$(uv)' = u'v + uv'$$
$$(u/v)' = (u'v - uv')/v^2$$

u, v the two factor functions (problem dependent)

u', v' their derivatives with respect to x (corresponding output per input unit)

A method that earns marks

1. Identify whether the expression is a product or quotient at the outermost level.
2. Name the component functions u and v and calculate their derivatives.
3. Insert all terms into the appropriate rule before simplifying.
4. For a quotient retain the original domain restriction and check the numerator subtraction order.

WORKED EXAMPLE

For $y = x^2 \sin x$, derivative is $2x \sin x + x^2 \cos x$. Both terms are needed.

1. For y = x^2 sin x, choose u = x^2 and v = sin x.
2. Their derivatives are u′ = 2x and v′ = cos x.
3. The product rule keeps each factor once unchanged: u′v + uv′.
4. Thus y′ = 2x sin x + x^2 cos x; multiplying the two derivatives alone would miss both contributions.

Common mistake
The quotient denominator is the square of the original denominator.

Exam habit: Write the entire quotient-rule numerator in brackets before expanding; the minus sign applies to the whole second product.

SELF-CHECK

For y = (x+1)/(x-1), what is the derivative for $x \neq 1$?

Answer: -2/$(x-1)^2$: the numerator is (x-1) - (x+1) = -2.

LESSON 04 / 12

Chain rule and trigonometric functions

After this lesson, you can

- Identify inner and outer operations in a composite function.
- Include the inner derivative in trigonometric and power derivatives.

Start with: Power rule; sine and cosine derivatives; radians.

THEORY

A composite function has an outer operation acting on an inner expression. Differentiate the outside while preserving the inside, then multiply by the derivative of the inside. Trigonometric derivative formulas use radians.

RULE

$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$$

$$(\sin x)' = \cos x$$

$$(\cos x)' = -\sin x$$

$u = g(x)$ inner expression (problem dependent)

$g'(x)$ inner derivative (inner-expression unit per input unit)

A method that earns marks

1. Identify the last operation applied to the expression: this is the outer function.
2. Differentiate that outer function while leaving the inner expression in place.
3. Multiply by the derivative of the inner expression.
4. Check the sign for cosine derivatives and use radians for trigonometric rules.

WORKED EXAMPLE

For $y = (3x+1)^4$, derivative = $4(3x+1)^3 \cdot 3 = 12(3x+1)^3$.

1. In y = $(3x+1)^4$, the inner expression is 3x+1 and the outer operation is raising to the fourth power.
2. The outer derivative gives $4(3x+1)^3$.
3. The inner derivative is 3, so it must multiply the outer result.

4. Combining the constants gives $y' = 12(3x+1)^3$.

Common mistake
Forgetting the inner derivative is the most common lost mark.

Exam habit: Before finishing, circle the inner expression and verify that its derivative appears as a factor.

SELF-CHECK

What is the derivative of $\cos(x^2)$?

Answer: $-2x \sin(x^2)$, combining the cosine derivative with the inner derivative 2x.

Four more trigonometric derivatives

The supplied final asks for a cotangent derivative. Learn these rules together with sine, cosine and the chain rule.

- Use radians. Here u is a function of x, and u' means its derivative.
- Apply the outside rule, then multiply by the inside derivative.
- Use a rule only where the original function is defined. Remember: $\sec u = 1/\cos u$, $\csc u = 1/\sin u$, and $\cot u = \cos u/\sin u$.

$$\frac{d}{dx}\tan u = \sec^2 u\, u'$$
$$\frac{d}{dx}\cot u = -\csc^2 u\, u'$$
$$\frac{d}{dx}\sec u = \sec u \tan u\, u'$$
$$\frac{d}{dx}\csc u = -\csc u \cot u\, u'$$

Differentiate $y = \cot(5x)$.

1. The inside is $u = 5x$, so $u' = 5$.
2. The cotangent rule gives $y' = -\csc^2(5x) \times 5$.

$y' = -5\csc^2(5x)$. The minus sign and the factor 5 both matter.

LESSON 05 / 12

Implicit differentiation

After this lesson, you can

- Differentiate y terms while treating y as a function of x.
- Collect and isolate all terms containing y′.

Start with: Chain and product rules; algebraic rearrangement.

THEORY

When x and y appear in one equation, y is treated as a function of x. Differentiating a y term therefore introduces y'. Collect all y-prime terms and solve algebraically. This can describe curves that are not single-valued functions globally.

RULE

$$\frac{d}{dx}(y^2) = 2yy'$$

$$\frac{d}{dx}(xy) = y + xy'$$

y' dy/dx, the local slope of the curve (y unit per x unit)

A method that earns marks

1. Differentiate both sides with respect to x.
2. Attach y′ when differentiating a function of y, and use the product rule for xy.
3. Move all terms containing y′ to one side.
4. Factor out y′, divide by its coefficient and state where that division is valid.

WORKED EXAMPLE

From $x^2 + y^2 = 25$, differentiation gives $2x + 2yy' = 0$. Thus $y' = -\frac{x}{y}$ wherever y is nonzero.

1. The curve $x^2 + y^2 = 25$ need not be solved globally for a single y value.
2. Differentiating x^2 gives 2x; differentiating y^2 gives 2y y′ by the chain rule.
3. The constant 25 differentiates to zero, so 2x + 2y y′ = 0.
4. Rearranging gives y′ = -x/y wherever $y \neq 0$; division by y requires this restriction.

Common mistake
Do not treat y as a constant or omit the product rule on xy.

Exam habit: For an xy term write y + xy′, not simply y or xy′; both factors depend on x.

SELF-CHECK

Why does the derivative of y^2 contain y′?

Answer: Because y depends on x; the chain rule multiplies 2y by dy/dx.

LESSON 06 / 12

Logarithmic, exponential and inverse-trig derivatives

After this lesson, you can

- Differentiate exponential and logarithmic composites.
- Use logarithmic differentiation for a positive variable base and exponent.

Start with: Chain rule; natural logarithms; product rule; positive-domain restrictions.

THEORY

Exponential functions reproduce themselves up to an inner derivative. Natural logarithms give a reciprocal factor. Logarithmic differentiation helps when variable powers or products are awkward. Inverse-trigonometric functions are different from reciprocal trigonometric functions.

RULE

$$(e^u)' = e^u u'$$
$$(\ln|u|)' = u'/u$$
$$(\arctan x)' = 1/(1+x^2)$$

e base of the natural exponential (dimensionless)

$\ln u$ natural logarithm of a positive dimensionless argument (dimensionless)

A method that earns marks

1. Identify the outer function and check its real domain.
2. For a logarithm use inner derivative divided by the inner expression.
3. For an exponential retain the exponential and multiply by the inner derivative.

4. For y = x^x with x > 0, take logs first, differentiate implicitly, then restore y.

WORKED EXAMPLE

For $y = \ln(x^2 + 1)$, derivative is $\frac{2x}{x^2+1}$. For y=e^{3x}, derivative is $3e^{3x}$.

1. For ln(x^2+1), the inner derivative is 2x and the inner expression is always positive.
2. The logarithmic chain rule therefore gives 2x/(x^2+1).
3. For the separate function e^(3x), the exponential is retained unchanged.
4. Multiplication by the inner derivative 3 gives 3e^(3x).

Common mistake
The derivative of ln u is u prime/u, not 1/ln u.

Exam habit: The power rule nx^(n-1) assumes a constant exponent; x^x needs another method.

SELF-CHECK

For x > 0, what is d(x^x)/dx?

Answer: x^x(ln x + 1). Take ln y = x ln x, then y′/y = ln x + 1.

Other exponential, logarithm and inverse-trig rules

The supplied final includes 3^x and $\log_3 x$. Their derivatives need a factor involving $\ln 3$.

- The base a is a fixed positive number, with $a \neq 1$. For the logarithm, require $u > 0$.
- For the displayed arcsine and arccosine derivatives, require $|u| < 1$. Angles are in radians.
- Find the inside derivative u' first. Inverse sine means $\arcsin$; it is different from $1/\sin$.

$$\frac{d}{dx}a^u = a^u \ln(a)\, u'$$
$$\frac{d}{dx}\log_a u = \frac{u'}{u \ln a}$$
$$\frac{d}{dx}\arcsin u = \frac{u'}{\sqrt{1-u^2}}$$
$$\frac{d}{dx}\arccos u = -\frac{u'}{\sqrt{1-u^2}}$$

Differentiate 3^x and $\log_3 x$.

1. In both cases the inside expression is $u = x$, so $u' = 1$.
2. For the exponential, multiply by $\ln 3$. For the logarithm, divide by $\ln 3$.

$(3^x)' = 3^x \ln 3$, and $(\log_3 x)' = 1/(x \ln 3)$ for $x > 0$.

LESSON 07 / 12

L Hopital and indeterminate forms

After this lesson, you can

- Check for a permitted indeterminate quotient before using L Hopital.
- Evaluate a simple limit by differentiating numerator and denominator.

Start with: Limits; exponential and trigonometric derivatives; chain rule.

THEORY

L Hopital rule can evaluate a quotient limit with the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$ under its differentiability and limit conditions. Differentiate numerator and denominator separately for this limit procedure; this is not the quotient rule. Other indeterminate forms must first be rewritten.

RULE

$$\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)}$$

when the rule applies

$0/0$ indeterminate limiting form, not an arithmetic value (not a measured quantity)

∞/∞ indeterminate quotient form (not a measured quantity)

When this applies

The numerator and denominator satisfy the differentiability and denominator-derivative conditions near the limiting point.

A method that earns marks

1. Substitute the limiting input to identify the quotient form.
2. Confirm 0/0 or infinity/infinity and the rule conditions before differentiating.
3. Differentiate numerator and denominator separately, not with the quotient rule.
4. Evaluate the new limit; if still indeterminate, check conditions again before repeating.

WORKED EXAMPLE

As x approaches zero, $\frac{e^x-1}{x}$ has form $\frac{0}{0}$. The derivative quotient is $\frac{e^x}{1}$, giving limit 1.

1. As x approaches zero, e^x - 1 approaches zero and x approaches zero.
2. The quotient therefore has the required 0/0 form.
3. Differentiate numerator and denominator separately to obtain e^x/1.
4. Its limit is e^0 = 1, so the original limit is 1.

Common mistake
Do not use the rule on a quotient whose substitution gives an ordinary finite value.

Exam habit: Show the initial indeterminate form; applying the rule without checking its conditions can invalidate the method.

SELF-CHECK

What is the limit of sin(3x)/x as x approaches zero?

Answer: 3, since the derivative quotient is 3 cos(3x)/1 and cos 0 = 1.

LESSON 08 / 12

Increasing, decreasing and concavity

After this lesson, you can

- Use a first-derivative sign chart to find monotonic intervals.
- Distinguish a stationary point from a proven extremum or inflection.

Start with: Factoring quadratics; sign tests; first and second derivatives.

THEORY

The sign of the first derivative controls increasing and decreasing intervals. The sign of the second controls concavity. A critical number lies in the domain where the first derivative is zero or undefined. An inflection point requires a concavity change, not merely a zero second derivative.

RULE

$$f' > 0 : \text{ increasing;}$$
$$f'' > 0 : \text{ concave up}$$

f' first derivative, controlling increase or decrease (output per input unit)

f'' second derivative, controlling concavity (output per input unit squared)

A method that earns marks

1. Find where f′ is zero or undefined within the original domain.
2. Split the domain at those critical numbers and test the sign on each interval.
3. Use sign changes to classify local maxima and minima.
4. Use f″ signs separately for concavity and require a concavity change for an inflection point.

WORKED EXAMPLE

For $f = x^3 - 3x$, derivative is $3(x - 1)(x + 1)$. The function increases outside $[-1, 1]$ and decreases between -1 and 1.

1. For f(x) = x^3 - 3x, f′ = $3x^2$ - 3 = 3(x-1)(x+1).
2. The critical numbers are -1 and 1.
3. Testing x = -2, 0 and 2 gives derivative signs positive, negative and positive.
4. The function increases on (-∞,-1) and (1,∞), and decreases on (-1,1); the sign changes identify a local maximum at -1 and minimum at 1.

Common mistake
A derivative equal to zero does not automatically mean a maximum.

Exam habit: Give open intervals for increasing and decreasing behaviour, then state extrema as points if asked.

SELF-CHECK

Is f″(c) = 0 alone enough to prove an inflection point?

Answer: No. Concavity must change across c.

LESSON 09 / 12

Graphing, asymptotes and absolute extrema

After this lesson, you can

- Find absolute extrema by comparing all candidates on a closed interval.
- Distinguish a removable hole from a vertical asymptote.

Start with: Domain restrictions; critical numbers; evaluating a function at endpoints.

THEORY

Graphing combines domain, intercepts, derivative signs and limiting behaviour. A cancelled factor can create a hole rather than an asymptote. On a closed interval a continuous function attains absolute extrema; compare endpoint and interior critical values.

RULE

$$\text{Candidates} = \text{endpoints plus interior critical numbers}$$

$[a, b]$ closed interval including both endpoints (input unit)

$f(c)$ function value at a candidate point (output unit)

A method that earns marks

1. Check the domain and whether the function is continuous on the stated closed interval.
2. Find all interior critical numbers.
3. Evaluate the original function at those numbers and at both endpoints.
4. Compare the values and report each extremum value together with where it occurs.

WORKED EXAMPLE

For x^2 on $[-1, 2]$, candidates are -1,0,2. Values are 1,0,4, so the absolute minimum is 0 and maximum is 4.

1. For f(x) = x^2 on [-1,2], the derivative 2x is zero at x = 0.
2. The candidate inputs are the two endpoints -1 and 2 and the interior critical number 0.
3. The original function values are f(-1)=1, f(0)=0 and f(2)=4.
4. The smallest is 0 at x=0; the largest is 4 at x=2.

Common mistake
Do not omit endpoints or compare derivative values instead of function values.

Exam habit: Use the original function for the comparison table; comparing derivative values does not locate the highest graph point.

SELF-CHECK

Can an endpoint be the absolute maximum even when its derivative is not zero?

Answer: Yes. Absolute extrema on a closed interval must include endpoint checks.

LESSON 10 / 12

Optimization and mean-value theorems

After this lesson, you can

- Translate a simple optimization constraint into one variable.
- Check mean-value-theorem assumptions and solve for the interior input.

Start with: Area and perimeter; derivatives; continuity and differentiability.

THEORY

Optimization translates a word problem into a one-variable function and an allowed domain. Rolle theorem requires equal endpoint values as well as continuity and interior differentiability. The mean-value theorem matches some instantaneous slope to the average slope.

RULE

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$
$$a < c < b$$

x one rectangle side length (length unit)

A rectangle area (square length unit)

c interior input guaranteed by the mean-value theorem (input unit)

A method that earns marks

1. For optimization name the objective, write the constraint and identify the feasible domain.
2. Eliminate one variable, differentiate and find interior critical candidates.
3. Check endpoints or derivative signs to justify the optimum.

4. For a theorem question verify its assumptions before setting instantaneous slope equal to average slope.

WORKED EXAMPLE

A rectangle with perimeter 20 has area $x(10-x)$. Its derivative is $10-2x$, so maximum area occurs at $x=5$: a square of area 25.

1. Perimeter 20 gives 2x + 2y = 20, so the other side is y = 10 - x.
2. Area becomes A(x) = x(10-x) with 0 < x < 10 for a nondegenerate rectangle.
3. Its derivative is 10 - 2x, which vanishes at x = 5.
4. The second derivative is -2, confirming a maximum; both sides are 5 and area is 25 square units.

Common mistake
A candidate outside the physical domain is not a valid optimization answer.

Exam habit: A critical number alone is not a maximum proof; include a sign change, second derivative or candidate comparison.

SELF-CHECK

For f(x)=x^2 on [1,3], what c satisfies the mean-value theorem?

Answer: c=2: the average slope is (9-1)/(3-1)=4 and f′(c)=2c=4.

LESSON 11 / 12

Antiderivatives and substitution

After this lesson, you can

- Choose a substitution whose differential matches the integrand.
- Return the antiderivative to x and include the integration constant.

Start with: Chain rule; power integration; constant factors.

THEORY

An indefinite integral represents a family of antiderivatives, so include an arbitrary constant. Substitution reverses the chain rule: select an inner expression whose derivative is present. Differentiate your answer as a fast correctness check.

RULE

$$\int x^n\,dx = x^{n+1}/(n+1) + C\ (n \neq -1)$$

$$\int dx/x = \ln|x| + C$$

u chosen inner expression (problem dependent)

du differential of the inner expression (same unit as u)

C arbitrary additive integration constant (antiderivative unit)

A method that earns marks

1. Choose an inner expression whose derivative is also present as a factor.
2. Write u and du and replace every part of the integral consistently.
3. Integrate in u, preserving any numerical factors.
4. Substitute back to x, add C and differentiate once to check.

WORKED EXAMPLE

For the integral of $2x(x^2+1)^3$, set $u = x^2+1$ and $du = 2x\,dx$. Answer is $\frac{(x^2+1)^4}{4} + C$.

1. For the integral of $2x(x^2+1)^3$, choose $u = x^2+1$.
2. Then $du = 2x\,dx$, so the complete integral becomes the integral of $u^3\,du$.
3. The antiderivative is $u^4/4 + C$.
4. Substitute back to get $(x^2+1)^4/4 + C$; differentiation restores $2x(x^2+1)^3$.

Common mistake
The power rule exception is n=-1; do not divide by zero.

Exam habit: Do not leave a mixture of u and x in the substituted integral; it hides missing differential factors.

SELF-CHECK

What is the integral of $6x(x^2+2)^2\,dx$?

Answer: $(x^2+2)^3 + C$. With $u=x^2+2$, the integral becomes 3 times the integral of $u^2\,du$.

Integrate sine, cosine and an exponential

These short rules reverse the chain rule. A sine integral appears in the supplied final.

- Here a and b are constants and $a \neq 0$. Trigonometric angles use radians.

- Divide by the coefficient a; differentiation would multiply by it.
- Add C for an indefinite integral. Differentiate your answer to check it.

$$\int \sin(ax+b)\,dx = -\frac{\cos(ax+b)}{a} + C$$

$$\int \cos(ax+b)\,dx = \frac{\sin(ax+b)}{a} + C$$

$$\int e^{ax+b}\,dx = \frac{e^{ax+b}}{a} + C$$

Find $\int \sin(3x)\,dx$.

1. The sine antiderivative is negative cosine.
2. The inside derivative is 3, so divide by 3.
3. Check: $\frac{d}{dx}[-\cos(3x)/3] = \sin(3x)$.

$$-\frac{1}{3}\cos(3x) + C.$$

LESSON 12 / 12

Definite integrals, signed area and the fundamental theorem

After this lesson, you can

- Distinguish signed integral from total geometrical area.
- Apply the fundamental theorem, including a variable-upper-limit chain factor.

Start with: Antiderivatives; positive and negative regions; substituting bounds.

THEORY

A definite integral gives signed accumulation: contributions below the axis are negative. Geometrical area requires positive contributions from each region. The fundamental theorem connects integration and differentiation; for a variable upper limit, include its derivative.

RULE

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

$$\frac{d}{dx}\int_a^{g(x)} f(t)\,dt = f(g(x))g'(x)$$

a, b lower and upper integration bounds (input unit)

F an antiderivative of f (f unit times input unit)

t dummy integration variable (input unit)

A method that earns marks

1. Identify whether the question asks for signed accumulation or total area.
2. Find an antiderivative and calculate upper value minus lower value.
3. For total area split at sign changes and make each region contribution positive.
4. For differentiation with upper bound g(x), evaluate f at g(x) and multiply by g′(x).

WORKED EXAMPLE

The $\int_{-1}^{1} x\,dx$ is zero by cancellation, but total geometrical area is 1. These answer different questions.

1. For y = x from -1 to 1, the region below the axis contributes negative signed area.
2. An antiderivative is $x^2/2$; evaluation at 1 and -1 gives 1/2 - 1/2 = 0.
3. Geometrically each half is a triangle with area 1/2.
4. Total geometrical area is 1/2 + 1/2 = 1, while the signed integral remains zero.

Common mistake
Never keep x-limits on an integral that has been fully changed to u.

Exam habit: For a definite integral evaluate both bounds; no arbitrary +C is needed in the final numerical answer.

SELF-CHECK

Does a zero definite integral prove the function is zero throughout the interval?

Answer: No. Positive and negative contributions can cancel.

Read a Riemann sum as rectangles

The course syllabus includes sigma notation and area as a limit. The symbol $\sum$ simply means add the terms.

- Split the interval from a to b into n equal widths.
- Choose a sample point x_i^* in each small interval. Its function value is that rectangle's signed height.
- Add height times width. More, narrower rectangles approach the definite integral when the function is continuous.
- Below-axis heights contribute negative signed area.

$$\Delta x = \frac{b-a}{n}$$

rectangle sum $= \sum_{i=1}^{n} f(x_i^*)\,\Delta x$

Use two right-end rectangles to estimate $\int_0^1 x\,dx$.

1. Each width is $\Delta x = (1-0)/2 = 1/2$.
2. The right endpoints are $1/2$ and 1, giving heights $1/2$ and 1.
3. The estimate is $(1/2)(1/2) + (1)(1/2) = 3/4$.

Estimate: $3/4$. Exact integral: $1/2$. The right-end estimate is too large because the function increases.

Before the tests

These original questions practise the lesson objectives. They are not predictions of an exam paper. Use the separate practice book if you want all papers together.

Calculus I / Test A

30 marks | Suggested time: 30 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. The derivative of x^2 at $x = 3$ is:

   A. 3

   B. 6

   C. 9

   D. 0

2. The derivative of a constant 7 is:

   A. 7

   B. 1

   C. x

   D. 0

3. The derivative of xe^x is:

   A. $e^x(1 + x)$

   B. 0

   C. e^x

   D. xe^x

4. The derivative of $\sin(5x)$ is:

   A. $-5\sin(5x)$

   B. $\sin(5)$

   C. $\cos(5x)$

   D. $5\cos(5x)$

5. For $x^2 + y^2 = 25$, the slope at (3,4) is:

   A. 0

   B. $\frac{3}{4}$

   C. $-\frac{3}{4}$

   D. $\frac{4}{3}$

6. The derivative of $\arctan x$ is:

   A. $\arctan(1)$

B. $\frac{1}{1+x^2}$

C. $\frac{1}{\cos^2 x}$

D. $-\frac{1}{1+x^2}$

Part II: Show your reasoning

1. Use the definition to find the derivative of f(x)=3x^2.

2. Find the tangent to f(x)=x cubed at $x = 2$.

3. Differentiate y=(x+1)/(x-1), for x not equal to 1.

4. Differentiate y=cos(x^2).

5. Find dy/dx for xy+y squared=6.

6. Differentiate y=x to the power x for x greater than zero.

Calculus I / Test B

30 marks | Suggested time: 30 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. Before applying L Hopital directly, the quotient should have form:

   A. $\frac{0}{0}$ or $\frac{\infty}{\infty}$

   B. $\frac{1}{1}$ only

   C. $\frac{0}{5}$ only

   D. Any fraction

2. If f' is negative on an interval, f is:

   A. Decreasing

   B. Constant

   C. Always concave down

   D. Increasing

3. To find absolute extrema on a closed interval, check:

   A. Only where f' is zero

   B. Only endpoints

   C. Endpoints and interior critical numbers

   D. Only where f is zero

4. Rolle theorem additionally requires:

   A. An infinite interval

   B. Equal endpoint function values

   C. A discontinuity

   D. A nonzero derivative everywhere

5. An indefinite integral must generally include:

   A. Only a numerical answer

   B. A fixed upper limit

   C. No variable

   D. An arbitrary constant

6. The $\int_{-2}^{2} x\,dx$ equals:

   A. 8

B. -4

C. 0

D. 4

Part II: Show your reasoning

1. Evaluate the limit of sin(3x)/x as x approaches zero.

2. For $f(x) = x^2$-4x, find increasing and decreasing intervals and the minimum point.

3. Find absolute maximum and minimum of $f(x) = x^2$-2x on [0,3].

4. Use the mean-value theorem for $f(x) = x^2$ on [1,3] to find c.

5. Evaluate the integral of 6x(x^2+2) squared with respect to x.

6. Evaluate the integral from 0 to 1 of 2x(x^2+1) dx.

Calculus I / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **B.** The derivative is 2x, so its value at 3 is 6.
2. **D.** A constant function has zero rate of change.
3. **A.** The product rule differentiates each factor in turn.
4. **D.** The inner derivative contributes a factor of 5.
5. **C.** Implicit differentiation gives slope $-\frac{x}{y}$.
6. **B.** This is the standard inverse-tangent derivative.

Part II: Written solutions

1. [3(x+h) squared -3x^2]/h = [6xh+3h squared]/h = 6x+3h. Taking the limit gives 6x. One mark each for setup, expansion, cancellation and limit.
2. f(2)=8; derivative=3x^2; slope at 2 is 12. Tangent y-8=12(x-2), equivalently y=12x-16. One mark per step.
3. Derivative = [(x-1)-(x+1)]/(x-1) squared = -2/(x-1) squared. Award marks for quotient rule, numerator order, denominator square and simplification.
4. Outer derivative is -sin(x^2); inner derivative is 2x. Multiply to obtain -2x sin(x^2). Credit both layers, sign and product.
5. Differentiate: y+x y'+2y y'=0. Collect: (x+2y)y'=-y. Hence y'=-y/(x+2y), where the denominator is nonzero. Four marks across these steps.
6. Take logarithms: ln y=x ln x. Differentiate: y'/y=ln x+1. Multiply by y: y'=x to the x times (ln x+1). Four marks for these stages.

Calculus I / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **A.** The rule concerns specified indeterminate quotient forms.
2. **A.** Negative slope means the function decreases as x increases.
3. **C.** The largest or smallest value may occur at either kind of candidate.
4. **B.** Equal endpoint heights make the average slope zero.
5. **D.** Differentiation loses constants, so all antiderivatives differ by a constant on an interval.
6. **C.** The odd integrand has equal and opposite signed areas.

Part II: Written solutions

1. The form is $\frac{0}{0}$. L Hopital gives limit 3 cos(3x)/1 = 3. Alternatively use the standard sine limit. Credit form, derivative, evaluation and answer.
2. Derivative = 2x-4, zero at 2. It is negative for x<2 and positive for x>2. Minimum is (2,-4). One mark each for derivative, intervals and point.
3. Critical point x=1. Values: f(0)=0, f(1)=-1, f(3)=3. Minimum -1 at x=1; maximum 3 at $x = 3$. Credit candidates, evaluations and both extrema.
4. The polynomial is continuous and differentiable. Average slope = (9-1)/(3-1)=4. Solve 2c=4 to get c=2, which is inside (1,3). Four marks for hypotheses, slope, solution and interval check.
5. Set u=x^2+2, $du = 2x\,dx$. Integral becomes 3 times integral u squared du = u cubed+C = (x^2+2) cubed+C. Credit substitution, factor, integration and constant.
6. Set $u = x^2 + 1$; limits become 1 and 2. Integral of u du from 1 to 2 = (4-1)/2 = 3/2. Credit substitution, new limits, antiderivative and subtraction.

Source and coverage notes

MATH 203: derivative rules, implicit/logarithmic/exponential derivatives, L Hopital, graphing and extrema, optimization, Rolle and mean-value theorems, integration and the fundamental theorem. The full lecture syllabus is the main source.

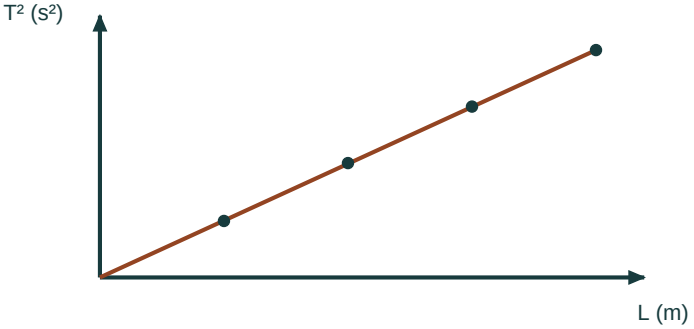
Source folder: Open original course folder The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. This revised study aid does not replace lecturer-specific derivations or instructions. All practice tests and suggested marking guidance are original study material.

05

Mechanics and Heat Lab

A field guide to understanding, working and checking.

Pendulum: choose axes that give a straight line



Slope = change in T² / change in L

Slope = 4 pi² / g

So g = 4 pi² / slope

Use small-amplitude oscillations.

All distinct experiments in the folder: pendulum, free fall, static and dynamic springs, force table, cooling, Boyle law, expansion, specific heat, latent heat and Joule equivalent. Duplicate versions are consolidated.

12 lessons / 2 practice tests / Worked solutions

English edition / November 2026

How to use this handout

1. Read the concept and check the symbol meanings before choosing a formula.
2. Cover the worked steps and reproduce the reasoning yourself.
3. Attempt the quick check and the two subject tests without notes.
4. Mark your work; record whether the error was concept, method, algebra, units or reading. Retry after a delay.

Your study route

1. Earn marks from tables, graphs and uncertainty
2. Simple pendulum
3. Free fall and axis choice
4. Static Hooke law
5. Dynamic spring and gravity
6. Force table and equilibrium
7. Newton cooling law
8. Boyle law
9. Linear thermal expansion
10. Specific heat, latent heat and Joule equivalent
11. Calorimetry: determine a solid specific heat
12. Latent heat and electrical-to-thermal conversion

LESSON 01 / 12

Earn marks from tables, graphs and uncertainty

After this lesson, you can

- Calculate a fitted-line slope with the correct axis units.
- Distinguish percentage error from measurement uncertainty.

Start with: Coordinate differences; division; raw and derived measurements.

THEORY

A lab result is a measured quantity with units and a justified method. Record raw observations before calculating derived columns. A slope is change in the vertical variable divided by change in the horizontal variable. Its units are essential and determine how to recover the desired constant.

RULE

$$m = \Delta y / \Delta x$$

$$\% \text{ error} = 100|x_{\text{measured}} - x_{\text{reference}}|/|x_{\text{reference}}|$$

m slope of the selected vertical variable against horizontal variable (vertical unit / horizontal unit)

x_{measured} experimental estimate (same as reference)

$x_{\text{reference}}$ accepted comparison value (same as measurement)

A method that earns marks

1. Record raw readings with units before calculating derived quantities.
2. Label axes with quantities and units and draw a suitable best-fit line.
3. Choose two widely separated points on that fitted line and calculate rise over run.
4. Use axis units to interpret the slope; compare with a reference using the stated percentage-error formula.

WORKED EXAMPLE

A fitted line rises by 0.8 square seconds over 0.2 m. Its slope is 4 square seconds per metre. It is not 0.25 m per square second unless the axes are reversed.

1. The example has vertical change 0.8 s^2 and horizontal change 0.2 m.
2. Slope is rise over run: 0.8/0.2 = 4 s^2/m.
3. The reciprocal, 0.25 m/s^2, belongs to a graph with reversed axes and is a different slope.
4. A slope should come from the fitted line, not necessarily from two adjacent raw observations.

Common mistake
Do not force a line through the origin unless the model and measurement offsets justify it.

Exam habit: Write slope units next to the numerical calculation; they help reveal an inverted axis choice.

SELF-CHECK

Can percentage error alone describe repeatability of a measurement?

Answer: No. It compares with a reference; repeated-reading spread or an uncertainty estimate is needed to discuss repeatability.

Uncertainty is different from percentage error

An uncertainty describes measurement quality. Percentage error compares a result with a reference value.

- Use the uncertainty method specified by your experiment or instrument; do not invent a precision from extra calculator digits.
- Give the uncertainty the same unit as the value. Round the value to the same decimal place as the reported uncertainty.
- Repeated measurements help estimate and reduce the effect of random scatter. They do not remove a consistent zero offset.
- An uncertainty range is not a guarantee that every possible error is inside it.

reported result $=$ value $\pm$ uncertainty

$$\% \text{ error} = 100\frac{|x_{\text{measured}} - x_{\text{reference}}|}{|x_{\text{reference}}|}$$

A result is $(9.6 \pm 0.2)\,\mathrm{m/s^2}$, and the reference is $9.8\,\mathrm{m/s^2}$. State the given uncertainty and calculate percentage error.

1. The stated uncertainty is $0.2\,\mathrm{m/s^2}$.
2. Percentage error is $100|9.6 - 9.8|/9.8 \approx 2.04\%$.

Uncertainty: $\pm 0.2\,\mathrm{m/s^2}$. Percentage error: about 2.04%. They answer different questions.

LESSON 02 / 12

Simple pendulum

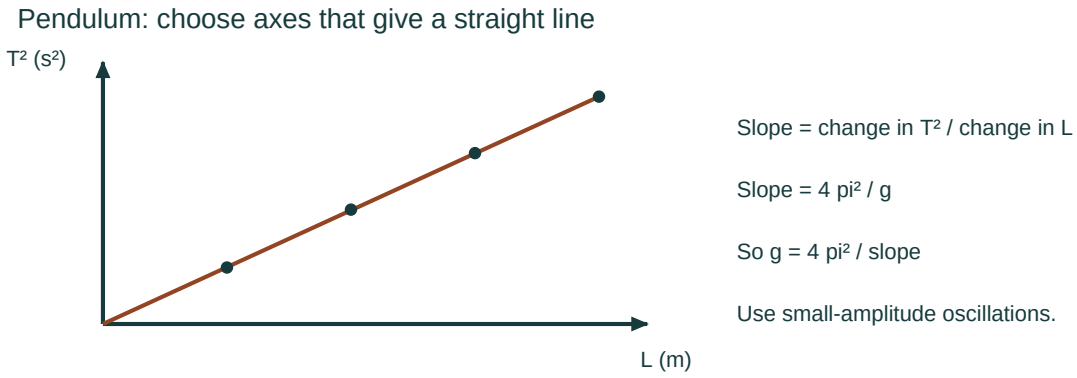
After this lesson, you can

- Measure pendulum length and period consistently.
- Recover g from one trial or a T^2-versus-L graph.

Start with: Complete oscillation; squaring; slope interpretation.

THEORY

For small angles a pendulum period depends mainly on length and gravity, not bob mass. Measure length from pivot to the centre of the bob. Timing several complete oscillations reduces the relative impact of reaction time. Keep the starting angle small and avoid pushing the bob.



Plot period squared vertically against pendulum length horizontally. The theoretical slope is $4\pi^2/g$. Illustrated points are ideal model values, not experimental observations.

RULE

$$T = t/N$$

$$T^2 = (4\pi^2/g)L$$

$$g = 4\pi^2/m \text{ for } T^2 \text{ versus } L$$

L pivot-to-bob-centre length (m)

t time for N complete oscillations (s)

T period of one oscillation (s)

g gravitational acceleration (m/s^2)

When this applies

The simple model assumes small angular displacement, negligible damping and a light inextensible suspension.

A method that earns marks

1. Measure length from pivot to the bob centre and release from a small angle without a push.
2. Time several complete oscillations and divide total time by their count.
3. Calculate T^2 and repeat at several lengths if graphing.
4. For T^2 on the vertical axis and L horizontally, use g = $4\pi^2$/slope.

WORKED EXAMPLE

For L=1 m and T=2 s, g = $4\pi^2$ times 1/4 = 9.87 $\mathrm{m/s^2}$.

1. For the example L = 1 m and T = 2 s.
2. Square the measured period: $T^2 = 4\ s^2$.
3. Rearranging $T^2 = 4\pi^2 L/g$ gives $g = 4\pi^2 L/T^2$.
4. Substitution gives g = $4\pi^2$ times 1/4 $\approx$ 9.87 m/s^2.

Common mistake
Using total timing t instead of the single-cycle period T makes g dramatically wrong.

Exam habit: State which variables are on which axes before converting a slope into g.

SELF-CHECK

A bob crosses equilibrium twice in one full cycle. Should each crossing be counted as a cycle?

Answer: No. Count a return to the same position moving in the same direction as one full cycle.

LESSON 03 / 12

Free fall and axis choice

After this lesson, you can

- Linearize free fall with a time-squared graph.
- Select the correct g conversion for either axis orientation.

Start with: Constant acceleration from rest; square of time; fitted-line slope.

THEORY

For release from rest with negligible air drag, fall distance grows with time squared. The course sheets mention both graph orientations; the equation must decide the slope conversion. A light sensor or electronic timer avoids large manual timing errors.

RULE

$$h = \tfrac{1}{2} g t^2;$$

$$m_{h \text{ vs } t^2} = g/2;$$

$$m_{t^2 \text{ vs } h} = 2/g$$

h vertical fall distance (m)

t fall time (s)

g gravitational acceleration (m/s^2)

m graph slope (depends on axes)

When this applies

The object starts from rest and air resistance is negligible over the measured fall.

A method that earns marks

1. Release from rest and measure fall time with the stated apparatus.
2. Calculate t^2 for each height.
3. State the graph axes and compare h = (g/2)t^2 with y = mx.
4. For h against t^2 use g = 2m; for t^2 against h use g = 2/m.

WORKED EXAMPLE

A plot of h against t^2 has slope 4.9 m per square second, so g=9.8 $\mathrm{m/s^2}$. Reversing axes gives slope about 0.204 square seconds per metre.

1. With h on the vertical axis, the fitted slope is 4.9 m/s^2.
2. The equation h = (g/2)t^2 shows this slope is half g, so g = 9.8 m/s^2.
3. Reversing the axes makes slope 2/g, about 0.204 s^2/m.
4. The reciprocal slopes describe the same ideal relationship; neither slope can be interpreted without its axes.

Common mistake
Do not use g=2 times slope for a t-squared-versus-height graph.

Exam habit: Do not memorize slope = g/2 without the graph orientation; use the displayed equation to decide.

SELF-CHECK

A t^2-versus-h slope is 0.20 s^2/m. What is g?

Answer: 10 m/s^2, using g = 2/0.20.

LESSON 04 / 12

Static Hooke law

After this lesson, you can

- Distinguish spring extension from loaded length.
- Find spring stiffness from a load or a force-extension slope.

Start with: Weight mg; length differences; centimetres to metres.

THEORY

Within the proportional range, spring extension is proportional to the applied load. Extension is loaded length minus unloaded length. The restoring spring force points opposite the extension, while the applied load magnitude is k times extension. An elastic spring returns to its original length after unloading, but that alone does not prove a straight-line force-extension relation.

RULE

$$F = k\Delta L$$

$$F = mg$$

$$k = m_{F \text{ vs } \Delta L}$$

F applied load magnitude (N)

ΔL loaded minus unloaded spring length (m)

k spring stiffness (N/m)

m total suspended mass (kg)

When this applies

Use Hooke law only over the proportional linear range; elastic recovery alone does not guarantee strict proportionality.

A method that earns marks

1. Record the unloaded spring length first.
2. Add the load safely and calculate extension as loaded minus unloaded length.
3. Convert extension to metres and load mass to force using mg.
4. Find k from F/ΔL or the slope of F against ΔL; check the graph remains linear.

WORKED EXAMPLE

A 2 N load produces 0.04 m extension, so k=50 N/m. A permanent change of unloaded length signals the elastic model may no longer apply.

1. The applied load is 2 N and the extension is 0.04 m, not the total spring length.
2. Using $F = k\Delta L$ gives $k = 2/0.04 = 50$ N/m.
3. This means an additional 1 m extension would require 50 N only if the same linear model remained valid.
4. If unloading leaves a changed length, the spring has not returned elastically; avoid extrapolating the original model.

Common mistake
Centimetres must be converted to metres before reporting k in N/m.

Exam habit: Include the hanger in total suspended mass if the apparatus instructions require it; otherwise the force scale can acquire an offset.

SELF-CHECK

A 0.30 kg total load extends the spring 6 cm with $g = 10$ m/s^2. What is k?

Answer: 50 N/m: the force is 3 N and extension is 0.06 m.

LESSON 05 / 12

Dynamic spring and gravity

After this lesson, you can

- Relate mass-spring period to load and spring stiffness.
- Use graph axes to distinguish a k experiment from a g experiment.

Start with: Static spring extension; period measurement; slope units.

THEORY

An oscillating mass-spring system has a period that increases with mass and decreases with stiffness. Combining static extension with dynamic period allows gravity to be estimated. Small oscillations and the same effective load model are needed for consistency.

RULE

$$T^2 = 4\pi^2 m/k$$
$$T^2 = (4\pi^2/g)\Delta L$$

T oscillation period (s)

m oscillating mass in the stated model (kg)

k spring stiffness (N/m)

ΔL static equilibrium extension (m)

g gravitational acceleration (m/s^2)

When this applies

Small oscillations stay in the spring proportional range; the static and dynamic models must treat hanger and effective spring mass consistently.

A method that earns marks

1. Measure small vertical oscillations about equilibrium and obtain period from several cycles.
2. For T^2 against mass compare the slope with $4\pi^2$/k.
3. For T^2 against static extension compare the slope with $4\pi^2$/g.
4. Apply the matching rearrangement and retain the effective-mass assumptions used by the apparatus.

WORKED EXAMPLE

A T-squared-versus-extension slope of 4.0 square seconds per metre gives g=$4\pi^2$/4.0=9.87 $\mathrm{m/s^2}$.

1. The example plots T^2 vertically against static extension horizontally.
2. Its slope is 4.0 s^2/m, corresponding to $4\pi^2$/g.
3. Rearrangement gives g = $4\pi^2$/4.0 $\approx$ 9.87 m/s^2.
4. A T^2-versus-mass slope would instead determine k; substituting it here would mix two different experiments.

Common mistake
The symbols for mass and graph slope can both be m; label them explicitly to avoid confusion.

Exam habit: Use slope units to identify the requested constant: s^2/kg for stiffness, s^2/m for gravity in these two plots.

SELF-CHECK

A T^2-versus-mass slope is 0.80 s^2/kg. What is k?

Answer: About 49.35 N/m, from $4\pi^2/0.80$.

LESSON 06 / 12

Force table and equilibrium

After this lesson, you can

- Calculate the equilibrant from perpendicular forces.
- Explain how the force-table ring indicates approximate equilibrium.

Start with: Vector components; Pythagoras; angles from positive x.

THEORY

The force table demonstrates vector addition. The resultant of two forces is their vector sum. The equilibrant has equal magnitude and opposite direction. A centred ring that does not press on the pin indicates approximate balance, subject to pulley friction.

RULE

$$\sum F_x = 0$$

$$\sum F_y = 0$$

$$F_i = m_i g$$

F_i tension force from hanging mass in the ideal setup (N)

m_i total hanging mass for a string (kg)

R resultant force magnitude (N)

When this applies

Mass-to-tension conversion neglects pulley friction and string mass; actual balance is approximate.

A method that earns marks

1. Convert each hanging mass to a force using the chosen g.
2. Resolve all forces into signed horizontal and vertical components.
3. Find the resultant, then reverse both components for the equilibrant.

4. Adjust the balancing force and check the ring is centred without pressing on the pin.

WORKED EXAMPLE

For 3 N east and 4 N north, the resultant is 5 N northeast. The equilibrant is 5 N southwest.

1. The 3 N east and 4 N north forces give resultant components (+3,+4) N.
2. Its magnitude is $\sqrt{(3^2+4^2)}$ = 5 N.
3. Its precise direction is 53.1° north of east; northeast here describes its quadrant.
4. The equilibrant has components (-3,-4) N and magnitude 5 N, directed 53.1° south of west.

Common mistake
The equilibrant is not the resultant in the same direction.

Exam habit: If an angle is requested, northeast alone is insufficient; give a numerical angle and reference axis.

SELF-CHECK

Does the equilibrant have the same direction as the resultant?

Answer: No. It has equal magnitude and opposite direction.

LESSON 07 / 12

Newton cooling law

After this lesson, you can

- Calculate temperature excess above the surroundings.
- Interpret an exponential cooling trend without confusing it with Celsius temperature.

Start with: Subtracting temperatures; halving; logarithm recognition.

THEORY

A hot object approaches room temperature as its temperature excess decays. The simple model assumes approximately constant surroundings and an effective cooling constant over the measured range. The temperature excess, not the absolute temperature, is the decaying quantity.

RULE

$$T - T_r = (T_0 - T_r)e^{-kt}$$
$$\ln[(T - T_r)/(T_0 - T_r)] = -kt$$

T object temperature (°C or K consistently)

T_r room temperature (same unit as T)

k effective cooling constant (inverse time)

t elapsed time (s or min consistently with k)

When this applies

Room temperature is approximately constant and the effective cooling constant applies over the measured range.

A method that earns marks

1. Measure room temperature and record it as the reference.
2. Subtract room temperature from each object temperature.
3. Compare excesses over time or use the stated logarithmic plot.
4. Add the room temperature back when converting a final excess into an object temperature.

WORKED EXAMPLE

If room temperature is 20 degrees C and the object cools from 80 to 50, the excess falls from 60 to 30 degrees, so it halves.

1. The room is at 20°C, so the initial 80°C object has excess 80 - 20 = 60°C.
2. At 50°C its excess is 50 - 20 = 30°C.
3. The excess has halved, even though 50 is not half of 80.
4. This distinction explains why the model approaches room temperature rather than cooling toward 0°C.

Common mistake
The object approaches room temperature, not zero degrees C.

Exam habit: When logging data, use a dimensionless excess ratio such as (T-Tr)/(T0-Tr), with temperatures consistently measured.

SELF-CHECK

At room temperature 25°C, an object starts at 85°C and its excess halves. What is its new temperature?

Answer: 55°C: initial excess 60°C becomes 30°C, then add 25°C.

LESSON 08 / 12

Boyle law

After this lesson, you can

- Use absolute pressure in a fixed-temperature gas comparison.
- Predict the inverse pressure-volume trend and check thermal settling.

Start with: Inverse proportion; multiplying and dividing; consistent pressure and volume units.

THEORY

At fixed temperature and fixed amount of gas, absolute pressure is inversely proportional to volume. A narrow uniform tube may use gas-column length as a proxy for volume. Let the gas settle thermally after changing pressure.

RULE

$$P_1V_1 = P_2V_2$$

$$P \propto 1/V$$

P absolute gas pressure (Pa or consistent pressure unit)

V gas volume (m^3 or consistent volume unit)

When this applies

Gas amount and temperature remain constant; the ideal Boyle-law approximation is suitable for the measured range.

A method that earns marks

1. Confirm the gas amount stays fixed and readings are taken at the same temperature.
2. Use absolute pressure rather than gauge pressure.
3. Write P1V1 = P2V2 and isolate the requested volume or pressure.
4. Check that larger pressure gives smaller volume and allow time for thermal settling between trials.

WORKED EXAMPLE

A gas at 100 kPa and 60 cubic centimetres has volume 30 cubic centimetres at 200 kPa, if temperature stays fixed.

1. Initially the gas has pressure 100 kPa and volume 60 cm^3.
2. At fixed temperature and amount, their product stays constant.
3. Doubling pressure to 200 kPa requires half the volume: V2 = 100 times 60/200 = 30 cm^3.
4. Keeping kPa and cm^3 is valid for this ratio because the corresponding units match on both sides.

Common mistake
Compressing quickly can warm the gas and violate the constant-temperature assumption.

Exam habit: Do not convert a gauge reading to absolute pressure by ignoring atmospheric pressure; use the apparatus definition.

SELF-CHECK

If volume is compressed quickly, why wait before recording a Boyle-law reading?

Answer: Compression can change temperature; waiting helps the gas return to the intended common temperature.

LESSON 09 / 12

Linear thermal expansion

After this lesson, you can

- Calculate a linear expansion coefficient with compatible length units.
- Extract the coefficient from a length-change-versus-temperature graph.

Start with: Small length changes; temperature differences; mm-to-m conversion.

THEORY

Heating changes the length of a rod slightly. The coefficient of expansion describes fractional length change per degree. A dial gauge often reads millimetres while the rod length is recorded in metres, so consistent units are critical.

RULE

$$\Delta L = \alpha L_0 \Delta T$$

$$\alpha = m_{\Delta L \text{ vs } \Delta T} / L_0$$

α linear expansion coefficient (K^{-1})

L_0 initial rod length (m)

ΔL length increase (m)

ΔT temperature increase (K)

When this applies

The temperature range is small enough to treat α as approximately constant and the rod temperature as sufficiently uniform.

A method that earns marks

1. Record original rod length and the initial and final temperatures.
2. Convert gauge displacement to the same length unit as the rod.
3. Rearrange $\Delta L = \alpha L_0 \Delta T$ to solve for α.
4. For a graph of ΔL against ΔT divide its slope by L_0.

WORKED EXAMPLE

A 1 m rod expands 0.5 mm over 50 K, giving alpha = 0.0005/(1 times 50)=0.00001 per K.

1. The 1 m rod expands by 0.5 mm, which is 0.0005 m.
2. The temperature rise is 50 K.
3. Divide extension by original length and temperature rise: α = 0.0005/(1 times 50).
4. The result is 0.00001 K^{-1}, or 1.0×10^{-5} K^{-1}.

Common mistake
A graph against absolute temperature may have an intercept; do not confuse it with a change-of-temperature plot.

Exam habit: Write the displacement conversion explicitly; the small numerical size of α is physically expected.

SELF-CHECK

What conversion error occurs if 0.5 mm is entered as 0.5 m?

Answer: The calculated expansion coefficient becomes 1000 times too large.

LESSON 10 / 12

Specific heat, latent heat and Joule equivalent

After this lesson, you can

- Separate sensible heating, phase-change energy and electrical input.
- Keep the joule-per-calorie conversion dimensional.

Start with: Energy balance; electrical power; grams-to-kilograms conversion.

THEORY

Mixing calorimetry balances heat lost by the hot sample against heat gained by water and calorimeter. Latent heat accounts for a phase change. Electrical heating provides IVt joules; comparing it with heat measured in calories gives joules per calorie, not a dimensionless ratio.

RULE

$$m_s c_s (T_s - T_f) = (m_w c_w + C_{\text{cal}})(T_f - T_w)$$

$$Q = mL$$

$$E = IVt$$

Q thermal energy transferred (J)

L specific latent heat (J/kg)

I heater current (A)

V heater voltage (V)

t heating duration (s)

When this applies

The latent-heat example assumes the sample starts at its melting point and all stated energy is used for melting.

A method that earns marks

1. Identify whether temperature changes, phase changes or both occur.

2. Calculate electrical input as IVt when heater readings are supplied.
3. Use mcΔT for temperature change and mL for a phase change.
4. If comparing joules with calories, divide joules by calories and state J/cal.

WORKED EXAMPLE

If 500 J melts 0.002 kg at its melting point with no other heating, latent heat is 250000 J/kg. If 418.4 J corresponds to 100 cal, the conversion is 4.184 J/cal.

1. The sample receives 500 J only for melting and the melted mass is 0.002 kg.
2. Rearranging Q = mL gives L = 500/0.002 = 250000 J/kg.
3. In the separate conversion example, 418.4 J is compared with 100 cal.
4. Their ratio is 418.4/100 = 4.184 J/cal; the unit is essential because the two energy scales differ.

Common mistake
Ignoring calorimeter heat capacity or heat loss can bias the inferred material property.

Exam habit: State where the energy goes before dividing by mass; not all heating energy necessarily produces the phase change.

SELF-CHECK

If both electrical input and thermal gain are already in joules, is their ratio a J/cal conversion?

Answer: No. It is dimensionless and tests energy accounting or an efficiency interpretation under the model.

LESSON 11 / 12

Calorimetry: determine a solid specific heat

After this lesson, you can

- Balance heat lost by a solid against heat gained by water and calorimeter.
- Solve for specific heat while using the correct temperature changes.

Start with: Heat capacity; mass in kg; a common final equilibrium temperature.

THEORY

A hot solid placed in cooler water transfers energy to the water and calorimeter. At equilibrium they share a final temperature. The ideal model assumes negligible heat loss to surroundings and no phase change. The calorimeter contribution must be included if it is specified.

RULE

$$c_s = \frac{(m_w c_w + C_{\text{cal}})(T_f - T_w)}{m_s(T_s - T_f)}$$

c_s solid specific heat capacity (J/(kg K))

m_s, m_w solid and water masses (kg)

C_{cal} calorimeter heat capacity (J/K)

T_f final equilibrium temperature (°C or K)

When this applies

The example neglects surroundings losses and calorimeter heat capacity and assumes no phase change.

A method that earns marks

1. Identify the initially hot body, initially cold water and common final temperature.
2. Write heat lost by the solid equal to heat gained by water plus calorimeter.
3. Use positive temperature drops and rises on opposite sides.
4. Solve for cs and state the heat-loss and calorimeter assumptions.

WORKED EXAMPLE

With a negligible calorimeter, 0.10 kg solid cooling from 80 to 30 degrees C heats 0.20 kg water from 25 to 30 degrees C. Taking water c=4200 gives solid c=840 J/(kg K).

1. Water warms from 25°C to 30°C, a rise of 5 K.
2. Its heat gain is 0.20 times 4200 times 5 = 4200 J.
3. The 0.10 kg solid cools from 80°C to 30°C, a drop of 50 K.
4. With negligible calorimeter heat capacity and losses, 0.10 times cs times 50 = 4200, giving cs = 840 J/(kg K).

Common mistake
Using the same temperature change for hot solid and water is generally wrong.

Exam habit: Use the calorimeter heat-capacity term when supplied; discarding it undercounts heat gained by the cold side.

SELF-CHECK

Why is the solid temperature change not the same as the water temperature change?

Answer: They have different starting temperatures, though they reach the same final equilibrium temperature.

LESSON 12 / 12

Latent heat and electrical-to-thermal conversion

After this lesson, you can

- List all energy stages for ice melting and subsequent warming.
- Interpret how lost heat biases a measured electrical-to-calorie ratio.

Start with: Specific heat versus latent heat; phase-change temperatures; units of energy.

THEORY

A phase-change experiment separates melting from subsequent warming of the melted material. In the Joule-equivalent experiment electrical energy is compared with heat gained by water and calorimeter. If both quantities are already in joules, their ratio tests energy accounting; it is not a joule-per-calorie conversion.

RULE

$$Q_{\text{ice}} = mL_f + mc_w(T_f - 0^\circ\text{C})$$

$$J_{\text{eq}} = E_{\text{joules}}/Q_{\text{calories}}$$

m ice mass (kg)

L_f specific latent heat of fusion (J/kg)

c_w liquid water specific heat (J/(kg K))

J_{eq} joules per calorie conversion estimate (J/cal)

When this applies

The ice formulas use an ordinary-pressure melting point of approximately 0°C and account for the stated final phase.

A method that earns marks

1. Identify whether ice starts below or at its melting point.
2. Include warming of ice if needed, then melting, then warming of meltwater to the final temperature.

3. For conversion, keep electrical energy in joules and measured thermal gain in calories.
4. Assess a loss by asking whether the recorded heat gain is too small or too large for the supplied energy.

WORKED EXAMPLE

If ice begins below zero degrees C, an additional warming-of-ice term is required before melting. If it begins at zero, that term is absent.

1. Ice below 0°C first needs energy to warm to 0°C; that stage uses ice specific heat.
2. At 0°C, energy mLf melts it without a temperature rise in the ideal fixed-pressure model.
3. If the final state is liquid above 0°C, add mcw(Tf - 0°C) for the meltwater.
4. When ice already starts at 0°C the first stage is absent; the melting and later warming stages remain if required.

Common mistake
Do not divide joules by joules and label the dimensionless answer J/cal.

Exam habit: Do not use the water specific heat for warming solid ice; the two phases have different heat capacities.

SELF-CHECK

If 836.8 J supplies a recorded heat gain of 200 cal, what is E/Q, and how would unrecorded heat loss bias it?

Answer: 4.184 J/cal. If heat escapes and measured Q is too small, E/Q tends to be too high.

Before the tests

These original questions practise the lesson objectives. They are not predictions of an exam paper. Use the separate practice book if you want all papers together.

Mechanics and Heat Lab / Test A

30 marks | Suggested time: 30 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. If y is force in N and x is extension in m, slope units are:

   A. N/m

   B. m/N

   C. N times m

   D. No units

2. Pendulum length should be measured to:

   A. The top of the table

   B. The lowest swing point

   C. The bottom of the bob

   D. The centre of the bob

3. For t^2 vertically versus h horizontally, g equals:

   A. Slope squared

   B. Slope/2

   C. 2 times slope

   D. 2/slope

4. The spring constant is the slope of:

   A. Time versus temperature

   B. Extension versus force

   C. Force versus extension

   D. Mass versus time

5. For fixed k, multiplying the oscillating mass by four makes the period:

   A. Four times

   B. Half

   C. Twice

   D. Unchanged

6. The equilibrant of a 7 N eastward resultant is:

   A. 7 N east

B. 14 N west

C. Zero

D. 7 N west

Part II: Show your reasoning

1. A measured g is 9.6 $\mathrm{m/s^2}$ and reference g is 9.8 $\mathrm{m/s^2}$. Calculate percentage error and name one possible systematic error.

2. A pendulum takes 30 s for 15 cycles at length 0.99 m. Calculate period and g.

3. The slope of t^2 against height is 0.20 square seconds per metre. Find g.

4. A 0.30 kg mass extends a spring by 6 cm. Use g=10 $\mathrm{m/s^2}$. Find k.

5. A T-squared-versus-mass graph has slope 0.80 square seconds per kg. Find spring constant.

6. Two forces are 6 N east and 8 N north. Find equilibrant magnitude and direction.

Mechanics and Heat Lab / Test B

30 marks | Suggested time: 30 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. In the simple cooling model, the cooling rate is proportional to:

   A. Mass only

   B. Zero

   C. Temperature excess above surroundings

   D. Absolute time only

2. Boyle law requires:

   A. Fixed temperature and gas amount

   B. Changing gas amount

   C. Fixed pressure

   D. Fixed volume

3. Two rods are made of the same material and have the same temperature rise. A rod twice as long has expansion:

   A. Twice as large

   B. Unchanged

   C. Four times as large

   D. Half as large

4. Electrical energy supplied by constant voltage and current for time t is:

   A. V/I

   B. IVt

   C. I/t

   D. Vt/I

5. In an ideal mixing experiment, the final equilibrium temperature lies:

   A. Below both

   B. Between the initial temperatures

   C. Always at zero

   D. Above both initial temperatures

6. To determine joules per calorie, divide electrical energy in joules by heat measured in:

A. Watts

B. Calories

C. Kelvin

D. Joules

Part II: Show your reasoning

1. Room temperature is 25 degrees C. A body starts at 85 degrees C and its temperature excess halves in 5 minutes. What is its temperature then?

2. A trapped gas has volume 40 cubic centimetres at absolute pressure 120 kPa. Find its volume at 160 kPa at the same temperature.

3. A 0.50 m rod expands 0.30 mm after warming by 40 K. Find its expansion coefficient.

4. A heater at 6 V carries 2 A for 100 s. If all energy melts a sample already at its melting point and the melted mass is 0.004 kg, find supplied energy and latent heat.

5. A 0.20 kg solid cools from 90 to 40 degrees C and warms 0.10 kg water from 30 to 40 degrees C. Ignore calorimeter heat capacity and losses; use water c=4200 J/(kg K). Find the solid specific heat.

6. An electrical experiment supplies 836.8 J and records heat gain of 200 cal. Find the conversion factor. If unaccounted heat escapes, does E/Q measured tend to be too high or too low?

Mechanics and Heat Lab / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **A.** Slope units are vertical units divided by horizontal units.
2. **D.** The ideal mass is represented at the bob centre.
3. **D.** Rearranging h = gt^2/2 gives t^2 = (2/g)h.
4. **C.** Force = k times extension matches y = mx.
5. **C.** Period is proportional to the square root of mass.
6. **D.** Equal magnitude and opposite direction produce zero net force.

Part II: Written solutions

1. Percentage error = 100 times 0.2/9.8 = 2.04 percent. A consistent height-scale zero offset is one possible systematic error. Two marks for the calculation and two for a specific error with its consistency explained.
2. T=30/15=2 s. g=$4\pi^2$ times 0.99/4=9.77 $\mathrm{m/s^2}$. Two marks per calculation.
3. g = 2/0.20 = 10 $\mathrm{m/s^2}$. Four marks for axis identification, rearrangement, substitution and unit.
4. Force = 3 N; extension = 0.06 m; k=3/0.06=50 N/m. Award marks for force, conversion, ratio and unit.
5. Slope = $4\pi^2$/k, so k=$4\pi^2$/0.80=49.35 N/m. Credit the selected axes, rearrangement, numerical value and units.
6. Resultant magnitude is 10 N at 53.1 degrees north of east. Equilibrant is 10 N at 233.1 degrees from positive x, or 53.1 degrees south of west. Two marks for magnitude and two for direction.

Mechanics and Heat Lab / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **C.** Newton cooling uses the difference between object and surrounding temperatures.
2. **A.** Only pressure and volume are varied in the ideal Boyle experiment.
3. **A.** Expansion is proportional to original length.
4. **B.** Electrical power is VI; multiplying by time gives energy.
5. **B.** Energy flows from the hotter body to the cooler one until equilibrium.
6. **B.** The denominator determines the per-calorie unit.

Part II: Written solutions

1. Initial excess = 60 degrees C. Halved excess = 30 degrees C. Add room temperature: 25+30=55 degrees C. Credit both differences and final interpretation.
2. V2 = 120 times 40/160 = 30 cubic centimetres. Credit controlled conditions, equation, rearrangement and unit.
3. Convert expansion to 0.00030 m. Alpha=0.00030/(0.50 times 40)=0.000015 per K. Credit conversion, formula, substitution and result.
4. Energy=6 times 2 times 100=1200 J. Latent heat=120$\frac{0}{0}$.004=300000 J/kg. Two marks per calculation; the no-loss and initial-phase-temperature assumptions matter.
5. Water gains 0.10 times 4200 times 10=4200 J. Solid loses 0.20 times cs times 50=10cs. Thus cs=420 J/(kg K). Two marks for each side and result.
6. Factor=836.8/200=4.184 J/cal. Heat escaping reduces measured Q for the supplied E, tending to make the ratio too high. Two marks for each part.

Source and coverage notes

All distinct experiments in the folder: pendulum, free fall, static and dynamic springs, force table, cooling, Boyle law, expansion, specific heat, latent heat and Joule equivalent. Duplicate versions are consolidated.

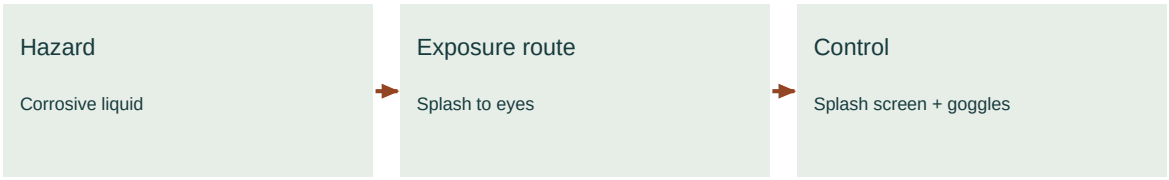
Source folder: Open original course folder The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. This revised study aid does not replace lecturer-specific derivations or instructions. All practice tests and suggested marking guidance are original study material.

06

Instruments and Lab Safety

A field guide to understanding, working and checking.

Make the safety reasoning visible



Controls must match the actual procedure and local safety instructions.

SCI 101 lecture chapters 1-3: accident prevention, chemical hazards, laboratory techniques, emergency principles and biosafety. This is exam study material; real laboratory work follows the current instructor procedure and chemical-specific safety information.

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8 lessons / 2 practice tests / Worked solutions

English edition / November 2026

How to use this handout

1. Read the concept and check the symbol meanings before choosing a formula.
2. Cover the worked steps and reproduce the reasoning yourself.
3. Attempt the quick check and the two subject tests without notes.
4. Mark your work; record whether the error was concept, method, algebra, units or reading. Retry after a delay.

Your study route

1. Hazard, risk and prevention
2. Personal protective equipment and contamination
3. Exposure routes and toxic effects
4. Labels, SDS and chemical properties
5. Glassware, measurement and safe apparatus
6. Hoods, heating, cooling and pressure
7. Waste, housekeeping and emergency priorities
8. Biosafety and the control hierarchy

LESSON 01 / 8

Hazard, risk and prevention

After this lesson, you can

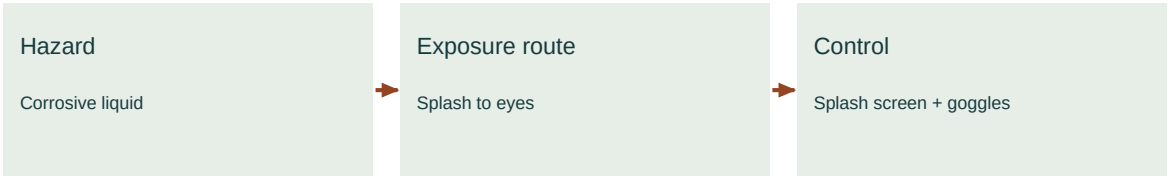
- Distinguish a hazard from the risk of harm in a particular situation.
- Choose a control that interrupts the actual exposure route.

Start with: Hazard means potential for harm; exposure means contact with the harmful agent.

THEORY

A hazard is a source of possible harm. Risk depends on exposure and the chance and severity of harm. Accident prevention is shared: prepare, follow approved procedures and report unsafe conditions. A familiar chemical or routine experiment can still present hazards.

Make the safety reasoning visible



Controls must match the actual procedure and local safety instructions.

Link the hazard to a specific exposure route and suitable controls. Eye protection is one layer, not a substitute for eliminating or reducing the hazard.

RULE

Good answer: hazard → exposure route → control

A method that earns marks

1. Name the harmful property and the activity creating exposure.
2. Identify who could be exposed, by which route and under what conditions.
3. Select a control that reduces the source or interrupts the route.
4. Explain the remaining need for the approved procedure and required personal protection.

WORKED EXAMPLE

A volatile harmful liquid on an open bench creates an inhalation hazard. Using the approved fume-hood procedure reduces exposure; goggles alone do not control vapour inhalation.

1. The example liquid is volatile and harmful, so vapour can leave an open container.
2. A person nearby may inhale that vapour; this connects the chemical hazard to an exposure route.
3. An approved fume-hood procedure controls vapour at its source when suitable for that substance and operation.
4. Goggles protect eyes from appropriate hazards but do not remove vapour from inhaled air.

Common mistake
Do not write only be careful; name a concrete control.

Exam habit: For a scenario answer write a linked chain: hazard, exposure route, suitable control and reason.

SELF-CHECK

Why is naming only a hazardous chemical insufficient to describe a risk scenario?

Answer: Risk also depends on how exposure can occur, how likely it is and how serious its consequences could be.

LESSON 02 / 8

Personal protective equipment and contamination

After this lesson, you can

- Trace how contaminated gloves transfer material to clean surfaces.
- Select PPE for the task and remove it without spreading contamination.

Start with: Contamination transfer; clean and contaminated work zones; task-specific protection.

THEORY

Splash goggles protect against chemical splashes; ordinary spectacles do not provide equivalent coverage. Gloves must be compatible with the chemical and inspected. No glove protects indefinitely. Contaminated gloves can carry chemicals to phones, door handles and notebooks.

RULE

Select PPE for the specific hazard and task

A method that earns marks

1. Identify the contaminant and which surfaces or body areas need protection.
2. Use PPE specified for the material and task rather than assuming one glove type protects against everything.
3. Keep contaminated gloves away from phones, door handles and other clean shared items.
4. Remove and dispose of PPE according to the approved procedure, then perform required hand hygiene.

WORKED EXAMPLE

A student touches a phone with contaminated gloves. The issue is transfer of contamination to a surface later touched without gloves. Remove gloves correctly before touching clean shared items.

1. The student glove is already contaminated before touching the phone.

2. Contact transfers material from glove to phone, creating a new contaminated surface.
3. Later bare-hand contact can move that material to skin, food or the face.
4. Removing gloves correctly before touching clean items breaks that transfer chain; the phone may still need the approved decontamination procedure.

Common mistake
More layers of unsuitable gloves do not establish chemical compatibility.

Exam habit: Explain what is transferred and to where; simply saying wear gloves does not solve a contamination scenario.

SELF-CHECK

Do gloves make it acceptable to touch your face during laboratory work?

Answer: No. Gloves can carry contamination directly to the face.

LESSON 03 / 8

Exposure routes and toxic effects

After this lesson, you can

- Identify inhalation, ingestion, skin or eye contact and injection exposures.
- Distinguish route of exposure from the timing of a toxic effect.

Start with: Contact with an agent can occur by several routes; visible vapour is not required.

THEORY

Chemicals can enter by inhalation, ingestion, skin or eye absorption, or injection through contaminated sharps. Acute exposure is short-term; chronic exposure is repeated or long-term. Effects can be local or systemic, immediate or delayed, reversible or irreversible.

RULE

Dose, duration, route and substance all influence harm

A method that earns marks

1. Read the activity and locate the point where the agent could enter or contact the body.
2. Name the route using the specific evidence in the scenario.

3. Distinguish a rapid acute effect from an effect associated with longer or repeated exposure.
4. Match prevention to the identified route and follow the material-specific procedure.

WORKED EXAMPLE

Eating in a laboratory can lead to ingestion of contamination. A contaminated needle can create injection exposure even without visible vapour.

1. Eating in the laboratory can move contamination from hands or surfaces to food.
2. Swallowing that material is ingestion, even when the contaminant is not visible.
3. A contaminated needle can breach the skin and introduce material directly: this is injection exposure.
4. The absence of visible vapour does not remove either risk because neither route depends on inhalation.

Common mistake
A lack of smell does not prove safety.

Exam habit: Name both the route and the evidence, for example ingestion because contaminated food is swallowed.

SELF-CHECK

A droplet splashes into an eye. Is the route inhalation?

Answer: No. It is eye or mucous-membrane contact.

LESSON 04 / 8

Labels, SDS and chemical properties

After this lesson, you can

- Use a label and SDS to connect chemical properties with controls.
- Distinguish corrosive and oxidizing hazards in an explanation.

Start with: Chemical identity; label symbols; an SDS gives substance-specific hazard and handling information.

THEORY

Read the identity and hazard information before use. Safety data sheets describe hazards, handling, storage, exposure controls, first aid and other substance-specific information. Flammable, oxidizing, corrosive and toxic refer to different hazards and can apply to one substance simultaneously.

RULE

$$\text{Identity} + \text{hazard} + \text{controls} + \text{incompatibilities}$$

A method that earns marks

1. Confirm the chemical identity from its label before selecting a procedure.
2. Read relevant SDS information for hazards, incompatibilities and required handling controls.
3. Explain how the named property causes harm in the scenario.
4. Use the approved storage or handling arrangement rather than inferring compatibility from appearance.

WORKED EXAMPLE

A corrosive liquid can damage tissue, while an oxidizer may intensify fire even when it is not itself a fuel. These require different control reasoning.

1. A corrosive liquid can damage living tissue or certain materials by contact.
2. An oxidizer can support or intensify combustion even when it is not itself the fuel.
3. The mechanisms differ, so the same general statement handle carefully does not explain either control.
4. A good answer links splash/contact controls to corrosivity and incompatibility or fire-prevention controls to oxidizing behaviour as specified by the SDS.

Common mistake
Do not infer that all colourless liquids are harmless or that one symbol captures every hazard.

Exam habit: Pair each property with its consequence; listing pictogram names alone gives less evidence of understanding.

SELF-CHECK

Does a chemical being nonflammable prove it cannot intensify a fire?

Answer: No. An oxidizer may intensify combustion without being a fuel.

LESSON 05 / 8

Glassware, measurement and safe apparatus

After this lesson, you can

- Select apparatus according to measurement purpose and safe operating limits.
- Explain why volumetric preparation and heating require different vessels.

Start with: Volume; calibration mark; damaged glassware can fail during use.

THEORY

Inspect glassware for chips and cracks and apparatus for damaged wiring before use. A beaker is useful for holding and mixing; a graduated cylinder gives more useful volume estimates; a volumetric pipette or flask is designed for a specified accurate volume. Read liquid levels at eye height using the appropriate meniscus.

RULE

Choose the instrument for required precision, not convenience

A method that earns marks

1. State whether the task requires approximate volume, accurate delivery or preparation to a final volume.
2. Choose the apparatus designed for that purpose and check its condition.
3. Follow the specified filling, meniscus and operating procedure.
4. Before heating, verify that the vessel and setup are approved for the process.

WORKED EXAMPLE

A volumetric flask prepares a solution to its marked final volume. It is not the appropriate vessel for directly heating a reaction.

1. The volumetric flask is calibrated to contain a stated final volume at its reference conditions.
2. The solution is brought to the mark as part of an approved preparation procedure.
3. Its main role is volume accuracy, not serving as a directly heated reaction vessel.
4. For heating a reaction, use the vessel and support specified by the laboratory method rather than substituting the flask.

Common mistake
Extra decimal places do not make an imprecise instrument accurate.

Exam habit: In apparatus questions state the purpose and limitation together: prepares final volume, unsuitable for directly heating the reaction.

SELF-CHECK

Does having a calibration mark mean a flask is suitable for direct heating?

Answer: No. Calibration and suitability for heating are different design requirements.

LESSON 06 / 8

Hoods, heating, cooling and pressure

After this lesson, you can

- Explain why heating a closed unsuitable vessel can create a pressure hazard.
- Recognize when an apparatus setup requires instructor review.

Start with: Heating can increase vapour production and gas pressure; equipment has specific operating limits.

THEORY

A fume hood controls airborne chemical exposure when used correctly. It is not a storage cabinet or a waste-evaporation device. Heating and cooling systems can introduce burns, pressure changes and material failure. Approved setups account for both temperature and pressure.

RULE

Control vapour, ignition, pressure and thermal contact separately

A method that earns marks

1. Identify heat source, volatile materials and any sealed volume.
2. Check the approved procedure and apparatus pressure and temperature limits.
3. Use the specified containment, heating and ventilation arrangements.
4. If the setup differs or becomes unsafe, stop and alert the responsible instructor instead of modifying it ad hoc.

WORKED EXAMPLE

A volatile solvent warmed in an unsuitable sealed container can build pressure. The correct response is to stop and consult the approved setup rather than improvise a vent.

1. Warming the volatile solvent increases vapour formation.
2. In an unsuitable sealed container, expanding gas and vapour can increase internal pressure.
3. The container or closure may fail because it was not designed for that operation.
4. The example therefore calls for stopping and consulting the approved setup; improvising a vent is not a validated correction.

Common mistake
A hood does not remove every fire, pressure or contact hazard.

Exam habit: Explain the mechanism, pressure buildup from heating, before giving the required stop-and-consult response.

SELF-CHECK

Does placing an unsuitable sealed heated vessel in a fume hood solve its pressure hazard?

Answer: No. A hood does not make a vessel pressure-rated or validate an unsafe heating setup.

LESSON 07 / 8

Waste, housekeeping and emergency priorities

After this lesson, you can

- Use identification and approved procedures to choose a waste or spill response.
- State safe priorities when a spill material is unknown.

Start with: Chemical incompatibility; designated waste streams; laboratory emergency instructions.

THEORY

Waste must follow the laboratory segregation and disposal procedure. Broken glass, sharps and chemical waste are different streams. Good housekeeping keeps exits and emergency equipment accessible. In an incident, alert others and follow the established response; do not improvise a hazardous cleanup.

RULE

Recognize → alert → follow trained response

A method that earns marks

1. Identify the material only if that can be done safely from available records or labels.
2. Keep others clear and alert the instructor or responsible responder.
3. Use the laboratory response plan for the identified hazard and your training level.
4. For routine waste, use the designated compatible container and labelling procedure.

WORKED EXAMPLE

An unknown spill is not an invitation to mix in a neutralizer. Alert the instructor, keep others clear and follow the spill response appropriate to the identified material.

1. An unknown spill has uncertain properties, so an assumed neutralizer may be incompatible.
2. Adding another chemical could cause heat, gas formation or another hazardous reaction.
3. Alerting the instructor and keeping people clear limits exposure while the material and response are assessed.
4. Cleanup then follows the approved spill procedure for the identified material and trained personnel.

Common mistake
Do not recommend that an untrained student fight every fire or clean every spill.

Exam habit: In an emergency scenario prioritize alerting and exposure prevention; do not invent a chemical cleanup recipe.

SELF-CHECK

Should an unknown laboratory liquid automatically be washed into a sink?

Answer: No. Its identity, compatibility and approved disposal route must be established.

LESSON 08 / 8

Biosafety and the control hierarchy

After this lesson, you can

- Classify controls within the hazard-control hierarchy.
- Distinguish engineering containment from training, work practices and PPE.

Start with: A biological hazard may be spread through aerosols, contact or sharps depending on the activity.

THEORY

Biosafety aims to prevent unintended exposure to biological agents and their release. The general control hierarchy prefers removing a hazard, then using a safer substitute, then engineering controls, administrative controls and work practices, and PPE. The suitable controls depend on the task: removing or replacing the biological material may not be possible. Training and procedures are administrative controls; suitable containment equipment is an engineering control. Safe work practices include hand hygiene and avoiding aerosol generation.

RULE

Eliminate → substitute → engineering → administrative/work practices → PPE

A method that earns marks

1. Identify the biological material and activity-specific exposure route.
2. Consider elimination or substitution where feasible before downstream controls.
3. Classify equipment containment as engineering, training and procedures as administrative or work-practice controls, and worn barriers as PPE.
4. Explain how the selected combination addresses the task and follows institutional biosafety procedures.

WORKED EXAMPLE

Training is administrative. A suitable biological safety cabinet is engineering containment. Gloves are PPE. A safe handling technique is a work practice.

1. Training changes how people work, so it is an administrative control.
2. A suitable biological safety cabinet provides engineering containment for its approved uses.
3. Gloves form a worn barrier and are PPE.

4. A safe handling technique is a work practice, generally grouped with administrative controls; the categories describe different protective mechanisms.

Common mistake
A device with airflow is not automatically suitable for every chemical or biological hazard.

Exam habit: Do not label every cabinet or hood as interchangeable containment; state suitable equipment for the defined biological task.

SELF-CHECK

What is the general hierarchy from source-focused to last-line protection?

Answer: Elimination, substitution, engineering controls, administrative controls and work practices, then PPE.

The control hierarchy in the correct order

The hierarchy ranks ways to control a hazard. A list of control types is not automatically in the preferred order.

- Eliminate: remove the hazard.
- Substitute: use a safer alternative after checking its risks.
- Engineering controls: separate people from the hazard using suitable equipment or containment.
- Administrative controls and work practices: use training, procedures and safe methods.
- PPE: use suitable worn protection. It adds protection but depends on correct selection and use.
- In a laboratory, eliminating or replacing the material may not be possible. The assessed task determines the suitable combination of controls.

Classify a suitable containment cabinet, safety training, gloves and correct handwashing.

1. The cabinet is an engineering control.
2. Training is administrative; gloves are PPE.
3. Handwashing is a safe work practice, often grouped with administrative controls.

Engineering; administrative; PPE; work practice, respectively. Engineering controls come before administrative controls and PPE in the general hierarchy.

Before the tests

These original questions practise the lesson objectives. They are not predictions of an exam paper. Use the separate practice book if you want all papers together.

Instruments and Lab Safety / Test A

20 marks | Suggested time: 20 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. Which best describes risk?

   A. The mass of any bottle

   B. The same as smell

   C. The label colour alone

   D. The chance and severity of harm under the exposure conditions

2. Which is appropriate before touching a clean door handle?

   A. Keep contaminated gloves on

   B. Wipe gloves on clothing

   C. Remove contaminated gloves correctly

   D. Smell the gloves

3. Eating with contaminated hands is mainly which exposure route?

   A. Inhalation

   B. Injection

   C. Noise exposure

   D. Ingestion

4. An oxidizer is especially associated with:

   A. Intensifying combustion

   B. Preventing all fires

   C. Measuring temperature

   D. Making all chemicals inert

Part II: Show your reasoning

1. A student uses an unfamiliar reagent without reading its safety information. Identify the problem and two appropriate preparatory actions.

2. Explain why ordinary prescription glasses and a lab coat alone are insufficient for a splash-risk task.

3. Classify inhaling vapour and a contaminated needle puncture by exposure route. Explain why no smell does not prove no hazard.

4. A bottle has both flammable and corrosive hazards. State two distinct risks and a control principle for each.

Instruments and Lab Safety / Test B

20 marks | Suggested time: 20 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. Which is designed to prepare one accurately specified solution volume?

   A. A beaker

   B. A watch glass

   C. A volumetric flask

   D. A test-tube rack

2. A fume hood should be used as:

   A. Permanent chemical storage

   B. A controlled work enclosure for appropriate vapour hazards

   C. A way to evaporate all waste

   D. A food cupboard

3. An unknown chemical spill should first prompt a student to:

   A. Alert the instructor and keep people clear

   B. Mix it with another chemical

   C. Ignore it

   D. Taste it

4. Which is an administrative control?

   A. Gloves

   B. Training and written procedures

   C. A sealed rotor

   D. Splash goggles

Part II: Show your reasoning

1. Name a suitable device to transfer an accurate fixed 10 mL volume and explain two good practices.

2. Give two reasons why heating a tightly sealed vessel containing a volatile liquid is hazardous.

3. A box blocks access to the eyewash station. Explain the risk and an appropriate preventive action.

4. Classify goggles, safety training, a suitable containment cabinet and correct handwashing.

Instruments and Lab Safety / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **D.** Risk depends on both the hazard and how people are exposed.
2. **C.** This prevents spreading contamination outside the work area.
3. **D.** Material can pass from the hands to food and the digestive system.
4. **A.** Oxidizers can support or intensify combustion.

Part II: Written solutions

1. Problem: unassessed chemical hazards. Actions: consult the reagent SDS and approved procedure; clarify required controls and waste handling with the instructor before use. One mark for hazard, one for exposure reasoning and one per specific action.
2. Prescription glasses do not seal against splashes; appropriate splash goggles are needed. A coat does not protect hands, so compatible gloves may be required by the assessed procedure. Award one mark per protection gap and suitable control.
3. Vapour: inhalation. Needle: injection. Some harmful substances are odourless, or odour detection is unreliable. One mark per route and two for the explanation.
4. Fire risk: control ignition sources using the approved setup. Tissue damage from splashes: prevent contact using appropriate containment and compatible protection. One mark per risk and per linked control.

Instruments and Lab Safety / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **C.** The calibration mark defines the specified contained volume under its calibration conditions.
2. **B.** It is an engineering control for relevant airborne exposures.
3. **A.** Unknown hazards require assessment and the established response.
4. **B.** Administrative controls organize how work is authorized and performed.

Part II: Written solutions

1. Use an appropriate 10 mL volumetric pipette with a pipette aid. Read the meniscus at eye level and follow the calibrated delivery procedure. Credit device, aid and two measurement practices.
2. Vapour generation can raise pressure, and thermal expansion can further stress the vessel. Rupture can release hot liquid or vapour. Credit two mechanisms and their resulting consequences.
3. It can delay emergency irrigation after exposure. Keep access clear and report the obstruction so it is corrected promptly under laboratory procedures. Credit emergency delay, possible worsening of injury, removal/reporting and prevention.
4. Goggles: PPE. Training: administrative control. Suitable containment cabinet: engineering control. Correct handwashing: safe work practice (often grouped with administrative controls). One mark each.

Source and coverage notes

SCI 101 lecture chapters 1-3: accident prevention, chemical hazards, laboratory techniques, emergency principles and biosafety. This is exam study material; real laboratory work follows the current instructor procedure and chemical-specific safety information.

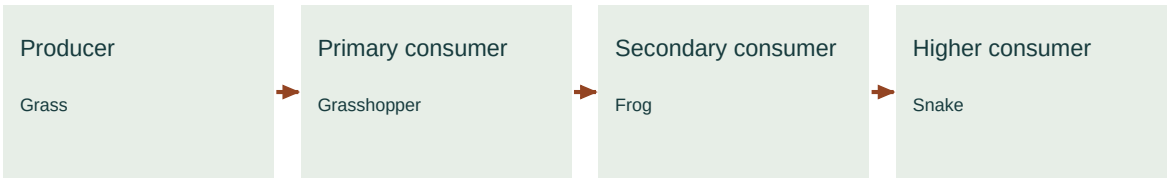
Source folder: Open original course folder The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. This revised study aid does not replace lecturer-specific derivations or instructions. All practice tests and suggested marking guidance are original study material.

07

Humans and the Environment

A field guide to understanding, working and checking.

Food-chain arrows show transfer to the consumer



Energy transfers along the chain; some is dissipated as heat at each stage.

ENV101 elective: ecosystems, relationships among organisms, pollution, ecological footprint, waste management and sustainable development. The Arabic source lectures are explained here in English.

8 lessons / 2 practice tests / Worked solutions

English edition / November 2026

How to use this handout

1. Read the concept and check the symbol meanings before choosing a formula.
2. Cover the worked steps and reproduce the reasoning yourself.
3. Attempt the quick check and the two subject tests without notes.
4. Mark your work; record whether the error was concept, method, algebra, units or reading. Retry after a delay.

Your study route

1. Environment and ecosystem components
2. Energy flow, food chains and decomposers
3. Population growth and limiting factors
4. Relationships among organisms
5. Pollution: sources, pathways and effects
6. Ecological footprint and biocapacity
7. Waste management and the hierarchy
8. Sustainable development and integrated decisions

LESSON 01 / 8

Environment and ecosystem components

After this lesson, you can

- Classify living and nonliving ecosystem components.
- Explain one interaction linking an organism with an abiotic factor.

Start with: An ecosystem includes organisms and the physical conditions around them.

THEORY

An ecosystem combines organisms with the physical and chemical surroundings with which they interact. Biotic factors are living components; abiotic factors include light, temperature, water and minerals. A population is one species in an area; a community includes interacting populations.

RULE

Organism → population → community → ecosystem

A method that earns marks

1. Name the ecosystem and define the area being considered.
2. Sort its organisms as biotic and its physical or chemical conditions as abiotic.
3. Choose one organism and explain how a named abiotic factor affects it.
4. Use a causal sentence rather than an unconnected list of examples.

WORKED EXAMPLE

Fish, algae and bacteria are biotic components of a pond. Water temperature and dissolved oxygen are abiotic factors affecting them.

1. Fish, algae and bacteria are organisms, so they are biotic pond components.
2. Water temperature and dissolved oxygen are abiotic conditions even though living things can affect them.
3. Fish depend on sufficient dissolved oxygen for respiration.
4. The example therefore shows both a classification and an interaction between the living community and its physical environment.

Common mistake
A habitat is where an organism lives; an ecosystem includes the interacting system.

Exam habit: For explain questions add an interaction after classifying: name the factor, the organism and the effect.

SELF-CHECK

Are bacteria abiotic because they are microscopic?

Answer: No. Bacteria are living organisms and therefore biotic components.

LESSON 02 / 8

Energy flow, food chains and decomposers

After this lesson, you can

- Assign producer and consumer roles within a stated food chain.
- Distinguish energy flow from nutrient cycling.

Start with: Producers capture energy; consumers obtain energy by feeding; decomposers break down organic material.

THEORY

Producers use light or chemical energy to make organic material from inorganic substances. Consumers obtain organic material by feeding. Decomposers break down dead material and help recycle nutrients. Energy passes through feeding levels and much becomes dispersed heat. Nutrients cycle, but usable energy is not recycled in the same way.

Food-chain arrows show transfer to the consumer



Energy transfers along the chain; some is dissipated as heat at each stage.

Arrows point from food to consumer. This is a simplified food chain; real ecosystems contain food webs. Nutrients cycle, while energy flows through the system.

RULE

Producer → primary consumer → secondary consumer

A method that earns marks

1. Start the chain with its producer and draw arrows in the direction of food-energy transfer.
2. Assign each consumer level from what that organism eats in the stated chain.
3. Explain that energy is used and dissipated at each transfer.

4. Describe decomposers returning nutrients to the environment without claiming energy is recycled.

WORKED EXAMPLE

In grass to grasshopper to frog, grass is the producer, grasshopper the primary consumer and frog the secondary consumer.

1. Grass captures light energy and makes organic material, so it is the producer.
2. The grasshopper eats the producer and is the primary consumer.
3. The frog eats that consumer and is the secondary consumer.
4. The arrows grass → grasshopper → frog show transfer toward the eater; dead material from all levels can enter decomposer pathways.

Common mistake
The 10 percent rule is a rough teaching approximation, not a universal exact constant.

Exam habit: Consumer levels are roles in the specified chain; do not assign a permanent level without looking at the feeding relationship.

SELF-CHECK

Does energy cycle back through the ecosystem in the same way as mineral nutrients?

Answer: No. Energy flows through and is eventually dissipated as heat; nutrients can be recycled.

LESSON 03 / 8

Population growth and limiting factors

After this lesson, you can

- Calculate net population change from four demographic processes.
- Explain how a limiting factor can constrain population growth.

Start with: Population means members of one species in a defined area and period; basic signed addition.

THEORY

Population size changes through births, deaths, immigration and emigration. Rapid exponential growth can occur when resources are abundant. Limited food, space and other conditions constrain growth. Carrying capacity describes a population size an environment can sustain under specified conditions.

RULE

$$\Delta N = \text{births} + \text{immigration} - \text{deaths} - \text{emigration}$$

A method that earns marks

1. Set the population boundary and time interval.
2. Treat births and immigration as additions and deaths and emigration as removals.
3. Add the gains and subtract the losses to obtain net change.
4. If explaining growth limits, link a specific resource or condition to survival or reproduction.

WORKED EXAMPLE

If births are 30, deaths 10, immigration 5 and emigration 8, the population increases by 17.

1. The example gives 30 births and 5 immigrants, so total additions are 35.
2. It gives 10 deaths and 8 emigrants, so total removals are 18.
3. Net change is 35 - 18 = +17 individuals.
4. This is the increase during the interval, not the final population; a starting population would be needed to calculate the latter.

Common mistake
Carrying capacity is not an unchanging property independent of habitat conditions.

Exam habit: Label a positive result increase and a negative result decrease; do not report net change as the final population size.

SELF-CHECK

If births equal deaths, must population size stay unchanged?

Answer: No. Immigration and emigration can still produce a net change.

LESSON 04 / 8

Relationships among organisms

After this lesson, you can

- Identify ecological interactions from their effects on each partner.
- Support an interaction label with scenario evidence.

Start with: Benefit, harm and little measurable effect are relative to the organisms in the interaction.

THEORY

Competition occurs when organisms use a limited shared resource. Predation involves one organism consuming another. In mutualism both benefit; in commensalism one benefits without a substantial effect on the other; in parasitism one benefits at the host expense.

RULE

Mutualism $(+, +)$

commensalism $(+, 0)$

parasitism $(+, -)$

A method that earns marks

1. Identify the two organisms and the resource or activity linking them.
2. Describe the effect on the first organism and on the second separately.
3. Match the two effects to the appropriate interaction category.
4. Use evidence from the case rather than assuming all close associations are mutualistic.

WORKED EXAMPLE

A parasite obtaining nutrients while harming its host is a plus-minus interaction. Two plants competing for limited light are both constrained by competition.

1. The parasite obtains nutrients, so it benefits from the interaction.
2. The host is harmed, giving the benefit/harm pattern commonly written plus/minus.
3. The two competing plants both face restricted access to limited light.
4. Competition is therefore a minus/minus interaction in the comparison with unrestricted resource access, even if one competitor does better than the other.

Common mistake
A relationship may be context-dependent; answer using the effects given in the question.

Exam habit: Write the effect on each partner before naming the interaction; this makes similar scenarios easier to distinguish.

SELF-CHECK

Does one competitor winning mean competition is automatically a benefit/harm interaction?

Answer: No. Both incur the effects of competing for a limited resource; one may still outperform the other.

LESSON 05 / 8

Pollution: sources, pathways and effects

After this lesson, you can

- Build a source-pathway-receptor explanation of pollution harm.
- Select prevention that interrupts a stated nutrient-runoff pathway.

Start with: Nutrients support growth; decomposition can consume dissolved oxygen.

THEORY

Pollution changes environmental conditions in ways that can cause harm or reduce usefulness. Sources can be natural or human-made. Air, water and soil pollution are connected through transport and deposition. A useful answer links a source to a pathway, an effect and a control.

RULE

Source → pathway → receptor → effect

A method that earns marks

1. Name the pollution source and the released material.
2. Trace how the material reaches the affected ecosystem.
3. Explain the intermediate process connecting exposure to the final effect.
4. Choose a preventive measure and state which link it reduces.

WORKED EXAMPLE

Nutrient-rich runoff can stimulate excessive algal growth. Subsequent decomposition can consume dissolved oxygen and harm aquatic organisms.

1. Excess nutrients from fertilizer can be carried by runoff into a water body.
2. Nutrient enrichment can stimulate excessive algal growth.
3. When organic material is decomposed, microbial respiration can consume dissolved oxygen.
4. Low oxygen can harm fish; reducing excess fertilizer or intercepting runoff addresses the source or pathway before that effect occurs.

Common mistake
Do not say algae always produce an immediate oxygen shortage without explaining decomposition and conditions.

Exam habit: Use connecting verbs such as carries, stimulates and consumes to make every step in the causal chain explicit.

SELF-CHECK

Why is algae cause fish death an incomplete explanation?

Answer: It omits the conditions and intermediate processes, especially decomposition and dissolved-oxygen depletion.

LESSON 06 / 8

Ecological footprint and biocapacity

After this lesson, you can

- Compare ecological demand with biocapacity on a common basis.
- Interpret a numerical balance as a deficit or reserve.

Start with: Demand versus available capacity; subtraction; per-person and total quantities must not be mixed.

THEORY

Ecological footprint estimates demand for biologically productive area needed to support consumption and absorb certain wastes. Biocapacity represents available biological production capacity. A deficit occurs when demand exceeds capacity. A global hectare is a productivity-adjusted unit, not simply any local hectare.

RULE

$$\text{Ecological deficit} = \text{footprint} - \text{biocapacity}$$

if positive

A method that earns marks

1. Identify footprint as demand and biocapacity as available biological production capacity.
2. Check that both figures use the same units, time basis and population basis.
3. Subtract biocapacity from footprint to test for a deficit.
4. Explain the result within the indicator scope rather than treating it as every environmental impact.

WORKED EXAMPLE

A population with footprint 3 global hectares per person and biocapacity 2 has a deficit of 1 per person.

1. The example footprint is 3 global hectares per person, describing demand.
2. Biocapacity is 2 global hectares per person, describing available capacity on the same basis.
3. Demand exceeds capacity by 3 - 2 = 1 global hectare per person.
4. The result is an ecological deficit, not an area measured in ordinary unadjusted local hectares.

Common mistake
Ecological footprint and carbon footprint are related but not identical measures.

Exam habit: Write global hectares per person in a per-person result; omitting the population basis changes its meaning.

SELF-CHECK

Can a region total footprint be compared directly with biocapacity per person?

Answer: No. Convert both to totals or both to per-person values first.

LESSON 07 / 8

Waste management and the hierarchy

After this lesson, you can

- Distinguish prevention, reuse, recycling, recovery and disposal.
- Apply the waste hierarchy to a concrete material scenario.

Start with: Reuse keeps an item in service; recycling reprocesses its material.

THEORY

Waste management involves preventing, collecting, sorting, treating and disposing of waste. Prevention and reduction usually come before reuse, recycling, recovery and final disposal. Hazardous items require appropriate separate handling; not every household discard belongs in an ordinary bin.

RULE

Prevent/reduce → reuse → recycle → recover → dispose

A method that earns marks

1. Identify whether the action avoids waste, keeps the item in use or processes discarded material.
2. Classify the action using the general hierarchy.
3. For a proposed choice consider prevention or reduction first where feasible.
4. Check material hazards and the approved local collection stream before actual disposal.

WORKED EXAMPLE

Refilling a reusable bottle is reuse. Turning sorted used glass into new glass is recycling. Avoiding unnecessary packaging is prevention.

1. Refilling a durable bottle keeps the same item in service, so it is reuse.
2. Processing separated used glass into new glass changes the material into a new product, so it is recycling.
3. Avoiding unnecessary packaging prevents that material becoming waste at all.
4. These actions belong at different stages of the hierarchy and should not all be labelled recycling.

Common mistake
Recycling does not erase all upstream resource use or justify unlimited consumption.

Exam habit: Use the action in the scenario as evidence: avoid means prevention, use again means reuse, reprocess means recycling.

SELF-CHECK

Is using a clean jar again to store suitable items recycling?

Answer: It is reuse, because the same item is used again without material reprocessing.

LESSON 08 / 8

Sustainable development and integrated decisions

After this lesson, you can

- Assess a project across environmental, social and economic dimensions.
- State measurable indicators and a trade-off for a sustainability decision.

Start with: Present and future needs; a benefit in one dimension can have costs in another.

THEORY

Sustainable development addresses present needs while preserving future ability to meet needs. Environmental, social and economic dimensions interact. A project can improve one dimension while creating costs in another, so evaluation should consider benefits, trade-offs and long-term effects.

RULE

Environment + society + economy

A method that earns marks

1. Define the project and the needs it is intended to meet.
2. Give one environmental, one social and one economic effect.
3. Select an indicator that could test each effect.
4. Explain one trade-off over the project life rather than declaring the project sustainable from a single benefit.

WORKED EXAMPLE

A public-transport project can reduce emissions, improve access and lower travel costs, but its construction and operation also require resources and funding.

1. The public-transport example can reduce transport emissions if its use changes travel patterns.
2. Improved access to education or work is a social benefit to evaluate.
3. Travel costs, operating costs and funding provide economic questions.
4. Construction also consumes resources and money, so a full assessment compares these costs with benefits over time.

Common mistake
Do not equate sustainable development with only economic growth or only nature protection.

Exam habit: Replace vague words such as greener with a measurable indicator such as electricity use, emissions, access or cost over a stated period.

SELF-CHECK

Is lower operating energy alone enough to prove a project is sustainable?

Answer: No. Social access, economic viability, construction impacts and long-term effects also require assessment.

Before the tests

These original questions practise the lesson objectives. They are not predictions of an exam paper. Use the separate practice book if you want all papers together.

Humans and the Environment / Test A

20 marks | Suggested time: 20 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. Which is an abiotic factor?

   A. Algae

   B. Bacteria

   C. Water temperature

   D. Fish

2. Decomposers mainly help by:

   A. Recycling nutrients from dead matter

   B. Creating sunlight

   C. Stopping all energy loss

   D. Removing all producers

3. Which directly reduces a local population?

   A. Reproduction

   B. Births

   C. Immigration

   D. Emigration

4. Both organisms benefit in:

   A. Parasitism

   B. Mutualism

   C. Predation

   D. Competition

Part II: Show your reasoning

1. Classify sunlight, grass, soil mineral content and a rabbit as biotic or abiotic.

2. A question specifies 10 percent transfer efficiency at each step. Producers have 10000 energy units. Find energy available to primary and secondary consumers.

3. A population begins with 200 individuals. During a period there are 25 births, 12 deaths, 8 immigrants and 6 emigrants. Find the final size.

4. Identify a relationship where both benefit and another where one benefits while the other is harmed. Explain the signs.

Humans and the Environment / Test B

20 marks | Suggested time: 20 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. Excess nutrient input into water is most closely associated with:

   A. Eutrophication

   B. Lens refraction

   C. Sterilization

   D. No biological change

2. A footprint larger than biocapacity indicates:

   A. Unlimited sustainability

   B. An ecological reserve

   C. An ecological deficit

   D. Zero resource use

3. Which is highest in the general waste hierarchy?

   A. Disposal after mixing

   B. Landfill

   C. Uncontrolled burning

   D. Waste prevention

4. A complete sustainability assessment considers:

   A. Only this month

   B. Environmental, social and economic effects

   C. Only profit

   D. Only tree numbers

Part II: Show your reasoning

1. Explain a three-step route from excess fertilizer use to fish harm and name one preventive measure.

2. A region has footprint 4.5 and biocapacity 2.8 global hectares per person. Identify and calculate the balance.

3. Classify avoiding excess packaging, reusing a jar, reprocessing aluminium cans and placing residual waste in an approved landfill.

4. Give one environmental, one social and one economic indicator for a campus energy-efficiency project, plus one trade-off to evaluate.

Humans and the Environment / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **C.** Temperature is a physical condition rather than a living organism.
2. **A.** They break down organic matter and return nutrients to the system.
3. **D.** Emigration removes individuals from the local population.
4. **B.** Mutualism is defined by benefits to both participants.

Part II: Written solutions

1. Sunlight and mineral content are abiotic. Grass and rabbit are biotic. One mark each.
2. Primary consumers receive 1000 units; secondary consumers receive 100 units. Two marks for each transfer. These values depend on the stated efficiency.
3. Net change =25-12+8-6=15. Final size=215. Credit signs of all four processes and final total.
4. Mutualism is (+,+): both gain a benefit. Parasitism is (+,-): parasite benefits and host is harmed. One mark for each name and each explanation.

Humans and the Environment / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **A.** Nutrient enrichment can drive excessive growth and subsequent oxygen depletion.
2. **C.** Demand exceeds the available biological capacity under the accounting model.
3. **D.** Avoiding waste reduces the need for later treatment and disposal.
4. **B.** The concept integrates all three dimensions and future consequences.

Part II: Written solutions

1. Runoff carries nutrients to water; algal growth increases; decomposition can lower dissolved oxygen and harm fish. Reduce excess nutrient application or intercept runoff. One mark for each causal step and one for prevention.
2. It has an ecological deficit of 4.5-2.8=1.7 global hectares per person. Credit comparison, subtraction, units and deficit interpretation.
3. Prevention, reuse, recycling and disposal, respectively. One mark for each classification.
4. Examples: electricity use/emissions; comfort or accessibility; cost savings/payback; upfront cost versus long-term savings. One mark for each relevant, measurable item.

Source and coverage notes

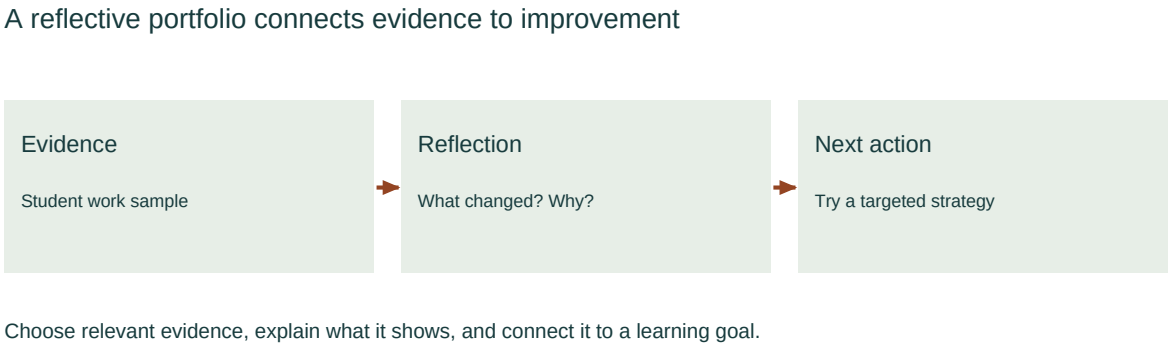
ENV101 elective: ecosystems, relationships among organisms, pollution, ecological footprint, waste management and sustainable development. The Arabic source lectures are explained here in English.

Source folder: Open original course folder The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. This revised study aid does not replace lecturer-specific derivations or instructions. All practice tests and suggested marking guidance are original study material.

08

Professional Development Seminar

A field guide to understanding, working and checking.



Elective lecture themes: teaching as a profession, teacher formation, professional learning, portfolios, standards and educational technology. This follows the teacher-development focus of the supplied file; it is not a generic career-skills course.

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8 lessons / 2 practice tests / Worked solutions

English edition / November 2026

How to use this handout

1. Read the concept and check the symbol meanings before choosing a formula.
2. Cover the worked steps and reproduce the reasoning yourself.
3. Attempt the quick check and the two subject tests without notes.
4. Mark your work; record whether the error was concept, method, algebra, units or reading. Retry after a delay.

Your study route

1. Teaching as a profession
2. Professionalization and the teacher role
3. Teacher formation: before and during service
4. Preparation, practice and professional standards
5. Plan continuing professional development
6. Portfolios and reflective evaluation
7. Ethics, collaboration and learning communities
8. Educational technology and evaluation of impact

LESSON 01 / 8

Teaching as a profession

After this lesson, you can

- Explain two criteria that make teaching a profession.
- Support each criterion with a distinct classroom example.

Start with: A professional criterion is a defining feature, not simply an employment benefit.

THEORY

The source distinguishes a profession through specialized knowledge, extended preparation, social purpose, ethical responsibilities and organized practice. Teaching is more than delivering facts: it develops learners and contributes to society. A strong exam response connects each professional criterion to a teaching example.

RULE

Criterion + explanation + teaching example

A method that earns marks

1. Define profession using expertise, preparation, social purpose and ethical responsibility.
2. Select two distinct criteria relevant to the question.
3. Explain each criterion in plain language.
4. Attach a concrete teaching example to each explanation.

WORKED EXAMPLE

Specialized knowledge includes subject expertise and pedagogy; ethical responsibility includes fair assessment and respect for learners.

1. Subject expertise concerns what is taught; pedagogy concerns how learners are helped to understand it.
2. Together they illustrate specialized professional knowledge rather than information delivery alone.
3. Fair assessment and respect for learners illustrate ethical responsibility.
4. The examples support two different criteria, so they provide more evidence than repeating expertise in several forms.

Common mistake
Do not assume being paid alone makes an activity a profession.

Exam habit: Use two complete criterion-explanation-example pairs when four marks are available, rather than a long unsupported list.

SELF-CHECK

Does being paid alone establish that an activity is a profession?

Answer: No. Professional status also involves specialized preparation, knowledge, responsibility and organized standards of practice.

LESSON 02 / 8

Professionalization and the teacher role

After this lesson, you can

- Describe teacher roles through observable professional actions.
- Explain how assessment evidence guides subsequent instruction.

Start with: A role describes a function; an action shows how that function appears in practice.

THEORY

The lecture describes development from pre-professional through semi-professional toward growing professionalization. Teacher roles extend from presenting knowledge to guiding learning, assessing progress, modelling values and supporting society. Historical labels describe development, not a reason to dismiss present expertise.

RULE

Role → observable professional action

A method that earns marks

1. Identify the role asked about, such as assessor or learning guide.
2. Describe a specific action the teacher takes in that role.
3. Explain how the action supports learners.
4. For historical comparisons, discuss changes in preparation or organization rather than dismissing current expertise.

WORKED EXAMPLE

A teacher acting as an assessor uses evidence from student work to adapt instruction. Acting as a guide, the teacher helps learners reason rather than supplying every answer.

1. As an assessor, the teacher examines student work to identify what learners understand.
2. If several answers show the same error, that evidence indicates a learning gap.
3. Adapting instruction in response connects the assessor role to better teaching decisions.
4. As a guide, the teacher supports reasoning through questions or strategies instead of supplying every answer.

Common mistake
A list of role names without explanation earns less evidence of understanding.

Exam habit: After every role name add a verb and object: diagnoses a misconception, gives targeted feedback or guides strategy choice.

SELF-CHECK

Why is gives a test alone a weak explanation of the assessor role?

Answer: It names an activity but does not explain how evidence is interpreted or used to support learning.

LESSON 03 / 8

Teacher formation: before and during service

After this lesson, you can

- Distinguish pre-service preparation, induction and continuing development.
- Match a support activity to its career stage and purpose.

Start with: Teachers develop before entering practice and throughout their careers.

THEORY

Teacher formation is a continuing process. Pre-service preparation establishes knowledge and skills; induction supports entry to actual teaching; continuing professional development addresses changing needs and improves practice. Training is one method of development, not the whole process.

RULE

Pre-service → induction → continuing development

A method that earns marks

1. Locate the person before employment, in early practice or in established service.
2. Name the matching formation stage.
3. Explain the specific knowledge or support needed at that stage.
4. Describe how the activity supports practice rather than treating attendance as proof of improvement.

WORKED EXAMPLE

A mentor supporting a newly appointed teacher belongs to induction. Later peer observation and inquiry can support continuing development.

1. The example teacher is newly appointed and is entering real classroom practice.
2. Mentor support at this point is induction, bridging initial preparation and workplace demands.
3. Later peer observation and inquiry are examples of continuing development.
4. The method may look similar across stages, but the career context and purpose determine the classification.

Common mistake
Graduation marks an entry point, not the end of professional learning.

Exam habit: In a comparison give both timing and purpose for each stage; stating before and after alone is incomplete.

SELF-CHECK

Is every teacher workshop automatically pre-service preparation?

Answer: No. A workshop during employment may be induction support or continuing development, depending on career stage and purpose.

LESSON 04 / 8

Preparation, practice and professional standards

After this lesson, you can

- Explain how supervised field experience links theory with practice.
- Construct a feedback cycle using observable evidence.

Start with: Preparation includes subject knowledge, pedagogy and supervised teaching; standards describe expected practice.

THEORY

Teacher preparation combines subject knowledge, pedagogical understanding and supervised practice. Field experience connects theory with real learners and classroom decisions. Professional standards make expectations explicit and support self-evaluation. Licensing is a recognition process whose current requirements must be checked with the responsible body.

RULE

Knowledge → supervised application → feedback → improvement

A method that earns marks

1. Identify the teaching expectation or intended learner outcome.
2. Collect relevant evidence from observation and student work.
3. Choose one specific change suggested by that evidence.
4. Apply the change and evaluate the next attempt against the same expectation.

WORKED EXAMPLE

After teaching a supervised lesson, a trainee reviews observations and student work, then revises the next lesson. This is a feedback cycle.

1. The trainee first teaches a supervised lesson, putting planned ideas into practice.
2. Observation notes and student work show what occurred and what learners achieved.
3. The trainee compares that evidence with the intended learning and revises the next lesson.
4. The cycle is completed by checking whether the revision improves the next observed outcome.

Common mistake
Do not invent current licence requirements from an older lecture.

Exam habit: When standards are mentioned, link one expectation to evidence; do not invent current licensing rules from a lecture.

SELF-CHECK

Why does feedback without a subsequent change and evaluation leave the cycle incomplete?

Answer: It identifies an issue but does not show whether the teacher acted on it or improved practice.

LESSON 05 / 8

Plan continuing professional development

After this lesson, you can

- Write a professional-development plan linking need, goal, activity and evidence.
- Distinguish participation evidence from evidence of impact.

Start with: A development goal should address an identified teaching or learning need.

THEORY

Professional development is purposeful and sustained. It begins with a learning need, sets a clear goal, chooses suitable activities and evaluates effects on practice and learners. Workshops, peer observation, professional reading and collaborative inquiry can all contribute.

RULE

Need → goal → activity → evidence of impact

A method that earns marks

1. Use classroom evidence to name one specific development need.
2. Write a focused goal and a realistic review period.
3. Choose a learning activity and classroom application that address the goal.
4. Compare relevant evidence before and after the application to judge progress.

WORKED EXAMPLE

A teacher notices weak student explanations, studies questioning strategies, tries them for four weeks and compares the quality of student responses.

1. The teacher notices weak student explanations, identifying a specific need.
2. Studying questioning strategies is the development activity, but study alone does not show impact.
3. Trying the strategies for four weeks puts the learning into classroom practice.
4. Comparing response samples before and after can reveal whether explanations became clearer and better supported.

Common mistake
Attendance certificates show participation, not necessarily improved teaching.

Exam habit: Keep all four plan parts linked to the same need; an unrelated workshop does not satisfy a coherent development plan.

SELF-CHECK

What does a workshop attendance certificate demonstrate directly?

Answer: Participation. Improved teaching or learning needs additional evidence from practice and outcomes.

LESSON 06 / 8

Portfolios and reflective evaluation

After this lesson, you can

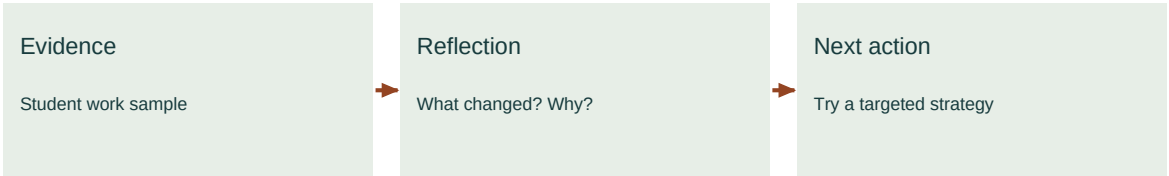
- Select portfolio evidence for a defined goal or standard.
- Explain how reflection turns an artifact into evidence of development.

Start with: An artifact is a document or sample; evidence requires an explanation of what it demonstrates.

THEORY

A professional portfolio is a purposeful selection of evidence linked to goals and standards, with reflection on what the evidence demonstrates. It can include plans, student work, feedback and records of improvement. A large unorganized collection is less useful than a selective, explained one.

A reflective portfolio connects evidence to improvement



Choose relevant evidence, explain what it shows, and connect it to a learning goal.

A useful portfolio explains the meaning of selected evidence and identifies a next action. Collecting many documents without reflection is not enough.

RULE

Evidence + reflection + link to a goal

A method that earns marks

1. State the portfolio purpose or professional goal.
2. Select a small number of relevant artifacts and organize them clearly.
3. Explain what each artifact shows and what its limitations are.
4. Reflect on the outcome and identify a next improvement, protecting learner privacy.

WORKED EXAMPLE

A lesson plan paired with student work and a reflection on changes shows more than a lesson plan alone. It connects intention, outcome and learning.

1. The lesson plan shows what the teacher intended learners to do.
2. Student work shows something about the resulting learning and where difficulties remained.
3. A reflection compares the plan with that outcome and explains why a change was made.
4. The three pieces together support a development claim more clearly than an unexplained plan alone.

Common mistake
Do not include identifiable student information unnecessarily; follow the institution privacy rules.

Exam habit: For each item finish the sentence this demonstrates… and connect it to a goal or standard; anonymize learner material as required.

SELF-CHECK

Does adding more unrelated documents necessarily make a portfolio stronger?

Answer: No. Relevance, clear organization and reflection matter more than volume alone.

LESSON 07 / 8

Ethics, collaboration and learning communities

After this lesson, you can

- Rewrite personal criticism as evidence-based constructive feedback.
- Connect ethical collaboration with learner respect and improvement.

Start with: Observation is what can be seen or recorded; interpretation and personal judgement are different.

THEORY

Professional learning can be collaborative: colleagues discuss evidence, share practice and solve problems. Ethical practice includes respect, fairness, honesty and responsibility. Feedback should focus on observable practice and improvement, rather than personal attacks.

RULE

Observation + effect + constructive next step

A method that earns marks

1. Identify an observable feature of teaching rather than a personal label.
2. Describe its apparent effect on learners with available evidence.
3. Suggest one feasible next step connected to that effect.
4. Use respectful language and follow confidentiality and reporting responsibilities.

WORKED EXAMPLE

A colleague gives feedback that a task confused learners and suggests clearer success criteria, rather than calling the teacher incompetent.

1. Calling a teacher incompetent attacks the person and gives no specific route to improvement.
2. The example instead identifies that the task confused learners.
3. Suggesting clearer success criteria connects the feedback to a teaching action the colleague can try.
4. A later check of learner understanding can show whether that change helped, completing the improvement conversation.

Common mistake
Collaboration is not simply agreeing with everyone or sharing private student details.

Exam habit: Structure scenario answers as observation, learner effect and next step; avoid unsupported claims about personal character.

SELF-CHECK

Is constructive feedback the same as agreeing with every colleague decision?

Answer: No. It can challenge practice, but uses evidence, respect and actionable suggestions.

LESSON 08 / 8

Educational technology and evaluation of impact

After this lesson, you can

- Choose an educational tool to serve a specific learning objective.
- Evaluate technology through diagnosis, teacher response and learner evidence.

Start with: A tool supports an activity; learning depends on how the activity and follow-up are designed.

THEORY

Educational technology should serve a learning goal. Teachers need to select suitable tools, design learning activities, assess outcomes and consider access and ethical use. Technical operation alone does not establish teaching value. The seminar mentions digital competencies; programming instruction is excluded from this pack as requested.

RULE

Learning need → tool choice → activity → evaluation

A method that earns marks

1. Name the learner objective and the misconception or difficulty to investigate.
2. Choose a suitable accessible tool and design questions that reveal reasoning.
3. Inspect response patterns and adapt teaching accordingly.
4. Use a new question or response sample to judge whether understanding improved.

WORKED EXAMPLE

A short digital quiz can reveal misconceptions immediately, but the teacher must use the results to adapt instruction for it to support learning.

1. The brief digital quiz can make response patterns visible quickly.
2. Plausible incorrect options help reveal which misconception may be present.
3. The teacher must interpret those results and respond, for example with targeted explanation or discussion.
4. A fresh question then checks learning; the number of clicks or time on screen alone does not establish educational value.

Common mistake
More screen time is not itself evidence of better learning.

Exam habit: State the learning goal before the tool name and finish with evidence of learning; programming content is outside this course pack.

SELF-CHECK

What is missing if a teacher runs a digital quiz but never uses its results?

Answer: The evidence-to-action step: diagnosis has not been used to adapt instruction or check improvement.

Before the tests

These original questions practise the lesson objectives. They are not predictions of an exam paper. Use the separate practice book if you want all papers together.

Professional Development Seminar / Test A

20 marks | Suggested time: 20 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. Which is a professional criterion emphasized in the lecture?

   A. No responsibility to society

   B. Learning stops at graduation

   C. No preparation

   D. Specialized knowledge and ethical standards

2. Using student work to improve instruction best illustrates the teacher as:

   A. A storage clerk only

   B. An assessor and learning guide

   C. A person who never changes plans

   D. A passive spectator

3. Support during a teacher first years of practice is mainly:

   A. Only pre-service admission

   B. Induction

   C. Retirement

   D. Waste management

4. The central value of supervised field experience is:

   A. Eliminating evaluation

   B. Replacing all theory

   C. Connecting theory to practice with feedback

   D. Avoiding learners

Part II: Show your reasoning

1. Give two reasons teaching is considered a profession, each with an example.

2. Describe two teacher roles beyond explaining subject content.

3. Compare pre-service preparation with induction in two clear points.

4. Give a four-step improvement cycle for a trainee after an observed lesson.

Professional Development Seminar / Test B

20 marks | Suggested time: 20 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. Which is the strongest development goal?

   A. Improve one identified practice and evaluate its effect

   B. Collect the most certificates

   C. Become better somehow

   D. Attend anything

2. What makes a portfolio stronger?

   A. No evaluation

   B. More unrelated documents

   C. Evidence linked to goals and reflection

   D. Only decorative pages

3. Constructive professional feedback should be:

   A. Only praise regardless of evidence

   B. Unrelated to teaching

   C. Personal and humiliating

   D. Based on observation and useful next steps

4. The best reason to choose an educational tool is:

   A. It supports a defined learning objective

   B. It has the most animations

   C. Everyone must use the same tool

   D. It is newest

Part II: Show your reasoning

1. Write a short development plan for a teacher whose students struggle to explain answers.

2. Name two portfolio items and explain what each can demonstrate.

3. Rewrite this feedback constructively: Your lesson was bad.

4. Propose a simple technology use to diagnose misconceptions, and state how you would evaluate its value.

Professional Development Seminar / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **D.** Professional practice combines expertise, preparation and ethical obligations.
2. **B.** Assessment information guides the next teaching decision.
3. **B.** Induction bridges initial preparation and independent professional practice.
4. **C.** Practice tests and develops professional knowledge in authentic settings.

Part II: Written solutions

1. Specialized preparation: teachers study subject knowledge and pedagogy. Ethical duty: teachers assess fairly and protect learner dignity. Two marks for each criterion-example pair.
2. Assessor: identifies learning gaps using evidence and feedback. Learning guide: supports students in choosing strategies and solving problems. Two marks per role with a concrete action.
3. Pre-service occurs before entry and develops foundational subject and pedagogical knowledge. Induction occurs during early practice and provides mentoring and support in the real workplace. One mark each for stage and purpose in both cases.
4. Review evidence; identify one specific weakness; plan and apply a targeted change; evaluate the next lesson using evidence. One mark per coherent step.

Professional Development Seminar / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **A.** A focused goal connects action to measurable change.
2. **C.** The reader can see what the evidence proves and how practice developed.
3. **D.** Evidence-based feedback supports improvement while respecting the colleague.
4. **A.** Pedagogical purpose should guide selection.

Part II: Written solutions

1. Need: weak explanations. Goal: improve evidence-based responses. Activity: learn and apply structured questioning. Evidence: compare student response samples before and after a defined period. One mark per linked component.
2. A lesson plan can demonstrate planning and alignment. An anonymized student-work sample with commentary can demonstrate learning and the effect of feedback. Two marks per item and explanation.
3. Example: Several students could not identify the task goal; stating the success criteria and checking one example before independent work may help. Credit observable evidence, effect, actionable suggestion and respectful language.
4. Use a brief concept quiz with plausible distractors. Review response patterns, reteach the misconception and check with a new question. Evaluate whether explanations or transfer improve, not only whether students clicked. One mark each for tool, diagnosis, response and evaluation.

Source and coverage notes

Elective lecture themes: teaching as a profession, teacher formation, professional learning, portfolios, standards and educational technology. This follows the teacher-development focus of the supplied file; it is not a generic career-skills course.

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